The Anatomy of Wage Compression: Locally Robust Inference for Conditional Inequality and Exact Decomposition

Vedant Bhardwaj*
Universidad Carlos III de Madrid

October 2026

Abstract

An aggregate decline in wage inequality may reflect convergence in group mean wages or reduced dispersion among comparable workers. This paper develops a methodology that allows for an exact decomposition of these two dimensions conditional on a large number of observable factors. In particular, I combine combines an exact within–between decomposition of the Theil index with conditional-inequality profiles, accommodating high-dimensional worker, job, and employer characteristics through regularized learners. I derive a Neyman-orthogonal signal for conditional inequality and locally robust estimators for the decomposition, using cross-fitting to provide valid inference. Applying the framework to Portuguese matched employer–employee data from 2011 to 2019, a period of substantial minimum-wage increases, I find that the between-component decline slightly exceeds the aggregate decline, with about 95 percent captured by conditioning on worker and job characteristics and hence a limited role for firm characteristics. Despite a nearly unchanged aggregate within component, conditional inequality generally declines among non-college workers, particularly among high-school and some-college women and men with greater initial exposure to the 2014 minimum-wage increase. College-educated men show little comparable decline. These findings reveal wage compression for specific population groups concealed by aggregate measures.

*Universidad Carlos III de Madrid, Department of Economics. Email: vbhardwa@eco.uc3m.es. I am deeply grateful to my advisor, Juan Carlos Escanciano, for his invaluable guidance, patience, and generous time. I am also indebted to Jesús Fernández-Huertas Moraga for his extensive comments and support throughout this project. I thank Carlos Velasco, Jesús Carro, and Jan Stuhler for helpful comments and suggestions.

1 Introduction

When wage inequality falls, who becomes more equally paid? Wage compression is widespread: the International Labour Organization reports declining wage inequality since the start of the century in roughly two-thirds of the countries it covers (International Labour Organization, 2024). Rising minimum wages have contributed to compression in Brazil (Engbom & Moser, 2022), Uruguay (Bergolo et al., 2025), and Germany (Dustmann et al., 2022; Liang, 2026). An aggregate decline, however, does not reveal where compression occurred. Mean wage gaps may narrow across groups defined by observed worker, job, and employer characteristics or dispersion may fall among workers sharing those characteristics, or both. The distinction matters because declining aggregate inequality does not necessarily imply that wage differences among otherwise comparable workers have narrowed. Distinguishing these dimensions requires a framework that measures inequality among comparable workers while accommodating rich conditioning information and preserving an exact connection to aggregate inequality.

This paper develops an econometric framework for estimating these different dimensions of wage inequality and conducting inference with high-dimensional covariates. I apply the methodology to Portugal during a period of substantial minimum-wage increases. After remaining unchanged during 2011–2013, the monthly minimum wage increased five times between 2014 and 2019, from 485 to 600 euros, a nominal increase of approximately 24 percent. Before each increase, roughly 15–23 percent of workers in the analysis samples earned monthly base pay between the old and new minimum wage. Over 2011–2019, the Theil index of hourly wages declined by approximately 16 percent. It changed little during the freeze and fell every year once the increases began (Figure 3). Previous research documents the importance of minimum-wage increases in Portugal's wage compression (Oliveira, 2023, 2024). My application addresses a complementary question: where in the wage structure compression occurred and whether the changes were concentrated in employers most exposed to the increases.

The main empirical finding is that the decline in aggregate inequality is concentrated in the between component, which measures inequality in conditional mean wages. Under the richest conditioning set, the between component declines by approximately 22.4 percent between 2011 and 2019, a decline that slightly exceeds the total decline in the Theil index in absolute terms, while the aggregate within component, a wage-share-weighted average of conditional inequality, changes little. Most of the decline in between inequality is already present when conditioning on worker and job characteristics alone. Adding industry, location, and observed employer characteristics raises the level of measured between inequality but accounts for very little of its decline.

The small change in the aggregate within component conceals substantial heterogeneity. Adapting the wage-gap approach of Machin et al. (2003) and Draca et al. (2011), I measure initial employer exposure as the additional base-pay shortfall implied by the 2014 minimum-wage increase, relative to the employer's 2013 base wage bill. Within the exposure samples, adding this measure to the richest conditioning set increases the between component by less than one percent of total Theil inequality. The fitted conditional-inequality profiles are nevertheless

generally lower in 2019 than in 2011 for workers without a college degree, with the largest differences between the two years at higher exposure among women and men with high school or some college education. College-educated men show little comparable decline over most of the exposure range (Figure 8). These profiles average over workers observed at each exposure level in each year, so their changes can reflect changes in both conditional dispersion and workforce composition.

Why is it difficult to measure these patterns using existing approaches? The central challenge is defining comparable workers. If groups are defined by education alone, age-related differences in mean wages enter measured within-group inequality. Existing studies examine demographic wage gaps (Oka & Yamada, 2023) or inequality across industries (Garnero et al., 2015), and conditioning on additional characteristics separates more of these systematic differences. However, constructing separate cells becomes impractical when many variables, including continuous ones, are jointly considered. Decompositions into within- and between-firm inequality (Bergolo et al., 2025; Liang, 2026; Song et al., 2019) address a different question: workers within the same firm can differ in their characteristics, while workers sharing the same observed characteristics can work for different firms.

These limitations motivate a regression-based approach to inequality measurement that accommodates rich conditioning information while preserving an exact connection to aggregate inequality. I use the Theil index (Theil, 1967), which decomposes exactly into between and within components (Bourguignon, 1979; Shorrocks, 1980). For a vector of observed characteristics X, the between component is the Theil index of conditional mean wages, $\mathbb{E}[Y \mid X]$, while the within component is a wage-share-weighted average of Theil inequality among workers sharing those characteristics. The two components sum to aggregate Theil inequality. This decomposition remains well defined with continuous or high-dimensional conditioning variables, without requiring a finite partition into groups. Holding the population fixed, comparisons across nested conditioning sets quantify how much inequality is reclassified from within to between when further characteristics are introduced.

The same framework permits a more detailed analysis of within-group dispersion. Let $T_0(x)$ denote the Theil index of the conditional distribution of wages Y given $X = x$. This function measures wage inequality among workers with the specified observed characteristics. Although useful conceptually, the full function is difficult to estimate and interpret when X contains many variables. I therefore consider lower-dimensional summaries that average conditional inequality over selected characteristics. In the empirical application, these summaries describe how average conditional wage dispersion varies with employer exposure to minimum-wage increases, separately by sex and education. Unlike an aggregate within component, which collapses these differences into a single wage-share-weighted number, the profiles reveal where conditional dispersion is relatively high or low and how the patterns evolve over time.

Estimating these quantities raises an important econometric problem. Both the decomposition components and conditional inequality depend nonlinearly on unknown conditional moments of wages. Flexible estimation accommodates many worker, job, and employer characteristics, but regularization and estimation errors can propagate into plug-in inequality estimates at first

order. In particular, errors in estimated conditional mean wages can distort the allocation of inequality between the within and between components. A first stage that shrinks conditional mean wages toward the overall mean, for example, may understate inequality across worker types and overstate the dispersion remaining within them. Such distortions matter directly for the economic question because they can alter the estimated contribution of each component to aggregate wage compression.

To address this problem, I develop estimators based on locally robust moments whose expectations have zero first-order sensitivity to errors in the estimated conditional wage functions (Chernozhukov et al., 2022). The methodology distinguishes two types of objects. First, I construct Neyman-orthogonal moment conditions for the aggregate within and between components and for the additional between inequality associated with nested conditioning sets. These scalar targets require only conditional mean wage functions as nonparametric inputs, so they can be estimated directly without estimating the full conditional inequality surface. With cross-fitting, the estimators admit root-n inference under suitable first-stage convergence rates and regularity conditions. The estimated within and between components also add exactly to the sample Theil index.

Second, I derive an orthogonal signal for conditional Theil inequality. Its conditional expectation equals the target inequality index when evaluated at the true conditional moments, while its expectation is locally insensitive to errors in those moments. Following Semenova and Chernozhukov (2021), the signal can be combined with cross-fitting and series regression to estimate conditional inequality and lower-dimensional profiles. The approach accommodates flexible first-stage estimators while controlling their first-order contribution to estimation bias. Importantly, orthogonality does not eliminate the nonparametric approximation bias of the second-stage regression. The paper provides conditions under which this additional bias is negligible for pointwise inference. For the empirical exposure analysis, I use a growing cubic B-spline basis to estimate conditional-average profiles, retaining rich conditioning information in the first stage while reporting a small number of interpretable relationships.

The empirical analysis uses *Quadros de Pessoal*, a large matched employer–employee administrative dataset for Portugal (Card et al., 2016; Cardoso & Portugal, 2005). I first estimate the within and between components under four nested conditioning sets, using a common sample within each year. Beginning with worker and job characteristics, I successively add industry and location, employer scale, and other observed employer characteristics. This exercise shows that about 95 percent of the decline in between inequality is already present when conditioning on worker and job characteristics alone. Additional employer characteristics account for a nontrivial level of inequality in conditional mean wages, but their incremental contribution changes relatively little between 2011 and 2019. Thus, the observed decline in between inequality is predominantly associated with the distribution of average wages across worker and job types, rather than a decline in the additional inequality revealed by observed employer characteristics.

I then examine employer exposure to minimum-wage increases, measured using the additional base-pay shortfall implied by each increase relative to the employer's pre-increase base wage bill (Draca et al., 2011; Machin et al., 2003). The conditional-inequality profiles reveal considerable

variation in dispersion across exposure levels and demographic groups. The longer-run analysis holds employer exposure to the 2014 increase fixed at its value measured from 2013 wages, while a complementary analysis recalculates exposure before each of the five increases. These comparisons show that the evolution of wage inequality among observationally comparable workers differs across demographic groups and employers with different levels of minimum-wage exposure.

The interpretation of these results is descriptive. Changes in the aggregate components can reflect changes in conditional wage distributions, workforce composition, or both (DiNardo et al., 1996; Lemieux, 2006). Likewise, the empirical conditional-average profiles describe workers represented at each exposure level in each year; their changes need not hold the distribution of all other characteristics fixed. The analysis therefore does not identify causal effects of minimum-wage increases or isolate particular wage-setting mechanisms. Its contribution is to provide a more detailed accounting of the observed changes in the wage distribution, distinguishing dimensions of compression that conventional aggregate or firm-level summaries combine.

The paper contributes to two related literatures. The first contribution is econometric. Building on the locally robust estimation framework of Chernozhukov et al. (2022) and the debiased-signal regression approach of Semenova and Chernozhukov (2021), I derive an orthogonal signal for the conditional Theil index and establish sufficient conditions for estimation and pointwise inference. I also develop direct orthogonal estimators for the scalar components and nested increments of the exact decomposition. These results complement existing methods for estimation of functionals of conditional means, including the debiased machine-learning approach to inequality in predicted outcomes of Escanciano and Terschuur (2026). The main advantage of the framework is that it connects flexible adjustment for observed heterogeneity with interpretable measures of aggregate and conditional inequality, rather than treating these as separate empirical exercises.

The second investigates the evolution of wage inequality and the role of minimum wages. Studies have examined minimum-wage contributions to changes in aggregate wage inequality (Autor et al., 2016; Engbom & Moser, 2022; Lee, 1999; Oliveira, 2023), changes in conditional wage gaps across demographic groups (Oka & Yamada, 2023), and the role of firms in wage inequality (Bergolo et al., 2025; Card et al., 2013; Liang, 2026; Song et al., 2019). For Portugal, Leitão et al. (2022) study changes in firm pay premia. Relative to this literature, I focus on an exact decomposition based on a rich set of observed worker, job, and employer characteristics, complemented by conditional-inequality profiles. This distinction matters because inequality among workers sharing observed characteristics is neither the same as within-firm inequality nor the residual variation obtained after removing latent worker and firm effects.

The broader lesson from the Portuguese application is that understanding a decline in aggregate inequality requires looking beyond the aggregate decline itself. The between component identifies where most of the overall reduction is recorded, but the conditional analysis shows different changes in fitted dispersion profiles across worker groups and employer exposure levels. The proposed framework makes it possible to examine both dimensions within a common system of inequality measurement and inference.

The remainder of the paper is organized as follows. Section 2 introduces the conditional Theil index and the exact decomposition. Section 3 develops estimation and inference. Section 4 presents Monte Carlo evidence. Section 5 describes the data and empirical implementation, Section 6 reports the results, and Section 7 concludes.

2 Conditional Theil inequality and exact decomposition

This section defines the population objects underlying the empirical analysis. I first define the conditional Theil index, which measures wage inequality among workers who share observed characteristics, and its associated surface. Next, I derive the exact decomposition linking conditional inequality to aggregate wage inequality. I then examine how the decomposition changes when the conditioning information is expanded, showing how additional characteristics change the measured allocation of inequality between the within and between components while holding the population fixed. Finally, I introduce two ways of summarizing conditional inequality along selected characteristics: fixed-reference partial means and conditional-average profiles. These summaries differ in how they average over the remaining characteristics, a distinction that matters for interpreting the empirical exposure profiles.

Section 3 develops estimation and inference, while Appendix B presents the corresponding results for the mean log deviation (MLD).

2.1 The conditional Theil surface

Let Y denote a worker's wage and $X \in \mathcal{X}$ a vector of observed worker, job, and employer characteristics. The conditioning variables may be discrete, continuous, or a combination of both. I impose the following assumption to ensure that the inequality measures are well defined.

Assumption 1 (Positivity and integrability)**.** $Y > 0$ almost surely and $\mathbb{E}[|Y \log Y|] < \infty$.

Under Assumption 1, the mean wage $\mu = \mathbb{E}[Y]$ is positive and finite, since splitting the expectation over $Y \leq 1$ and $Y > 1$ gives $\mathbb{E}[Y] \leq 1 + \mathbb{E}[|Y \log Y|] < \infty$. The unconditional Theil index (Theil, 1967) is

$$T_0 = \mathbb{E}\left[\frac{Y}{\mu} \log\left(\frac{Y}{\mu}\right)\right] = \frac{\mathbb{E}[Y \log Y]}{\mu} - \log \mu.$$

The index depends only on wages relative to the mean: it is zero when all wages are equal and is unchanged when every wage is multiplied by the same positive factor. The Theil index belongs to the generalized entropy family and admits an exact within–between decomposition, with the between component measuring inequality in conditional mean wages.[1]

To measure inequality among workers sharing the same observed characteristics, I apply the Theil index to the conditional wage distribution given $X = x$. Define the two conditional

[1]The variance of log wages also decomposes exactly, but its between component is defined on mean log wages and it does not satisfy the Pigou–Dalton transfer principle (Foster & Ok, 1999). The Gini decomposition generally includes an additional overlap term; in companion work, Bhardwaj and Díaz-Villarejo (2026) develop a regression framework for the conditional Gini index.

moments

$$\mu_0(x) = \mathbb{E}[Y \mid X = x], \qquad \nu_0(x) = \mathbb{E}[Y \log Y \mid X = x].$$

Assumption 1 ensures that these moments are finite and $\mu_0(x) > 0$ for P_X-almost every x, where P_X denotes the distribution of X. At such values, the conditional Theil index is

$$T_0(x) = \mathbb{E}\left[\frac{Y}{\mu_0(x)} \log\left(\frac{Y}{\mu_0(x)}\right)\middle| X = x\right] = \frac{\nu_0(x)}{\mu_0(x)} - \log \mu_0(x).$$

The index $T_0(x)$ measures inequality among workers with characteristics x, that is, dispersion among comparable workers, and is zero when all of them earn the same wage. The second expression in (1) shows that it depends on the conditional wage distribution only through μ_0 and ν_0. These two conditional moments are the nuisance functions estimated in Section 3. I refer to the function $x \mapsto T_0(x)$ as the *conditional Theil surface*, and write $T_0(X)$ for its evaluation at the random covariate vector X.

2.2 Exact within–between decomposition

For a given set of conditioning variables X, the unconditional Theil index decomposes exactly into a within component T_W and a between component T_B. For a fixed population, the choice of conditioning variables determines the allocation between these components but leaves total inequality unchanged. I suppress their dependence on X unless comparing different conditioning sets.

Proposition 1 (Exact Theil decomposition)**.** *Under Assumption 1, the unconditional Theil index satisfies*

$$T = T_W + T_B,$$

where

$$\begin{aligned} T_W &= \mathbb{E}\left[\frac{\mu_0(X)}{\mu} T_0(X)\right], \\ T_B &= \mathbb{E}\left[\frac{\mu_0(X)}{\mu} \log\left(\frac{\mu_0(X)}{\mu}\right)\right]. \end{aligned} \tag{1}$$

The proof is provided in Appendix A.1.

The within component averages the conditional indices $T_0(x)$ over the distribution of X, weighting each by the relative conditional mean $\mu_0(x)/\mu$. Since $\mathbb{E}[\mu_0(X)] = \mu$, these weights have mean one. When X defines discrete groups, a group with population share p_g and mean wage μ_g receives weight $p_g\mu_g/\mu$, its share of total wages (Bourguignon, 1979; Shorrocks, 1980; Theil, 1967).[2] Thus, the contribution of a group to within inequality depends on both its wage share and its conditional inequality.

The between component measures the inequality that would remain if each worker's wage were replaced by the corresponding conditional mean $\mu_0(X)$, while retaining the distribution of observed characteristics. It is therefore the Theil index of conditional mean wages.

The conditional-expectation formulation also accommodates continuous characteristics without requiring a finite partition into groups. Section 3 develops a regression-based implementation

[2]The within component of the mean log deviation instead uses population weights; see Appendix B.

for rich conditioning information.

2.3 Nested conditioning sets

Adding characteristics to the conditioning set can reclassify inequality from the within to the between component. To make this precise, let $X^{(a)}$ and $X^{(b)}$ be two conditioning vectors for the same population such that $X^{(b)}$ contains all the information in $X^{(a)}$, either because it adds variables or because it refines the categories of existing ones:

$$\sigma(X^{(a)}) \subseteq \sigma(X^{(b)}), \tag{2}$$

where $\sigma(\cdot)$ denotes the generated sigma-algebra. For $r \in \{a, b\}$, define the conditional mean wage function

$$\mu_r(x_r) = \mathbb{E}[Y \mid X^{(r)} = x_r].$$

For notational simplicity, write $\mu_r = \mu_r(X^{(r)})$ when evaluated at the random covariates. By the nesting condition, $\mathbb{E}[\mu_b \mid X^{(a)}] = \mu_a$. Let $T_W^{(r)}$ and $T_B^{(r)}$ denote the corresponding components of Proposition 1.

Corollary 1 (Monotonicity under nested conditioning)**.** *Under Assumption 1 and (2), with both decompositions evaluated under the same population distribution,*

$$\begin{aligned} \Delta T_B := T_B^{(b)} - T_B^{(a)} &= T_W^{(a)} - T_W^{(b)} \\ &= \mathbb{E}\left[\frac{\mu_b}{\mu} \log\left(\frac{\mu_b}{\mu_a}\right)\right] \geq 0. \end{aligned} \tag{3}$$

Moreover, $\Delta T_B = 0$ if and only if

$$\mathbb{E}[Y \mid X^{(b)}] = \mathbb{E}[Y \mid X^{(a)}] \quad \textit{almost surely.} \tag{4}$$

The proof is provided in Appendix A.2. The last expression in (3) can be read as inequality in the finer conditional mean wages among workers who share the coarser characteristics. To see this, take the expectation in two steps, first given $X^{(a)}$ and then over $X^{(a)}$, and use $\mathbb{E}[\mu_b \mid X^{(a)}] = \mu_a$:

$$\Delta T_B = \mathbb{E}\left[\frac{\mu_a}{\mu}\, \mathbb{E}\left[\frac{\mu_b}{\mu_a} \log\left(\frac{\mu_b}{\mu_a}\right) \middle| X^{(a)}\right]\right].$$

The inner expectation is the Theil index of μ_b among workers with a given value of $X^{(a)}$, that is, inequality in mean wages across the finer groups within one coarser group. The increment averages these indices over the coarser groups with relative-mean weights μ_a/μ.

The increment is zero precisely when the added information leaves conditional mean wages unchanged. The added characteristics may nevertheless distinguish wage distributions with different conditional Theil indices. The monotonicity result concerns the aggregate within and between components and does not imply that conditional inequality falls at every covariate value.

For a sequence of nested conditioning sets evaluated on the same population, individual

increments depend on the order in which characteristics are introduced, while their sum depends only on the initial and final information sets. Within a given year, these increments measure how additional conditioning information reallocates inequality between the within and between components. Comparing the same increments across years describes how the additional inequality in conditional mean wages evolves over time. Such changes can reflect changes in conditional mean wages, workforce composition, or both.

2.4 Profiles of conditional inequality

When the conditioning vector contains many covariates, the surface $x \mapsto T_0(x)$ is not a useful reporting object, and a series second stage for all of X may be infeasible. Partial means can be used to provide a low-dimensional summary. Following Newey (1994b), I hold selected covariates fixed and average the surface over the remaining ones; see also Linton and Nielsen (1995) for the related marginal-integration approach.

Partition $X = (A, Z)$, where $A \in \mathcal{A}$ contains the covariates of interest and Z the remaining conditioning variables. For a reference distribution $F_Z^\star$ on the support of Z, define

$$\theta_0(a) = \int T_0(a, z)\, dF_Z^\star(z). \tag{5}$$

I refer to $a \mapsto \theta_0(a)$ as a *partial-mean profile*. At each a, it averages inequality measured separately within the conditional wage distributions indexed by (a, z). Because the same $F_Z^\star$ is used at every a, differences along the profile reflect differences in the conditional indices $T_0(a, z)$, not changes in the weight given to the remaining characteristics.

The reference distribution is part of the estimand. Natural choices are the population marginal distribution F_Z, which the simulation in Section 4 uses with known weights, or the empirical distribution of Z in a base-year sample, which fixes a common composition when profiles are compared across years. In either case, each reported a must satisfy a common-support requirement. With $\mathcal{S}_X = \mathrm{supp}(P_X)$,

$$F_Z^\star(\{z : (a, z) \in \mathcal{S}_X\}) = 1, \qquad \int |T_0(a, z)|\, dF_Z^\star(z) < \infty,$$

so that the reference assigns weight only to combinations (a, z) at which the surface is defined.[3]

A second method to summarize would be to compute the conditional-average profile

$$\tau_0(a) = \mathbb{E}[T_0(X) \mid A = a] = \int T_0(a, z)\, dF_{Z|A=a}(z), \tag{6}$$

which averages the same surface under the distribution of Z among workers observed at $A = a$. It describes the conditional inequality faced by those workers, but its variation in a mixes differences in $T_0(a, \cdot)$ with differences in the composition of Z across values of a. Corollaries 2 and 3 give pointwise inference for both θ_0 and τ_0, respectively. For θ_0, the inference results

[3] Conditional expectations are defined only P_X-almost surely. Under Assumption 3(ii), T_0 has a version that is continuous in the continuous covariates within each discrete stratum, and pointwise evaluations in (5) use that version.

treat the reference distribution $F_Z^\star$ as fixed, either because it is known or because inference is conditional on an independent reference sample. When the reference distribution is estimated from the same sample used to estimate conditional inequality, its sampling uncertainty must also be accounted for, which requires additional analysis.

These profiles generally differ from the Theil index calculated for all workers with $A = a$, which includes both within- and between-group inequality across values of Z. Its between component reflects differences in conditional mean wages $\mu_0(a, Z)$, while its within component averages the conditional Theil indices $T_0(a, Z)$ using each group's share of total wages rather than its share of workers, giving relatively greater weight to groups with higher mean wages. By contrast, $\tau_0(a)$ averages the conditional Theil indices using the distribution of Z given $A = a$, whereas $\theta_0(a)$ uses a fixed reference distribution $F_Z^\star$. Neither profile, when averaged over the distribution of A, generally recovers the aggregate within component T_W, which weights $T_0(X)$ by $\mu_0(X)/\mu$.

3 Locally robust estimation and inference

This section develops locally robust estimators and inference for the quantities defined in Section 2. The aggregate within and between components depend on the conditional mean wage, while the conditional Theil index depends on the conditional means of Y and $Y \log Y$. A natural estimation approach replaces these unknown functions with estimated counterparts, yielding plug-in estimators. However, bias in nonparametric first-stage estimates can affect these estimators at first order and invalidate conventional inference. I therefore construct Neyman-orthogonal moments, whose expectations have zero first-order sensitivity to errors in the first-stage functions at their true values, and combine them with cross-fitting.

For the conditional Theil surface, I derive an orthogonal signal based on the conditional means of Y and $Y \log Y$. When these moments are evaluated at their true values, the conditional expectation of the signal equals $T_0(x)$ and has zero first-order sensitivity to errors in the estimates of either conditional mean. I then regress the cross-fitted signal on a series basis by ordinary least squares. The aggregate within and between components and their nested increments require only the conditional mean wage as a nonparametric input.

Sections 3.1 and 3.2 construct the signal and the cross-fitted estimator. Section 3.3 states the conditions under which first-stage estimation error is asymptotically negligible. Sections 3.4 and 3.5 develop pointwise inference for the conditional Theil surface and its profiles. Section 3.6 develops the aggregate estimators and their root-n inference.

3.1 Signal and orthogonality

Let $W_i = (Y_i, X_i)$, $i = 1, \ldots, n$, be independent copies of $W = (Y, X)$, and write $\mathbb{E}_n[f(W_i)] = n^{-1} \sum_{i=1}^n f(W_i)$ for a sample average[4]. Throughout, $\|v\|$ denotes the Euclidean norm for vectors, and $\|M\|$ the spectral norm for matrices. For functions of X, $\|f\|_2$ and $\|f\|_\infty$ denote the $L_2(P_X)$

[4]The asymptotic results in this section are established under independent sampling. The empirical implementation with employer-level cross-fitting and clustered covariance estimation is described in Section 5.4.

and supremum norms, respectively.

Recall that $T_0(x) = \nu_0(x)/\mu_0(x) - \log \mu_0(x)$, where μ_0 and ν_0 are the conditional means of Y and $Y \log Y$ given X. The conditional Theil surface is estimated by series regression: T_0 is approximated by a linear combination of K basis functions $p_K : \mathcal{X} \to \mathbb{R}^K$. For example, one can use a tensor-product spline expansion in the continuous characteristics, interacted with indicators for the discrete ones. The least-squares projection of T_0 on p_K, with coefficient β_K and approximation error r_K, satisfies

$$T_0(x) = p_K(x)^\top \beta_K + r_K(x), \qquad \mathbb{E}[p_K(X) r_K(X)] = 0_K. \tag{7}$$

The theory allows $K = K_n$ to grow with the sample size. Under Assumptions 3(iii) and 4(i), the approximation error r_K vanishes as K grows.

Plug-in estimation. By the definition in Section 2.1, the conditional Theil surface can be expressed as the conditional mean of an outcome that is observable once $\mu_0(X)$ is known: $\mathbb{E}[\, Y \log Y/\mu_0(X) - \log \mu_0(X) \mid X \,] = T_0(X)$. For a candidate conditional mean function μ, define the uncorrected, or plug-in, moment

$$g_K(W; \beta, \mu) = p_K(X) \left\{ \frac{Y \log Y}{\mu(X)} - \log \mu(X) - p_K(X)^\top \beta \right\}. \tag{8}$$

At the true μ_0 it has mean zero at β_K,

$$\mathbb{E}[g_K(W; \beta_K, \mu_0)] = 0_K, \tag{9}$$

and since it is linear in β and the second-moment matrix of the basis, $Q_K = \mathbb{E}[p_K(X) p_K(X)^\top]$, is nonsingular under Assumption 3, it identifies $\beta_K = Q_K^{-1} \mathbb{E}[p_K(X) T_0(X)]$ uniquely. If μ_0 were known, β_K could be estimated by regressing $Y_i \log Y_i/\mu_0(X_i) - \log \mu_0(X_i)$ on $p_K(X_i)$.

Since it is unknown, the natural two-step procedure first estimates it and then regresses the generated outcome $\widetilde{s}_i = Y_i \log Y_i/\widehat{\mu}(X_i) - \log \widehat{\mu}(X_i)$, with $\widehat{\mu}(X_i) > 0$, on the basis:

$$\widetilde{\beta}_K = \widehat{Q}_K^{-1} \frac{1}{n} \sum_{i=1}^{n} p_K(X_i) \widetilde{s}_i, \qquad \widetilde{T}_K(x) = p_K(x)^\top \widetilde{\beta}_K, \tag{10}$$

where $\widehat{Q}_K = \mathbb{E}_n[p_K(X_i) p_K(X_i)^\top]$.

The plug-in moment can inherit bias from first-stage estimation. For a bounded perturbation direction h_μ, let $\mu_\tau = \mu_0 + \tau h_\mu$. The sensitivity of the moment to small changes in the conditional mean is captured by

$$\frac{d}{d\tau} \mathbb{E}[g_K(W; \beta_K, \mu_\tau)] \Big|_{\tau=0} \neq 0_K \quad \text{in general.}$$

Thus, the plug-in moment is not Neyman-orthogonal with respect to errors in the first-stage conditional mean. Such errors can enter at first order and invalidate conventional inference.

The correction. Local robustness (Chernozhukov et al., 2022; Newey, 1994a) addresses the first-order sensitivity of the plug-in moment by adding an influence-function correction. For the conditional Theil moment, the correction is

$$\phi_K(W;\mu,\nu) = -\frac{p_K(X)}{\mu(X)}\{Y-\mu(X)\}\left\{\frac{\nu(X)}{\mu(X)}+1\right\}. \tag{11}$$

The following proposition states the resulting locally robust moment. Its proof in Appendix A.3 derives the correction term and establishes Neyman orthogonality with respect to both nuisance functions.

Proposition 2 (Locally robust moment for the conditional Theil surface)**.** *Suppose μ_0 is bounded away from zero, ν_0 is bounded, and $\mathbb{E}[Y^2 + (Y\log Y)^2] < \infty$. For candidate functions $\mu : \mathcal{X} \to (0,\infty)$ and $\nu : \mathcal{X} \to \mathbb{R}$, define the locally robust conditional Theil moment*

$$\begin{aligned}\psi_K(W;\beta,\mu,\nu) = p_K(X)\left[\frac{Y\log Y}{\mu(X)} - \log\mu(X) - p_K(X)^\top\beta\right] \\ - \frac{p_K(X)}{\mu(X)}\{Y-\mu(X)\}\left\{\frac{\nu(X)}{\mu(X)}+1\right\}.\end{aligned} \tag{12}$$

This is the sum of the plug-in moment (8) and the correction (11). At the true conditional moments, $\mathbb{E}[\psi_K(W;\beta_K,\mu_0,\nu_0)] = 0_K$. For each finite K with bounded p_K and nonsingular Q_K, and for bounded perturbations (h_μ, h_ν) that preserve positivity of the conditional mean,

$$\frac{d}{d\tau}\mathbb{E}[\psi_K(W;\beta_K,\mu_0+\tau h_\mu,\nu_0+\tau h_\nu)]\Big|_{\tau=0} = 0_K. \tag{13}$$

The orthogonal signal. The locally robust moment can be written as $\psi_K(W;\beta,\mu,\nu) = p_K(X)\{s(W;\mu,\nu) - p_K(X)^\top\beta\}$, where

$$\boxed{s(W;\mu,\nu) = \frac{Y\log Y}{\mu(X)} - \log\mu(X) - \frac{Y-\mu(X)}{\mu(X)}\left\{\frac{\nu(X)}{\mu(X)}+1\right\}} \tag{14}$$

is the *orthogonal signal*, a corrected outcome constructed from each worker's wage and the two conditional moment functions. Since $\mathbb{E}[Y-\mu_0(X) \mid X] = 0$, the correction has conditional mean zero at the true moments. Consequently, by the definition in Section 2.1, the conditional expectation of the signal equals $T_0(X)$:

$$\mathbb{E}[s(W;\mu_0,\nu_0) \mid X] = T_0(X) \quad \text{almost surely.} \tag{15}$$

Equation 15 is a fundamental result. It means that, locally, one can treat inference on conditional inequality as a regression problem with dependent variable $s(W,\cdot)$ and regressor (X). The locally robust estimator therefore retains the least-squares second stage of (10), replacing the plug-in outcome $\widetilde{s}_i$ with the orthogonal signal. The next subsection describes its cross-fitted implementation.

3.2 Cross-fitted estimator

The estimator of the conditional Theil surface combines the orthogonal moment of Proposition 2 with cross-fitting (Chernozhukov et al., 2018, 2022). The correction removes the first-order effect of first-stage estimation errors, while cross-fitting eliminates own-observation (overfitting) bias, which is common with many machine learning methods.

Partition the observations into a fixed number J of folds $I_1, \ldots, I_J$. For each fold j, let $\mathcal{D}_{-j} = \{W_i : i \notin I_j\}$ be the training sample, which excludes the observations in fold j. The nuisance functions for fold j are estimated using only this training sample, giving $\widehat{\mu}_{-j}$ and $\widehat{\nu}_{-j}$. For each observation $i \in I_j$, construct the pseudo-outcome

$$\widehat{T}_i = s(W_i; \widehat{\mu}_{-j}, \widehat{\nu}_{-j}). \tag{16}$$

The estimator solves the sample analogue of the orthogonal moment (12),

$$\frac{1}{n}\sum_{i=1}^{n} p_K(X_i)\{\widehat{T}_i - p_K(X_i)^\top \widehat{\beta}_K\} = 0_K,$$

so the second stage is a least-squares regression of the pseudo-outcome on the basis. On the event that $\widehat{Q}_K$ is nonsingular,

$$\widehat{\beta}_K = \widehat{Q}_K^{-1}\frac{1}{n}\sum_{i=1}^{n} p_K(X_i)\widehat{T}_i, \qquad \widehat{T}_K(x) = p_K(x)^\top \widehat{\beta}_K.$$

This locally robust (LR) estimator differs from the uncorrected estimator (10) only through the outcome. The assumptions required for its consistency and asymptotic normality are given in Section 3.3.

3.3 Oracle equivalence and regularity conditions

The orthogonality in (13) removes first-order sensitivity of the population moment to errors in the conditional moments for each fixed K. The conditions below control the remaining bias and stochastic effects as K grows with n. Their role is summarized by the following result, which compares the feasible estimator with an oracle estimator that uses the true nuisance functions.

Lemma 1 (Oracle equivalence of the cross-fitted coefficients)**.** *Under Assumptions 2–4 stated below, define the oracle coefficient estimator using the same observations and empirical Gram matrix as the feasible estimator:*

$$\widehat{\beta}_K^o = \widehat{Q}_K^{-1}\mathbb{E}_n[p_K(X_i)s(W_i; \mu_0, \nu_0)].$$

Then

$$\sqrt{n}\,\|\widehat{\beta}_K - \widehat{\beta}_K^o\| = o_p(1). \tag{17}$$

The proof is in Appendix A.5. Together with the variance conditions of Theorem 1, this makes first-stage estimation error negligible for the pointwise inference results on the conditional

Theil surface. Oracle equivalence concerns only the first stage. The series regression focuses on the projection $p_K(x)^\top \beta_K$, which differs from $T_0(x)$ by the approximation error $r_K(x)$. Inference on $T_0(x)$ additionally requires that approximation error to be negligible relative to the standard error, as specified in Theorem 1.

The following assumptions cover sampling and first-stage estimation, the basis and the smoothness of the target, and the rates governing basis growth and first-stage convergence.

Sampling and first stage.

Assumption 2 (Sampling, moments, and fitted means)**.** (i) The observations $W_i = (Y_i, X_i)$ are independent and identically distributed, with $Y_i > 0$ almost surely.

(ii) For fixed constants, $0 < c_\mu \le \mu_0(x) \le C_\mu < \infty$ for almost every x, and

$$\mathbb{E}[Y^4 + |Y \log Y|^4 \mid X] \le M < \infty \quad \text{almost surely.} \tag{18}$$

(iii) For a fixed $\underline{\mu} \in (0, c_\mu]$ and every j, the fitted means satisfy $\widehat{\mu}_{-j}(x) \ge \underline{\mu}$ for almost every x, with probability tending to one.

(iv) The folds $I_1, \dots, I_J$ are formed independently of the data, J is fixed, and $n_j/n \to \pi_j \in (0,1)$. The nuisance estimates used on fold j are functions of the training sample $\mathcal{D}_{-j}$ and of randomization that is independent of the observations in I_j.

Condition (ii) bounds the conditional mean wage above and away from zero, ensuring that $\log \mu_0$ and $1/\mu_0$ are bounded. Condition (iii) imposes a positive lower bound on the fitted conditional means, which can be enforced by truncation: $\widehat{\mu}_{-j}(x) = \max\{\underline{\mu}, \widehat{\mu}^{\text{raw}}_{-j}(x)\}$. All subsequent convergence rates refer to these truncated estimates. The fourth-moment restrictions in (ii) are used to establish pointwise normality and consistency of feasible variance estimators for the growing-series surface and profile estimators (Appendix A.7); the scalar component estimators and first-stage oracle-equivalence argument require only second-moment bounds. Condition (iv) formalizes the cross-fitting requirement that each observation is evaluated using first-stage estimates constructed without that observation.

Basis and smoothness. Let $\zeta_K = \sup_{x \in \mathcal{X}} \|p_K(x)\|$ denote the envelope of the basis, and write $X = (X_c, X_d)$ for the continuous and discrete coordinates of X, where X_c takes values in a bounded rectangle normalized to $[0,1]^{d_x}$ and X_d may be empty.

Assumption 3 (Second-stage basis, smoothness, and approximation)**.** (i) **Identification.** The eigenvalues of $Q_K = \mathbb{E}[p_K(X)p_K(X)^\top]$ are bounded above and away from zero uniformly in K.

(ii) **Target smoothness.** T_0 belongs to a Hölder ball $\mathcal{H}^{s_T}$ of finite radius in X_c, uniformly over the values of X_d: with $s_T = r + \gamma$, $r \in \mathbb{N}_0$, and $\gamma \in (0,1]$, derivatives through order r are bounded and the order-r derivatives are γ-Hölder continuous.

(iii) **Approximation.** The span of p_K contains constants, and the projection error in (7) satisfies $\|r_K\|_\infty \lesssim a_{T,K}$ for a deterministic sequence $a_{T,K}$.

Assumption 3 restricts the series basis, the smoothness of T_0, and its approximation error. It does not require wages to be bounded or continuously distributed. The wage distribution may have mass points, such as those induced by a minimum wage. Condition (i) imposes eigenvalue bounds directly on the population Gram matrix and does not require X to have a density with respect to a prespecified reference measure. The distribution of X also determines the population projection underlying the approximation error in (iii). Proposition 3 provides sufficient conditions for (i) and (iii) for standard normalized series bases. It imposes additional density restrictions on the continuous characteristics, and gives the corresponding bounds for ζ_K and $a_{T,K}$.

Growth and rates. Define the maximal foldwise $L_2(P_X)$ estimation errors of the two conditional moments:

$$r_{\mu,n} = \max_j \|\widehat{\mu}_{-j} - \mu_0\|_2, \qquad r_{\nu,n} = \max_j \|\widehat{\nu}_{-j} - \nu_0\|_2.$$

The norms are evaluated over the population distribution of X, conditional on the corresponding training samples.

Assumption 4 (Series growth and first-stage nuisance rates)**.** (i) The basis dimension and approximation envelope satisfy

$$K \to \infty, \qquad \frac{K\zeta_K^2 \log K}{n} \to 0, \qquad a_{T,K} \to 0. \tag{19}$$

(ii) The truncated, cross-fitted nuisances satisfy

$$\sqrt{n}\,\zeta_K(r_{\mu,n}^2 + r_{\mu,n}r_{\nu,n}) = o_p(1), \qquad \zeta_K(r_{\mu,n} + r_{\nu,n}) = o_p(1). \tag{20}$$

Part (i) restricts the growth of the series basis relative to the sample size and requires the approximation error to vanish. Part (ii) imposes convergence-rate conditions on the cross-fitted first-stage estimators without prescribing a particular estimation method. The first restriction in (20) controls the second-order bias remaining after orthogonalization. Both restrictions depend on the basis envelope ζ_K. If both first-stage functions converge at a common rate r_n, then $r_n = o_p(n^{-1/4}\zeta_K^{-1/2})$ suffices for both restrictions under part (i). These conditions are sufficient but not necessarily sharp. The rate conditions in (20) ensure oracle equivalence: inference proceeds asymptotically as if the true conditional moments μ_0 and ν_0 were known.

Standard bases. The following proposition verifies Assumption 3(i) and (iii) for standard series bases and provides the corresponding envelope and approximation bounds.

Proposition 3 (Admissible second-stage series bases)**.** *Let Assumption 3(ii) hold. Suppose X_d takes a fixed finite number of values, each with probability bounded away from zero, and that the conditional density of X_c on $[0,1]^{d_x}$ is uniformly bounded above and away from zero*

across these values. Let p_K be a balanced tensor-product basis in X_c, fully interacted with indicators for X_d. Denote by $\mathcal{P}_K$ the population $L_2(P_X)$ projection onto the span of p_K, and let $L_K = 1 + \|\mathcal{P}_K\|_{\infty\to\infty}$. Then Assumption 3(i) and (iii) hold with $a_{T,K} = L_K K^{-\bar{s}_T/d_x}$ for the following bases:

(a) ***Spline and wavelet series.*** *For appropriately normalized B-splines of order s_0 with uniformly bounded mesh ratio, or compactly supported boundary-corrected wavelets of approximation order s_0,*

$$\bar{s}_T = \min\{s_T, s_0\}, \qquad \zeta_K \asymp \sqrt{K}, \qquad L_K = O(1).$$

(b) ***Legendre polynomial series.*** *For tensor products of shifted Legendre polynomials orthonormal under Lebesgue measure,*

$$\bar{s}_T = s_T, \qquad \zeta_K \asymp K, \qquad L_K \lesssim 1 + \zeta_K.$$

For either construction, $\|p_K(x)\|^2 \asymp K$ at each fixed interior point of a discrete stratum. The proof is provided in Appendix A.4.

Spline and wavelet bases have smaller envelopes than Legendre polynomial bases, making the growth and first-stage rate conditions easier to satisfy. For splines and wavelets, the basis-growth requirement becomes $K^2 \log K/n \to 0$, compared with $K^3 \log K/n \to 0$ for Legendre polynomials. If both first-stage functions converge at a common rate, the corresponding sufficient rates are $o_p((nK)^{-1/4})$ and $o_p(n^{-1/4}K^{-1/2})$, respectively. The proposition bounds the approximation error; Assumption 4(i) separately requires the bound to vanish, which for Legendre series with only the general bound on L_K holds when $s_T > d_x$.

With oracle equivalence in hand, the next subsection derives the limiting distribution of the oracle series estimator and the feasible variance estimator, which together give pointwise inference for the conditional Theil surface.

3.4 Inference for the conditional Theil surface

This subsection establishes pointwise consistency and confidence intervals for $T_0(x)$. The results combine the oracle equivalence in Lemma 1 with conditions on the variance and the series approximation error.

The oracle regression uses the signal evaluated at the true conditional moments. Its error is

$$\varepsilon_K = s(W; \mu_0, \nu_0) - p_K(X)^\top \beta_K.$$

By (15) and the projection condition, $\mathbb{E}[\varepsilon_K \mid X] = r_K(X)$ and $\mathbb{E}[p_K(X)\varepsilon_K] = 0_K$. Thus, the regression error includes both the sampling noise of the signal and the approximation error of the basis. Under the conditions stated below, the asymptotic variance of $\widehat{T}_K(x)$ is $V_K(x)/n$. Its feasible estimator, $\widehat{V}_K(x)/n$, uses the OLS residuals $\widehat{\varepsilon}_{Ki} = \widehat{T}_i - p_K(X_i)^\top \widehat{\beta}_K$ from regressing the

cross-fitted signal on the series basis. Specifically,

$$\begin{aligned} \Omega_K &= \mathbb{E}[p_K(X)p_K(X)^\top \varepsilon_K^2], \qquad V_K(x) = p_K(x)^\top Q_K^{-1}\Omega_K Q_K^{-1} p_K(x), \\ \widehat{\Omega}_K &= \mathbb{E}_n[p_K(X_i)p_K(X_i)^\top \widehat{\varepsilon}_{Ki}^2], \quad \widehat{V}_K(x) = p_K(x)^\top \widehat{Q}_K^{-1}\widehat{\Omega}_K \widehat{Q}_K^{-1} p_K(x). \end{aligned} \tag{21}$$

Theorem 1 (Consistency and pointwise inference for the conditional Theil surface)**.** *Let Assumptions 2–4 hold.*

(a) Consistency and rate. *For any fixed $x \in \mathcal{X}$,*

$$\widehat{T}_K(x) - T_0(x) = O_p\left(\frac{\|p_K(x)\|}{\sqrt{n}}\right) + O(a_{T,K}), \tag{22}$$

so $\widehat{T}_K(x) \to_p T_0(x)$. At fixed interior points of the bases in Proposition 3, the stochastic term is $O_p(\sqrt{K/n})$.

(b) Asymptotic normality and feasible variance. *At a fixed reported point x, suppose $V_K(x) \asymp \|p_K(x)\|^2 > 0$ and*

$$\sqrt{\frac{n}{V_K(x)}}\, a_{T,K} \to 0. \tag{23}$$

Then

$$\frac{\widehat{T}_K(x) - T_0(x)}{\sqrt{V_K(x)/n}} \Rightarrow N(0,1), \tag{24}$$

$\widehat{V}_K(x)/V_K(x) \to_p 1$, and

$$\frac{\widehat{T}_K(x) - T_0(x)}{\sqrt{\widehat{V}_K(x)/n}} \Rightarrow N(0,1). \tag{25}$$

The proofs are in Appendices A.6 and A.7.

Under part (b), an asymptotically valid pointwise 95 percent confidence interval for $T_0(x)$ is $\widehat{T}_K(x) \pm 1.96\sqrt{\widehat{V}_K(x)/n}$. It has the usual heteroskedasticity-robust OLS form, using residuals from the regression of the cross-fitted signal on the series basis. Section 3.5 develops the corresponding inference results for the profiles.

Condition (23) controls approximation bias. For each K, the regression targets the population projection $p_K(x)^\top \beta_K$, which approaches $T_0(x)$ as the approximation error vanishes. Confidence intervals for $T_0(x)$ additionally require this error to be of smaller order than the standard error $\sqrt{V_K(x)/n}$. The condition uses the approximation envelope supplied by Proposition 3. The weaker pointwise requirement $\sqrt{n/V_K(x)}\,|r_K(x)| \to 0$ suffices. Expressing this condition in terms of $V_K(x)$ accommodates differences in the variance of series estimators across evaluation points.

3.5 Partial-mean and conditional-average profiles

The two profiles of Section 2.4 summarize the surface along a subvector A of X. The partial-mean estimator averages the fitted surface over a fixed reference distribution whereas the conditional-average estimator regresses the cross-fitted signal conditional on X on basis functions of A alone. The latter is used for the empirical exposure profiles.

Fixed-reference partial means. For the target $\theta_0(a)$ in (5), averaging the fitted surface over $F_Z^\star$ gives

$$\widehat{\theta}_K(a) = \bar{p}_K(a)^\top \widehat{\beta}_K, \qquad \bar{p}_K(a) = \int p_K(a,z)\, dF_Z^\star(z).$$

Its approximation error $\bar{r}_K(a) = \int r_K(a,z)\, dF_Z^\star(z)$ satisfies $|\bar{r}_K(a)| \lesssim a_{T,K}$. The asymptotic variance and its feasible estimator are $\bar{V}_K(a)/n$ and $\widehat{\bar{V}}_K(a)/n$, where

$$\bar{V}_K(a) = \bar{p}_K(a)^\top Q_K^{-1} \Omega_K Q_K^{-1} \bar{p}_K(a),$$
$$\widehat{\bar{V}}_K(a) = \bar{p}_K(a)^\top \widehat{Q}_K^{-1} \widehat{\Omega}_K \widehat{Q}_K^{-1} \bar{p}_K(a).$$

Averaging the fitted surface over the reference distribution changes the variance of the estimator but does not necessarily reduce its approximation error. The averaged approximation error remains bounded by $a_{T,K}$, while the relevant standard error is $\sqrt{\bar{V}_K(a)/n}$. Pointwise inference therefore requires the approximation error to be negligible relative to the standard error of the partial-mean estimator.

Corollary 2 (Fixed-reference partial-mean inference)**.** *Let Assumptions 2–4 hold. Let $F_Z^\star$ be deterministic or the empirical distribution of a reference sample independent of the estimation sample. In the latter case, $\theta_0(a)$ is defined using that empirical distribution, and the statements below hold conditionally on the reference sample.*

Fix a reported a such that $(a,z) \in \mathcal{X}$ for $F_Z^\star$-almost every z. Suppose $\bar{V}_K(a) \asymp \|\bar{p}_K(a)\|^2 > 0$ and

$$\sqrt{\frac{n}{\bar{V}_K(a)}}\, a_{T,K} \to 0. \tag{26}$$

Then $\widehat{\theta}_K(a) \to_p \theta_0(a)$, $\widehat{\bar{V}}_K(a)/\bar{V}_K(a) \to_p 1$, and

$$\frac{\widehat{\theta}_K(a) - \theta_0(a)}{\sqrt{\widehat{\bar{V}}_K(a)/n}} \Rightarrow N(0,1).$$

The proof is in Appendix A.8.

The weaker pointwise requirement $\sqrt{n/\bar{V}_K(a)}\,|\bar{r}_K(a)| \to 0$ can replace (26). When the reference distribution is constructed from an independent sample, inference is conditional on that sample and treats its empirical distribution as fixed. The resulting confidence intervals, therefore, apply to the partial-mean profile defined using the observed reference distribution. Inference for a profile based on the underlying population distribution, or for a reference distribution estimated from the same sample as the conditional moments, requires accounting for additional estimation uncertainty.

Conditional-average profiles. For the target $\tau_0(a) = \mathbb{E}[T_0(X) \mid A = a]$ in (6), the signal identity and iterated expectations give

$$\mathbb{E}[s(W; \mu_0, \nu_0) \mid A] = \tau_0(A). \tag{27}$$

Thus, the conditional-average profile is estimated by regressing the cross-fitted signal on a series basis in A alone. The signal is constructed using first-stage estimates of μ_0 and ν_0 conditional on the full vector X, rather than on A alone.

Let $q_L(A) \in \mathbb{R}^L$ be a reporting basis with $L = L_n \to \infty$, and write $Q_{A,L} = \mathbb{E}[q_L(A)q_L(A)^\top]$. The estimator is

$$\begin{aligned}\widehat{Q}_{A,L} &= \mathbb{E}_n[q_L(A_i)q_L(A_i)^\top],\\ \widehat{\gamma}_L &= \widehat{Q}_{A,L}^{-1}\mathbb{E}_n[q_L(A_i)\widehat{T}_i],\\ \widehat{\tau}_L(a) &= q_L(a)^\top\widehat{\gamma}_L.\end{aligned}$$

Write the population projection as $\tau_0(a) = q_L(a)^\top\gamma_L + r_L(a)$, with $\mathbb{E}[q_L(A)r_L(A)] = 0$. The regression error and variance factor are

$$\begin{aligned}\varepsilon_L &= s(W;\mu_0,\nu_0) - q_L(A)^\top\gamma_L,\\ \Omega_L &= \mathbb{E}[q_L(A)q_L(A)^\top\varepsilon_L^2],\\ V_L(a) &= q_L(a)^\top Q_{A,L}^{-1}\Omega_L Q_{A,L}^{-1}q_L(a).\end{aligned}$$

Define $\widehat{V}_L(a)$ by the same sandwich formula as (21), using q_L, $\widehat{\gamma}_L$, and the residuals $\widehat{T}_i - q_L(A_i)^\top\widehat{\gamma}_L$.

Corollary 3 (Inference for conditional-average profiles)**.** *Let A be an observed subvector of X, and let Assumption 2 hold. Suppose Assumption 3 holds for the reporting regression, with (A, τ_0, q_L) replacing (X, T_0, p_K), and let $a_{\tau,L}$ bound its approximation error. Suppose Assumption 4 holds with $(L, \zeta_{A,L}, a_{\tau,L})$ replacing $(K, \zeta_K, a_{T,K})$, where $\zeta_{A,L} = \sup_a \|q_L(a)\|$. The first-stage rates continue to refer to the conditional moments given X.*

Then $\widehat{\tau}_L(a) \to_p \tau_0(a)$ at each fixed a. At a reported point satisfying

$$V_L(a) \asymp \|q_L(a)\|^2 > 0, \qquad \sqrt{\frac{n}{V_L(a)}}\, a_{\tau,L} \to 0,$$

we also have $\widehat{V}_L(a)/V_L(a) \to_p 1$ and

$$\frac{\widehat{\tau}_L(a) - \tau_0(a)}{\sqrt{\widehat{V}_L(a)/n}} \Rightarrow N(0,1). \tag{28}$$

The proof is in Appendix A.9.

The reporting regression therefore requires smoothness and approximation conditions on τ_0, while the first stage still uses all characteristics in X. Its rate conditions involve the reporting-basis envelope $\zeta_{A,L}$.

An extension to conditional inequality treatment effects is provided in Remark 1 in Appendix A.11.

3.6 Theil components and nested increments

The within and between components require only the conditional mean wage as a nuisance function. Their second stage is a sample average, with no series regression. Substituting (1) into

the within component of Proposition 1 and using iterated expectations gives

$$\begin{aligned} T_W &= \frac{1}{\mu}\mathbb{E}[\mu_0(X)T_0(X)] \\ &= \frac{1}{\mu}\mathbb{E}[\nu_0(X) - \mu_0(X)\log\mu_0(X)] \\ &= \frac{1}{\mu}\mathbb{E}\left[Y\log\left(\frac{Y}{\mu_0(X)}\right)\right]. \end{aligned} \tag{29}$$

The between component can likewise be written as

$$T_B = \frac{1}{\mu}\mathbb{E}[\mu_0(X)\log\mu_0(X)] - \log\mu. \tag{30}$$

These representations avoid estimating ν_0 or the conditional Theil surface.

Orthogonal moments. As with the plug-in signal in Section 3.1, replacing μ_0 with a first-stage estimate when estimating the within and between components can introduce first-order bias and invalidate conventional inference. Let $m : \mathcal{X} \to (0, \infty)$ denote a candidate conditional mean function, using m to distinguish it from the unconditional mean wage $\mu = \mathbb{E}[Y]$. The plug-in moments for the within and between components are

$$\begin{aligned} g_W(W; m(\cdot), \mu) &= \frac{Y}{\mu}\log\left(\frac{Y}{m(X)}\right) - T_W, \\ g_B(W; m(\cdot), \mu) &= \frac{m(X)}{\mu}\log\left(\frac{m(X)}{\mu}\right) - T_B. \end{aligned} \tag{31}$$

Both moments have mean zero at $m(\cdot) = \mu_0(\cdot)$. The following proposition gives the locally robust moments and establishes their Neyman orthogonality.

Proposition 4 (Locally robust moments for the Theil components)**.** *Suppose Assumption 2(i)–(ii) holds. The corrections for* (31) *are* $\varphi_W(W; m(\cdot), \mu) = -\{Y - m(X)\}/\mu$ *and* $\varphi_B(W; m(\cdot), \mu) = \{1 + \log(m(X)/\mu)\}\{Y - m(X)\}/\mu$*. The resulting locally robust moments are*

$$\begin{aligned} \psi_W(W; m(\cdot), \mu) &= \frac{Y}{\mu}\log\left(\frac{Y}{m(X)}\right) - \frac{Y - m(X)}{\mu} - T_W, \\ \psi_B(W; m(\cdot), \mu) &= \frac{Y}{\mu}\log\left(\frac{m(X)}{\mu}\right) + \frac{Y - m(X)}{\mu} - T_B. \end{aligned} \tag{32}$$

For $c \in \{W, B\}$, $\mathbb{E}[\psi_c(W; \mu_0, \mu)] = 0$*. For a bounded perturbation* h*, let* $m_t(\cdot) = \mu_0(\cdot) + th$*, which remains positive for sufficiently small* $|t|$*. Holding the scalar* μ *fixed,*

$$\frac{d}{dt}\mathbb{E}[\psi_c(W; m_t(\cdot), \mu)]\Big|_{t=0} = 0.$$

The proof is in Appendix A.12.

Estimators and exact adding-up. Using the fold construction of Section 3.2, write $\widehat{\mu}(X_i) = \widehat{\mu}_{-j}(X_i)$ for $i \in I_j$ and $\bar{Y} = \mathbb{E}_n[Y_i]$. Replacing μ by $\bar{Y}$ and adding the sample correction term to

the plug-in estimators yields the following locally robust estimators:

$$\widehat{T}_W^{LR} = \frac{1}{\bar{Y}}\mathbb{E}_n\left[Y_i \log\left(\frac{Y_i}{\widehat{\mu}(X_i)}\right) - \{Y_i - \widehat{\mu}(X_i)\}\right], \tag{33}$$

and

$$\widehat{T}_B^{LR} = \frac{1}{\bar{Y}}\mathbb{E}_n\left[Y_i \log \widehat{\mu}(X_i) + \{Y_i - \widehat{\mu}(X_i)\}\right] - \log \bar{Y}. \tag{34}$$

When both components use the same cross-fitted conditional means, all terms involving $\widehat{\mu}(X_i)$ cancel upon addition:

$$\widehat{T}_W^{LR} + \widehat{T}_B^{LR} = \frac{\mathbb{E}_n[Y_i \log Y_i]}{\bar{Y}} - \log \bar{Y}. \tag{35}$$

The estimated components therefore add exactly to the sample Theil index for any positive first-stage predictions.

Exact adding-up does not guarantee that the within and between components are estimated accurately. Errors in the estimated conditional mean can change how inequality is allocated between the two components without changing their sum. Section 4.4 examines the finite-sample bias of each component estimator.

Nested increments. For nested conditioning sets $X^{(a)}$ and $X^{(b)}$, let $\mu_r(x_r) = \mathbb{E}[Y \mid X^{(r)} = x_r]$ denote the conditional mean wage under set $r \in \{a, b\}$, as in Section 2.3. Compute $\widehat{T}_B^{(r),LR}$ using (34) with a cross-fitted estimate of μ_r. Both component estimates use the same observations, fold partition, and sample mean $\bar{Y}$. The increment of Corollary 1 is estimated by

$$\widehat{\Delta T}_B^{LR} = \widehat{T}_B^{(b),LR} - \widehat{T}_B^{(a),LR}. \tag{36}$$

By (35), this also equals $\widehat{T}_W^{(a),LR} - \widehat{T}_W^{(b),LR}$.

Inference. Under the conditions stated below, the estimators of T_W, T_B, and ΔT_B admit asymptotic linear representations with the following influence functions. These functions account for estimating the unconditional mean wage μ by $\bar{Y}$ and determine the asymptotic variances:

$$\begin{aligned}
\chi_W(W; X) &= \frac{Y}{\mu}\left\{\log\left(\frac{Y}{\mu_0(X)}\right) - T_W\right\} - \frac{Y - \mu_0(X)}{\mu}, \\
\chi_B(W; X) &= \frac{Y}{\mu}\left\{\log\left(\frac{\mu_0(X)}{\mu}\right) - T_B\right\} - \frac{\mu_0(X) - \mu}{\mu}, \\
\chi_\Delta(W) &= \chi_B(W; X^{(b)}) - \chi_B(W; X^{(a)}) \\
&= \frac{Y \log(\mu_b/\mu_a) - (\mu_b - \mu_a) - \Delta T_B\, Y}{\mu}.
\end{aligned} \tag{37}$$

For $c \in \{W, B\}$, let $\sigma_c^2 = \mathrm{Var}\{\chi_c(W; X)\}$. For each observation $i \in I_j$, construct the estimated influence function $\widehat{\chi}_{c,i}$ by evaluating χ_c at $\bar{Y}$, $\widehat{\mu}_{-j}(X_i)$, and $\widehat{T}_c^{LR}$ in place of μ, $\mu_0(X_i)$, and T_c, respectively. Since the estimated influence functions have zero sample mean, the feasible variance estimator is

$$\widehat{\sigma}_c^2 = \mathbb{E}_n[\widehat{\chi}_{c,i}^2].$$

The corresponding estimated standard error of $\widehat{T}_c^{LR}$ is $\widehat{\sigma}_c/\sqrt{n}$.

For the nested increment, define

$$\widehat{\chi}_{\Delta,i} = \widehat{\chi}_{B,i}^{(b)} - \widehat{\chi}_{B,i}^{(a)}, \qquad \widehat{\sigma}_\Delta^2 = \mathbb{E}_n[\widehat{\chi}_{\Delta,i}^2].$$

The corresponding estimated standard error is $\widehat{\sigma}_\Delta/\sqrt{n}$. Because the two between-component estimators use the same observations, taking the variance of their influence-function differences accounts for the covariance between them.

Theorem 2 (Estimation and inference for the Theil components)**.** *Let Assumption 2 hold, and suppose the cross-fitted conditional mean satisfies*

$$r_{\mu,n} = \max_{1\le j\le J} \|\widehat{\mu}_{-j} - \mu_0\|_2 = o_p(n^{-1/4}). \tag{38}$$

Then, for $c \in \{W, B\}$, $\widehat{T}_c^{LR} \to_p T_c$,

$$\sqrt{n}\{\widehat{T}_c^{LR} - T_c\} = \frac{1}{\sqrt{n}} \sum_{i=1}^{n} \chi_c(W_i; X) + o_p(1), \tag{39}$$

and $\widehat{\sigma}_c^2 \to_p \sigma_c^2$. If $\sigma_c^2 > 0$, then

$$\frac{\sqrt{n}\{\widehat{T}_c^{LR} - T_c\}}{\widehat{\sigma}_c} \Rightarrow N(0,1). \tag{40}$$

The proofs are in Appendices A.13 and A.14.

Corollary 4 (Paired inference for a nested between increment)**.** *Let $\sigma(X^{(a)}) \subseteq \sigma(X^{(b)})$. Suppose Assumption 2 and (38) hold for each conditional mean, using a common sample and fold partition. Then $\widehat{\Delta T}_B^{LR} \to_p \Delta T_B$,*

$$\sqrt{n}(\widehat{\Delta T}_B^{LR} - \Delta T_B) = \frac{1}{\sqrt{n}} \sum_{i=1}^{n} \chi_\Delta(W_i) + o_p(1), \tag{41}$$

and $\widehat{\sigma}_\Delta^2 \to_p \sigma_\Delta^2 := \mathrm{Var}(\chi_\Delta)$. If $\sigma_\Delta^2 > 0$, then

$$\frac{\sqrt{n}(\widehat{\Delta T}_B^{LR} - \Delta T_B)}{\widehat{\sigma}_\Delta} \Rightarrow N(0,1). \tag{42}$$

The proof is in Appendix A.15.

The component estimators require no second-stage series regression, so the rate restriction (38) on first stage estimators contains no factor involving a series dimension or basis envelope.

At the nested null $\Delta T_B = 0$, Corollary 1 implies $\mu_a(X^{(a)}) = \mu_b(X^{(b)})$ almost surely. Consequently, $\chi_\Delta = 0$ and $\sigma_\Delta^2 = 0$, so the first-order normal approximation does not justify a Wald test at this null. Inference at the null or along sequences approaching it requires a separate argument. The estimate $\widehat{\Delta T}_B^{LR}$ can also be negative in finite samples, although the population increment is nonnegative.

Table 1 summarizes the locally robust estimators and corresponding standard-error formulas developed in this section. It covers the conditional Theil surface, its partial-mean and conditional-average profiles, the aggregate within and between components, and the nested between increment.

Table 1: Locally robust Theil estimators and standard errors

Estimand	**Locally robust estimator**	**Estimated standard error**
Conditional inequality and profiles		
Conditional Theil surface: $T_0(x)$	$p_K(x)^\top \widehat{Q}_K^{-1}\mathbb{E}_n[p_i\widehat{s}_i]$	$\sqrt{\dfrac{p_K(x)^\top \widehat{\Sigma}_K p_K(x)}{n}}$
Fixed-ref. partial mean: $\theta_0(a)$	$\bar{p}_K(a)^\top \widehat{Q}_K^{-1}\mathbb{E}_n[p_i\widehat{s}_i]$	$\sqrt{\dfrac{\bar{p}_K(a)^\top \widehat{\Sigma}_K \bar{p}_K(a)}{n}}$
Conditional-avg. profile: $\tau_0(a)$	$q_L(a)^\top \widehat{Q}_{A,L}^{-1}\mathbb{E}_n[q_i\widehat{s}_i]$	$\sqrt{\dfrac{q_L(a)^\top \widehat{\Sigma}_L q_L(a)}{n}}$
Aggregate components and nested increment		
Within component: T_W	$\dfrac{\mathbb{E}_n[Y_i \log(Y_i/\widehat{\mu}_i) - Y_i + \widehat{\mu}_i]}{\bar{Y}}$	$\dfrac{\sqrt{\mathbb{E}_n[(Y_i\{\log(Y_i/\widehat{\mu}_i) - 1 - \widehat{T}_W^{\mathrm{LR}}\} + \widehat{\mu}_i)^2]}}{\bar{Y}\sqrt{n}}$
Between component: T_B	$\dfrac{\mathbb{E}_n[Y_i \log(\widehat{\mu}_i/\bar{Y}) + Y_i - \widehat{\mu}_i]}{\bar{Y}}$	$\dfrac{\sqrt{\mathbb{E}_n[(Y_i\{\log(\widehat{\mu}_i/\bar{Y}) - \widehat{T}_B^{\mathrm{LR}}\} - \widehat{\mu}_i + \bar{Y})^2]}}{\bar{Y}\sqrt{n}}$
Nested between increment: ΔT_B	$\dfrac{\mathbb{E}_n[Y_i \log(\widehat{\mu}_{b,i}/\widehat{\mu}_{a,i}) - \widehat{\mu}_{b,i} + \widehat{\mu}_{a,i}]}{\bar{Y}}$	$\dfrac{\sqrt{\mathbb{E}_n[(Y_i\{\log(\widehat{\mu}_{b,i}/\widehat{\mu}_{a,i}) - \widehat{\Delta T}_B^{\mathrm{LR}}\} - \widehat{\mu}_{b,i} + \widehat{\mu}_{a,i})^2]}}{\bar{Y}\sqrt{n}}$

Notes. Write $\mathbb{E}_n[f_i] = n^{-1}\sum_{i=1}^n f_i$, $\bar{Y} = \mathbb{E}_n[Y_i]$, $p_i = p_K(X_i)$, $q_i = q_L(A_i)$, and $\bar{p}_K(a) = \int p_K(a,z)\, dF_Z^\star(z)$. The out-of-fold moment predictions are $\widehat{\mu}_i$ and $\widehat{\nu}_i$, and the orthogonal signal is $\widehat{s}_i = Y_i \log Y_i/\widehat{\mu}_i - \log\widehat{\mu}_i - (Y_i - \widehat{\mu}_i)(\widehat{\nu}_i/\widehat{\mu}_i + 1)/\widehat{\mu}_i$. Define $\widehat{Q}_K = \mathbb{E}_n[p_i p_i^\top]$, $\widehat{Q}_{A,L} = \mathbb{E}_n[q_i q_i^\top]$, $\widehat{\beta}_K = \widehat{Q}_K^{-1}\mathbb{E}_n[p_i\widehat{s}_i]$, and $\widehat{\gamma}_L = \widehat{Q}_{A,L}^{-1}\mathbb{E}_n[q_i\widehat{s}_i]$. The covariance matrices are $\widehat{\Sigma}_K = \widehat{Q}_K^{-1}\widehat{\Omega}_K\widehat{Q}_K^{-1}$ and $\widehat{\Sigma}_L = \widehat{Q}_{A,L}^{-1}\widehat{\Omega}_L\widehat{Q}_{A,L}^{-1}$, with $\widehat{\Omega}_K = \mathbb{E}_n[p_i p_i^\top(\widehat{s}_i - p_i^\top\widehat{\beta}_K)^2]$ and $\widehat{\Omega}_L = \mathbb{E}_n[q_i q_i^\top(\widehat{s}_i - q_i^\top\widehat{\gamma}_L)^2]$. The last three standard errors directly substitute the influence functions from (37).

Section 4 evaluates the finite-sample performance of the estimators in a design with known population targets. It compares the locally robust and plug-in estimators across sample sizes and first-stage learners, assessing how the orthogonal correction affects estimation bias and confidence-interval coverage.

4 Simulation study

I use a Monte Carlo experiment to evaluate the finite-sample performance of the proposed estimators and assess the contribution of the locally robust (LR) correction. The design combines a smooth conditional Theil surface with jumps in conditional mean wages. I compare LR and plug-in estimators of the conditional surface, fixed-reference partial-mean profiles, and aggregate within- and between components across sample sizes and three first-stage learners: series, Lasso, and XGBoost. The LR surface and profile estimators have small biases across

all three learners, including the boosting learner used in the application, and their pointwise confidence intervals achieve reasonable coverage.

4.1 Design and target objects

The simulated outcome Y follows a conditional log-normal distribution, which allows its conditional mean and conditional Theil index to be specified separately. For covariates $X = (A, Z)$, defined below, I generate i.i.d. observations according to

$$Y = \mu_0(X) \exp\left\{ \sqrt{2t_0(X)}\, \varepsilon - t_0(X) \right\}, \qquad \varepsilon \sim N(0,1), \quad \varepsilon \perp X.$$

The corresponding conditional moments and Theil index satisfy

$$\begin{aligned} \mathbb{E}[Y \mid X] &= \mu_0(X), \\ \nu_0(X) = \mathbb{E}[Y \log Y \mid X] &= \mu_0(X)\{\log \mu_0(X) + t_0(X)\}, \\ T_0(X) &= t_0(X), \end{aligned}$$

where the last equality follows because the Theil index of a log-normal distribution equals half the variance of its logarithm. In the specification below, μ_0 jumps at thresholds in A. The jumps pass to ν_0 but not to $T_0 = t_0$, which is smooth in A within each value of Z.

Here A is a continuous covariate with $A \sim U[0,1]$, and $Z = (Z_1, Z_2)$ consists of two binary covariates. Writing $\Lambda(v) = (1 + e^{-v})^{-1}$, I generate the binary covariates sequentially as

$$\begin{aligned} Z_1 \mid A = a &\sim \text{Bernoulli}\{\Lambda(-0.2 + 1.1(a - 0.5))\}, \\ Z_2 \mid A = a, Z_1 = z_1 &\sim \text{Bernoulli}\{\Lambda(0.3 - 0.8a + 0.7z_1)\}. \end{aligned}$$

The probabilities of the four possible combinations of (Z_1, Z_2) vary with A, and each combination has positive conditional probability for all $a \in [0,1]$. I write $Z_{12} = Z_1 Z_2$ to denote the interaction between the two binary covariates.

The design introduces jumps in the nuisance functions μ_0 and ν_0 while keeping the target surface T_0 smooth in A for each combination of (Z_1, Z_2), consistent with the smoothness condition imposed on the target in the theory.

I specify the conditional mean as

$$\begin{aligned} \mu_0(a,z) &= \exp\{c + s\, h_\lambda(a,z)\}, \\ h_\lambda(a,z) &= 0.25z_1 - 0.15z_2 + \lambda\big[0.35\, \mathbf{1}\{a > 0.35\} - 0.30\, \mathbf{1}\{a > 0.70\} \\ &\qquad + 0.40z_1\, \mathbf{1}\{a > 0.55\} - 0.30z_2\, \mathbf{1}\{a < 0.25\}\big], \end{aligned}$$

where $\lambda = 0.1$, $c \approx 0.49$, and $s \approx 2.2124$; full-precision values are available on request. The constant c determines the scale of Y without affecting the Theil targets, s scales variation in $\log \mu_0$, and λ controls the size of the jumps.

I specify the conditional Theil surface as

$$t_0(a,z) = 0.1\big[0.04 + 0.08a^2 + 0.08\sin^2(\pi a) + 0.025z_1 + 0.035z_2 \\ + 0.02z_1z_2 + 0.025az_1 - 0.02az_2\big].$$

The surface $T_0 = t_0$ is smooth in A for each value of Z and satisfies the smoothness condition in Assumption 3(ii).

I evaluate the conditional surface $T_0(a,1,1)$, corresponding to the combination of binary covariates for which the conditional mean contains all the jumps. The tables report results at $a \in \{0.10, 0.50, 0.90\}$, the 10th, 50th, and 90th percentiles of A. The true surface values at these points are 0.01289, 0.02225, and 0.01969, respectively.

Following Section 2.4, I also consider the fixed-reference partial-mean profile

$$\theta_0(a) = \sum_{z\in\{0,1\}^2} w_z T_0(a,z), \qquad w_z = P(Z=z) = \int_0^1 P(Z=z \mid A=u)\,du.$$

Unlike $T_0(a,1,1)$, which measures conditional inequality at $Z=(1,1)$, the profile $\theta_0(a)$ averages conditional inequality over all four combinations of Z using the fixed weights w_z. The resulting profile values are 0.00848, 0.01765, and 0.01490 at $a = 0.10$, 0.50, and 0.90, respectively. The reference weights are computed from the known population distribution rather than estimated from the simulation sample, so there is no reference-distribution estimation error.

The scalar targets are the within and between components of Proposition 1, which in this design are

$$T_W = \frac{\mathbb{E}[\mu_0(X)t_0(X)]}{\mu}, \qquad T_B = \frac{\mathbb{E}[\mu_0(X)\log\mu_0(X)]}{\mu} - \log\mu, \qquad \mu = \mathbb{E}[Y].$$

Their population values are $T_W = 0.01512$ and $T_B = 0.05489$, yielding total inequality $T = 0.07001$, of which the between component accounts for 78 percent.

4.2 Implementation

I use $R = 500$ Monte Carlo replications for sample sizes $n \in \{1{,}000, 3{,}000, 5{,}000\}$, with two-fold cross-fitting. The second-stage regression uses a cubic B-spline basis in A with $J_n \asymp n^{0.33}$ interior knots, fully interacted with $(1, Z_1, Z_2, Z_{12})$, so that the total number of regressors is $K_n = 4(J_n + 4)$. The nuisance functions μ_0 and ν_0 are estimated using Series, Lasso, and XGBoost. Series and Lasso use a common spline dictionary, while XGBoost uses (A, Z_1, Z_2, Z_{12}) as input. Hyperparameter tuning is performed within each training fold. The loss functions and tuning procedures are available on request.

Let $\widehat{\mu}_i$ and $\widehat{\nu}_i$ denote out-of-fold predictions. The LR estimator regresses the pseudo-outcome in (16) on $p_{K_n}(X_i)$. The plug-in (PI) estimator uses the same conditional-mean predictions and basis, but omits the LR correction:

$$\widehat{T}_i^{\mathrm{PI}} = \frac{Y_i \log Y_i}{\widehat{\mu}_i} - \log\widehat{\mu}_i.$$

This PI estimator therefore projects the uncorrected signal, rather than evaluating the direct plug-in surface $\widehat{\nu}(x)/\widehat{\mu}(x) - \log \widehat{\mu}(x)$. For the within and between components, the PI estimators are obtained from the sample analogues of (31), using the same cross-fitted conditional mean estimates and sample mean wage. Standard errors for the LR estimators of the conditional Theil surface and the partial-mean profile are computed following Theorem 1 and Corollary 2, respectively. Following Escanciano and Terschuur (2026), I also report empirical coverage for the PI estimators using the corresponding LR standard errors.

Table 2: Conditional Theil surface at $Z_1 = Z_2 = 1$

			Locally robust			Plug-in (PI)	
n	a	First stage	Bias	RMSE	Cov. (%)	Bias	Cov. (%)
1,000	0.10	Series	0.001	0.004	93.6	0.002	38.8
		Lasso	0.001	0.003	93.8	−0.015	18.0
		XGBoost	0.000	0.003	94.0	−0.013	18.6
	0.50	Series	0.001	0.004	93.0	0.002	50.8
		Lasso	0.001	0.004	93.2	0.003	30.6
		XGBoost	0.001	0.004	92.8	0.031	8.2
	0.90	Series	0.001	0.005	92.8	0.002	46.0
		Lasso	0.001	0.005	92.8	0.001	26.8
		XGBoost	0.000	0.004	91.6	−0.009	20.4
3,000	0.10	Series	0.000	0.002	92.6	0.000	53.6
		Lasso	0.000	0.002	92.4	0.001	22.8
		XGBoost	0.000	0.002	92.8	−0.017	9.4
	0.50	Series	0.001	0.003	92.6	0.000	69.8
		Lasso	0.001	0.003	93.6	0.007	23.4
		XGBoost	0.000	0.003	94.0	0.015	20.0
	0.90	Series	0.000	0.003	93.6	0.000	61.8
		Lasso	0.000	0.003	93.6	0.003	25.0
		XGBoost	0.000	0.003	93.6	−0.017	13.8
5,000	0.10	Series	0.000	0.002	94.4	0.000	61.2
		Lasso	0.000	0.002	94.6	0.006	22.0
		XGBoost	0.000	0.002	95.4	−0.019	7.2
	0.50	Series	0.001	0.002	93.4	0.000	70.6
		Lasso	0.001	0.002	93.4	0.003	33.8
		XGBoost	0.000	0.002	94.0	0.015	10.6
	0.90	Series	0.000	0.002	94.2	0.001	66.2
		Lasso	0.000	0.002	94.4	0.001	28.6
		XGBoost	0.000	0.002	94.6	−0.007	19.4

Notes: Simulation results with 500 replications, 95% pointwise confidence intervals. Bias and RMSE are in Theil units. For the plug-in estimator, I report coverage using the same standard-error formula as for the locally robust estimator.
The target is $T_0(a, 1, 1)$. Since $A \sim U[0, 1]$, the reported a values are its 10th, 50th, and 90th percentiles.

4.3 Conditional surface and partial-mean profile

Tables 2 and 3 report bias, root mean squared error, and coverage of 95 percent pointwise confidence intervals for the LR and PI estimators of the conditional Theil surface and the fixed-reference partial-mean profile at $a \in \{0.10, 0.50, 0.90\}$. Figures 1 and 2 show pointwise

coverage over the 10th–90th percentile range of A, with LR coverage in the upper panels and PI coverage in the lower panels.

The LR estimators exhibit small biases across all three first-stage learners and sample sizes, whereas the PI estimators often exhibit biases comparable in magnitude to the targets when Lasso or XGBoost is used in the first stage. LR coverage is generally close to nominal, although some undercoverage remains, particularly for the partial-mean profile in smaller samples. PI coverage, by comparison, is substantially lower in every tabulated case. These results highlight the contribution of the LR correction to reducing first-stage estimation bias and improving coverage, particularly with regularized learners.

Table 3: Partial-mean Theil profile under a fixed reference distribution

			Locally robust			Plug-in (PI)	
n	a	First stage	Bias	RMSE	Cov. (%)	Bias	Cov. (%)
1,000	0.10	Series	0.001	0.004	92.6	0.001	22.2
		Lasso	0.000	0.001	93.0	−0.015	6.0
		XGBoost	0.000	0.002	91.8	0.000	19.8
	0.50	Series	0.001	0.002	91.4	0.002	37.4
		Lasso	0.000	0.002	93.8	0.015	3.6
		XGBoost	0.000	0.002	90.8	0.022	1.2
	0.90	Series	0.001	0.003	90.0	0.002	37.8
		Lasso	0.000	0.002	91.8	−0.013	13.6
		XGBoost	0.000	0.002	91.0	−0.007	22.6
3,000	0.10	Series	0.000	0.001	95.4	0.000	42.2
		Lasso	0.000	0.001	94.4	0.004	19.4
		XGBoost	0.000	0.001	94.6	−0.001	22.0
	0.50	Series	0.000	0.001	93.4	0.000	61.6
		Lasso	0.000	0.001	93.8	0.014	1.2
		XGBoost	0.000	0.001	94.2	0.014	2.0
	0.90	Series	0.000	0.001	91.2	0.000	51.6
		Lasso	0.000	0.001	91.2	0.002	28.4
		XGBoost	0.000	0.001	90.4	−0.007	17.2
5,000	0.10	Series	0.000	0.001	94.4	0.000	44.6
		Lasso	0.000	0.001	93.6	0.006	11.0
		XGBoost	0.000	0.001	94.8	−0.003	13.2
	0.50	Series	0.000	0.001	91.4	0.000	65.6
		Lasso	0.000	0.001	92.4	0.011	0.6
		XGBoost	0.000	0.001	94.4	0.012	0.8
	0.90	Series	0.000	0.001	93.6	0.000	57.6
		Lasso	0.000	0.001	93.8	0.002	25.8
		XGBoost	0.000	0.001	93.4	−0.003	22.4

Notes: Simulation results with 500 replications, 95% pointwise confidence intervals for the locally robust estimator. Bias and RMSE are in Theil units. For the plug-in estimator, I report coverage using the same standard-error formula as for the locally robust estimator.
The target is $\theta_0(a) = \sum_z P(Z = z)T_0(a, z)$ with known, fixed marginal weights. Since $A \sim U[0, 1]$, the reported a values are its 10th, 50th, and 90th percentiles.

4.4 Within- and between component results

Table 4 compares the LR and PI estimators of the within and between components. The LR estimators exhibit small biases across all three first-stage learners and sample sizes. For the between component, coverage is close to the nominal 95 percent level at the two larger sample sizes. For the within component, coverage depends more strongly on the first-stage learner. With XGBoost, it ranges from 91.4 to 94.2 percent, while greater undercoverage remains with Series and Lasso.

Table 4: Within and Between Theil components

			Locally robust			Plug-in (PI)	
n	First stage	Target	Bias	RMSE	Cov. (%)	Bias	Cov. (%)
1,000	Series	T_W	0.001	0.001	82.0	0.001	54.6
		T_B	−0.001	0.003	90.6	0.000	74.8
	Lasso	T_W	0.001	0.001	90.4	0.001	65.6
		T_B	−0.001	0.003	92.2	−0.003	69.8
	XGBoost	T_W	0.000	0.001	93.0	0.001	91.2
		T_B	0.000	0.003	93.2	−0.002	87.2
3,000	Series	T_W	0.000	0.001	85.2	0.000	66.2
		T_B	−0.001	0.002	94.4	0.000	87.6
	Lasso	T_W	0.000	0.001	89.2	0.000	76.6
		T_B	−0.001	0.002	94.6	−0.002	74.2
	XGBoost	T_W	0.000	0.001	91.4	0.000	89.2
		T_B	0.000	0.001	95.4	−0.002	80.4
5,000	Series	T_W	0.000	0.001	84.0	0.000	68.6
		T_B	0.000	0.001	94.8	0.000	89.2
	Lasso	T_W	0.000	0.000	87.2	0.000	74.0
		T_B	0.000	0.001	95.2	−0.001	79.8
	XGBoost	T_W	0.000	0.000	94.2	0.000	91.8
		T_B	0.000	0.001	94.0	−0.001	81.6

Notes: Simulation results with 500 replications, 95% pointwise confidence intervals for the locally robust estimator. Bias and RMSE are in Theil units. For the plug-in estimator, I report coverage using the same standard-error formula as for the locally robust estimator.
True values for within and between components are $T_W \approx 0.015$ and $T_B \approx 0.055$.

The coverage of the LR estimator exceeds PI coverage in every tabulated comparison, although the gains vary across component and learners. The improvement is particularly pronounced for the between component with regularized first stages and for the within component with Series and Lasso. For the within component with XGBoost, PI bias is already small and the coverage improvement is more modest. The favorable performance of XGBoost provides additional motivation for its use in the empirical application.

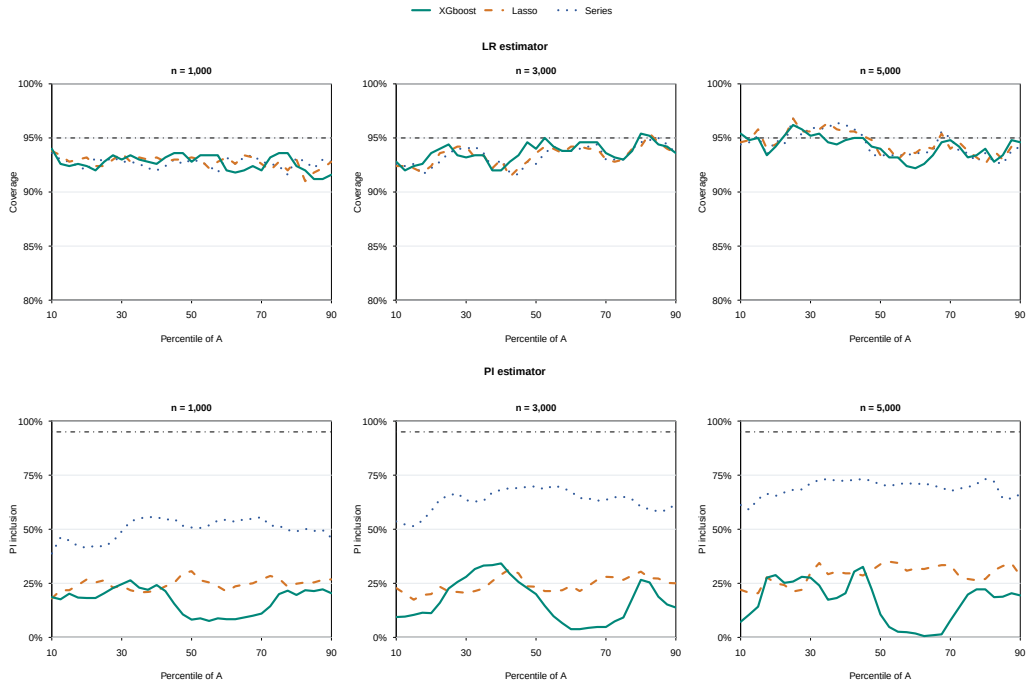


Figure 1: LR and plug-in coverage for the conditional Theil surface $T_0(a, 1, 1)$. For the plug-in estimator, I report coverage using the same standard-error formula as for the locally robust estimator. Results are based on 500 Monte Carlo replications.

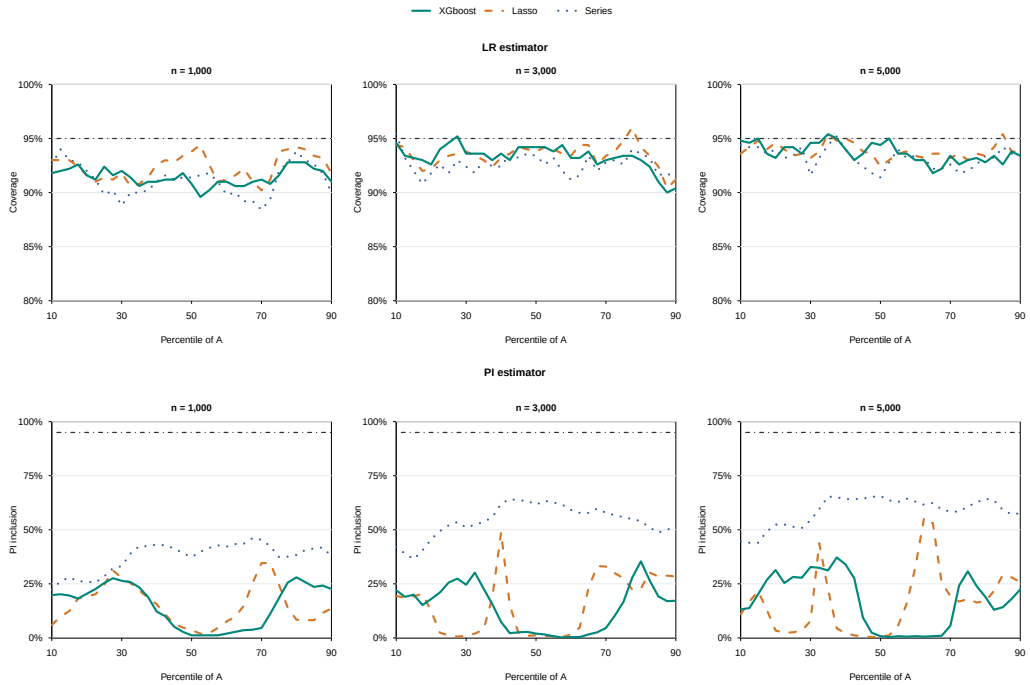


Figure 2: LR and plug-in coverage for the fixed-reference partial mean $\theta_0(a) = \sum_z P(Z = z)T_0(a, z)$. For the plug-in estimator, I report coverage using the same standard-error formula as for the locally robust estimator. Results are based on 500 Monte Carlo replications.

5 Data and empirical implementation

The application studies the conditional structure of hourly-wage inequality among native wage earners in mainland Portugal from 2011 to 2019, with particular attention to its evolution around successive minimum-wage increases. It consists of two complementary exercises. The first examines the evolution of aggregate wage inequality and its within and between components over the full period. Using the exact decomposition in Proposition 1, it quantifies how changes in the two components account for the change in total inequality. Same-sample comparisons across nested conditioning sets show how the allocation between within and between inequality changes as additional worker and employer characteristics are taken into account. The analysis also incorporates employers' initial exposure to the 2014 minimum-wage increase. Exposure is defined as the increase in eligible workers' base-pay shortfalls when the old minimum is replaced by the new one, evaluated at their 2013 wages. For each employer, this incremental shortfall is expressed as a percentage of the total monthly base pay of all its eligible workers in 2013, and the resulting measure is held fixed across years. Conditional-Theil profiles complement the decomposition by showing how conditional dispersion varies across initial exposure levels and how these differences evolve over the period.

The second examines changes around each of the five minimum-wage increases between 2014 and 2019. Because employers' wage structures can change between increases, exposure is recalculated from the wage distribution preceding each increase rather than kept fixed at its 2013 value. For each increase, the same employer-level exposure measure is then used to compare the wage distributions before and after the increase. Within–between decompositions describe how aggregate inequality changes across each comparison, while profiles reported separately by sex and education show how conditional inequality varies with exposure in each of the two years and how it changes across the increase. Wage growth by workers' pre-increase pay band describes adjustment among workers with valid wage observations in both years, while subsequent observation rates are calculated for the full group employed before the increase.

The two exercises therefore provide complementary views of the same distributional changes over the same period of time. The first traces the longer-run evolution of aggregate and conditional inequality using a common initial exposure classification, whereas the second updates exposure before each increase to study the individual transitions. They ask whether wage adjustment near the minimum wage coincided with changes in dispersion among workers with the same observed characteristics, and how those patterns differed across demographic groups and exposure levels. Both exercises are descriptive and do not identify the causal effects of the minimum-wage increases.

5.1 Data sources, sample, and wage measurement

The analysis uses the Portuguese *Quadros de Pessoal* (QP), an annual linked employer–employee administrative dataset compiled by the Gabinete de Estratégia e Planeamento (GEP) from mandatory employer reports submitted through Annex A of the *Relatório Único* (Card et al., 2016; Cardoso & Portugal, 2005). Public Administration employees under civil-service regimes

are not covered (Card et al., 2016; Cardoso & Portugal, 2005).[5] Each annual file reports employment and remuneration information for October. Worker records contain wages, hours, demographic characteristics, occupation, tenure, and contract information, while linked firm and establishment records provide employer characteristics and location. Persistent worker and employer identifiers permit observations to be linked across years. The analysis covers the period of 2011–2019.

The unit of observation is one selected job per worker and year. When a worker has multiple records, priority is given to a wage-earner job with valid pay and hours and then to the job with the greatest reported normal monthly hours; remaining ties are resolved using remuneration and employer identifiers. The complete ranking rule is reported in Appendix F.

After selecting one job per worker-year, I retain wage earners aged 18–64 working full time in mainland Portugal, with complete remuneration, positive regular pay and normal monthly hours, and a matched employer record. Workers in agriculture and those identified as apprentices are excluded. Conditional-inequality estimation additionally requires complete information on the conditioning variables in Table 5.

The main outcome is real regular hourly pay,

$$Y_{it} = c_t \frac{w_{it}^{\text{base}} + w_{it}^{\text{reg}}}{h_{it}},$$

where w_{it}^{base} denotes monthly base pay, w_{it}^{reg} regular monthly supplements, and h_{it} reported normal monthly hours. The factor c_t expresses nominal pay in October-2019 euros using the October all-items consumer price index for Portugal from Statistics Portugal (INE). Overtime pay and irregular bonuses are excluded. By construction, $Y_{it} > 0$, satisfying the positivity requirement in Assumption 1.

The Theil index and its exact within–between decomposition are invariant to a common price adjustment within each year. Real wages are nevertheless used to compare wage levels and wage growth across years.

5.2 Conditioning information

The analysis uses four nested conditioning sets (Table 5). Starting from worker and job characteristics in $X^{(1)}$, each step adds one block of information: industry and location in $X^{(2)}$, employer scale in $X^{(3)}$, and other observed employer characteristics in $X^{(4)}$, the set used for the main decomposition. The exposure-augmented set $(X^{(4)}, B)$ adds the employer's minimum-wage exposure, defined in Section 5.3. This ordering is motivated by evidence on wage differences associated with worker qualifications (Katz & Murphy, 1992), industry and location (Krueger & Summers, 1988; Moretti, 2011), and employer scale and heterogeneity (Brown & Medoff, 1989; Card et al., 2018). It allows the employer blocks to be assessed after accounting for observed worker and job characteristics, industry, and location.

Each step has the same accounting interpretation; two comparisons are central to the application. The comparison of $X^{(2)}$ with $X^{(4)}$ asks how much inequality among workers with

[5] The precise institutional coverage of QP is described in Appendix F.

the same observed worker and job characteristics, industry, and location is associated with differences in mean wages across employer types. Under $X^{(2)}$, these mean wage differences are part of the within component, because the employer characteristics that distinguish them have not yet been taken into account. Conditioning on $X^{(4)}$ separates these mean wage differences from the dispersion that remains among workers who also share the observed employer characteristics. The resulting increase in the between component measures the inequality accounted for by these differences. By Corollary 1, it equals the corresponding reduction in the within component.

The comparison of $X^{(4)}$ with $(X^{(4)}, B)$ asks whether workers sharing the characteristics in $X^{(4)}$ earn different average wages at employers with different levels of minimum-wage exposure. Adding exposure moves the inequality associated with these average wage differences from the within to the between component. The remaining within component summarizes wage dispersion among workers who share both the characteristics in $X^{(4)}$ and their employer's exposure level. Conditional-Theil profiles show how this remaining dispersion varies with exposure.

The increments associated with employer characteristics do not separate differences in wage-setting from the sorting of workers with different unobserved characteristics across employers. Repeating the decomposition over time shows how much of the change in aggregate inequality is accounted for by changes in the components associated with employer characteristics and exposure. These components depend on both conditional mean wages and the distribution of worker and employer characteristics, so their evolution can reflect changes in wages, composition, or both.

Table 5: Nested conditioning sets

Set	Additional information	Variables added
$X^{(1)}$	Worker and job characteristics	Sex, age, education, tenure, qualification level, occupation, permanent-contract status, and normal weekly hours.
$X^{(2)}$	Sector and location	Industry and NUTS-2 region of the establishment.
$X^{(3)}$	Employer scale	Log employment and number of establishments.
$X^{(4)}$	Other employer characteristics	Log real sales per worker, firm-age class, legal form, and ownership in three categories.
$(X^{(4)}, B)$	Minimum-wage exposure	Employer exposure B.

Notes: Each conditioning set contains all variables in the preceding set.

5.3 Minimum-wage exposure

The monthly minimum wage in mainland Portugal remained at 485 euros from 2011 to 2013. It increased to 505 euros in October 2014 and to 530, 557, 580, and 600 euros in January of 2016, 2017, 2018, and 2019, respectively.[6] The 2011–2013 observations provide a reference period for examining the evolution of wage inequality before this sequence of increases.

Employer exposure is constructed from the additional gap between workers' base pay and the minimum wage created by each increase. For each comparison $e = (r, s)$, r denotes the year before the increase and $s = r + 1$ the year in which it takes effect, with $r \in \{2013, 2015, 2016, 2017, 2018\}$.

[6] Amounts and effective dates follow the historical series published by the Portuguese Directorate-General for Employment and Labour Relations (DGERT).

Let m_r and m_s denote the old and new monthly minimum wages, and let w be a worker's nominal monthly base pay in year r. Holding this pay fixed, the amount by which it falls below the new minimum is $(m_s - w)_+$, where $(x)_+ = \max\{x, 0\}$. This amount can include a shortfall that already existed under the old minimum. To isolate the additional shortfall associated with the increase, I subtract that pre-existing gap.[7]

$$\kappa_e(w) = (m_s - w)_+ - (m_r - w)_+.$$

Employer exposure is the sum of these additional shortfalls, expressed as a percentage of the total monthly base pay of the workers included in the calculation. Let $\mathcal{E}_{j,e}$ contain the workers at employer j in year r who satisfy the restrictions in Section 5.1. Writing their nominal monthly base pay as w_{ir}, define

$$B_{j,e} = 100 \frac{\sum_{i \in \mathcal{E}_{j,e}} \kappa_e(w_{ir})}{\sum_{i \in \mathcal{E}_{j,e}} w_{ir}}.$$

The construction adapts the firm-level wage-gap approach used in the minimum-wage literature (Draca et al., 2011; Machin et al., 2003) to an increase in an existing minimum. It measures exposure in base pay, not the change in total employer labor costs.

Exposure is assigned to employers with at least ten workers in $\mathcal{E}_{j,e}$; this threshold applies only to the year before the increase. Among these employers, exposure is zero when every worker included in the calculation earned at least the new minimum before the increase.

The magnitude of $B_{j,e}$ measures the intensity of employer exposure, not the percentage of employers or workers exposed. For example, $B_{j,e} = 1$ means that the incremental base-pay shortfall equals one percent of the employer's total monthly base bill among the eligible workers. The denominator includes the base bill of all these workers, whereas the numerator includes only their additional shortfalls. Consequently, a small exposure value can coexist with a substantial share of workers earning below the new minimum.[8]

Across the five increases, 64–75 percent of employers with measured exposure have positive exposure, that is, at least one eligible worker paid below the new minimum, while the remaining 25–36 percent have zero exposure. Among employers with positive exposure, the median ranges from 0.45 percent in 2013–2014 to 0.98 percent in 2016–2017, while the upper quartile ranges from 1.26 to 2.29 percent. These percentages describe the size of the incremental shortfall relative to the base wage bill, rather than the proportion of employers exposed.

Exposure enters the two exercises differently. The first follows inequality from 2011 to 2019 using *initial exposure* to the 2014 increase, $B_{j,(2013,2014)}$, held fixed throughout. For 2011 and 2012, this is therefore a retrospective classification. The second compares consecutive annual

[7] Recorded base pay need not coincide with the remuneration used to assess minimum-wage compliance. Article 274 of the Portuguese Labour Code (Law No. 7/2009) allows specified payments in kind, sales commissions, and production bonuses to count towards guaranteed minimum monthly remuneration. Article 275 also provides for reduced minimum rates for specified categories of workers in training and workers with reduced working capacity, subject to statutory conditions. A recorded base-pay shortfall against the general minimum is therefore not, by itself, evidence of noncompliance.

[8] For illustration, consider an employer with 100 eligible workers in 2013, half earning monthly base pay of 485 euros and half earning 1,000 euros. The increase in the minimum from 485 to 505 euros generates an incremental shortfall of 1,000 euros relative to a base wage bill of 74,250 euros. Exposure is therefore approximately 1.35 percent, although half of the workers earn below the new minimum.

wage distributions around each increase using *rolling exposure*, $B_{j,e}$, recalculated from wages in the year before the increase and held fixed across the two observation years in comparison $e = (r, s)$.

In both exercises, each worker is assigned the exposure of their current employer; workers at employers without measured exposure are excluded. The samples need not contain the same workers over time.

The annual initial-exposure samples contain 1.10–1.16 million workers per year, representing 63–77 percent of workers with complete conditioning information. Workers outside these samples are employed by firms without a qualifying 2013 exposure measure, including firms absent in 2013 and firms with fewer than ten workers satisfying the construction restrictions in that year. Excluded workers have lower mean real hourly wages, ranging from 5.0 to 5.9 euros, compared with 7.5 to 8.1 euros in the annual analysis samples.

The separate wage-adjustment analysis follows workers in the pre-increase estimation sample who contributed to the exposure calculation. They retain their pre-increase employer's exposure even if they change employer. Subsequent observation does not require complete conditioning information in the following year. Observation rates use the full pre-increase cohort, whereas wage growth compares wages among the same workers with valid observations in both years.

5.4 Estimation and inference

Within each year, the nested decompositions use the sample of workers with complete information on $X^{(4)}$, so that all four conditioning sets are compared on the same observations. The exposure comparisons additionally require measured initial or rolling exposure for each worker's current employer, as defined in Section 5.3. This requirement reduces the sample by excluding workers at employers without sufficient pre-increase wage records to construct exposure. To retain the broader study population for the aggregate decomposition, this additional restriction is imposed only in the exposure analysis.

The within and between components are estimated using (33) and (34). These estimators require only the conditional mean wage, estimated separately for each year and conditioning set. The contribution of each additional block of conditioning information is the difference between the corresponding between components, using (36). In the aggregate decomposition using exposure, the comparison is between $T_B(X^{(4)}, B)$ and $T_B(X^{(4)})$, with both components computed on the same exposure sample.

For the exposure profiles, I summarize conditional inequality separately by sex and education. Let D denote the six groups formed by sex and three education categories: below high school, high school or some college, and college or above. Within each year, defining $X = (X^{(4)}, B)$, the profile is

$$\bar{T}_0(b, d) = \mathbb{E}[T_0(X) \mid B = b, D = d],$$

the average conditional Theil index among workers in group d whose employer has exposure b. This is the conditional-average profile τ_0 of Section 2.4, with $A = (B, D)$. By (15) and iterated expectations, $\mathbb{E}[s(W; \mu_0, \nu_0) \mid B, D] = \bar{T}_0(B, D)$. The nuisance parameters used to construct the

signal are estimated given the full X, while the series regression uses B and D.

I estimate the profiles by ordinary least squares, regressing the cross-fitted signals in (16) on a basis consisting of a natural cubic spline in exposure and a separate indicator for zero exposure, each interacted with the six groups. The number of interior knots satisfies $J_n \asymp n^{0.3}$, where n is the estimation-sample size, and knots are placed at quantiles of positive exposure in the year being fitted. For a given exposure comparison, let $q_t(B, D)$ denote this basis in year t. The fitted profile is $\widehat{\bar{T}}_t(b, d) = q_t(b, d)^\top \widehat{\beta}_t$. Under the conditions of Corollary 3, including independent sampling, $\widehat{\bar{T}}_t(b, d)$ consistently estimates $\bar{T}_{0,t}(b, d)$, and the pointwise confidence intervals defined in that corollary are asymptotically valid. The clustered covariance calculations used in the empirical application are described at the end of this subsection. Appendix E provides further implementation details.

For each rolling comparison $e = (r, s)$, both annual profiles are defined using the exposure measure $B_{j,e}$ constructed from wages in year r. The estimated change for group d at exposure b is

$$\widehat{\bar{T}}_{e,s}(b, d) - \widehat{\bar{T}}_{e,r}(b, d).$$

The long-run comparison between 2011 and 2019 is constructed analogously, using exposure to the 2014 increase measured from 2013 wages in both years. Because the spline knots are selected separately in each year, the comparison is between fitted profile values evaluated at the same (b, d), rather than between the estimated coefficient vectors. The profiles average conditional inequality over the workers observed at each exposure level in each year. Changes can therefore reflect both changes in conditional inequality and changes in worker composition at that exposure level.

Conditional moments are estimated separately for each year using gradient-boosted regression trees (XGBoost; T. Chen & Guestrin, 2016), a choice informed by their performance in the simulations. Models are tuned by cross-validation within each training sample, and cross-fitting uses employer-level folds. Within each sample, fold assignments are shared across conditioning sets, keeping the training partition fixed when assessing the contribution of additional information. Further implementation details and supplementary results are provided in Appendix E.

Standard errors for annual estimates are clustered by employer, while for estimates of changes between years use two-way clustering by employer and worker, with the employer–worker intersection correction of Cameron et al. (2011).[9]

6 Empirical results

The literature has shown that minimum-wage increases can reduce wage inequality, with effects extending beyond workers initially paid at the minimum (Engbom & Moser, 2022; Oliveira, 2023). Wage spillovers above the floor and changes in the allocation of workers across employers can contribute to this adjustment (Engbom & Moser, 2022). An aggregate decline in inequality therefore leaves an important question unanswered: did wage dispersion fall among workers with

[9] For related results on cross-fitting and inference under clustered sampling in double/debiased machine learning, see Chiang et al. (2022).

the same observed characteristics, or did average wage differences across worker and employer groups narrow? The decomposition and conditional-Theil profiles distinguish these patterns in the observed distribution.

Portugal provides an informative setting because a substantial share of workers earned base pay close to the national minimum. In the year preceding each of the five increases in 2014–2019, approximately 15.3–23.3 percent of workers in the samples had monthly base pay at or above the current minimum but below the new one.

Figure 3 presents the evolution of real hourly-wage inequality alongside the minimum-wage sequence. The monthly minimum remained at 485 euros during 2011–2013, increased to 505 euros in 2014, and reached 600 euros in 2019, a nominal increase of 23.7 percent relative to 2013. Over 2011–2019, the Theil index falls from approximately 0.239 to 0.201, a decline of about 16 percent, while the Gini index falls from approximately 0.345 to 0.312, a decline of about 10 percent. Both indices change little during 2011–2013 and then decline steadily from 2014 onward, with the largest annual declines in the Theil index in 2018 and 2019.

Figure 3: Aggregate hourly-wage inequality and the minimum wage, 2011–2019



Notes: The left panel reports annual Theil and Gini indices of hourly wages for workers satisfying the basic sample restrictions in Section 5.1. The decompositions in Table 6 additionally require complete information on $X^{(4)}$, so their total Theil indices refer to a more restricted sample. The right panel reports the monthly minimum wage in mainland Portugal in force in October of each year, expressed in nominal euros.

Section 6.1 reports the nested decompositions and the contribution of observed employer characteristics. The exposure analysis then reports the additional between inequality associated with minimum-wage exposure and presents the initial- and rolling-exposure profiles. Finally, I examine wage growth among workers initially paid near the minimum wage.

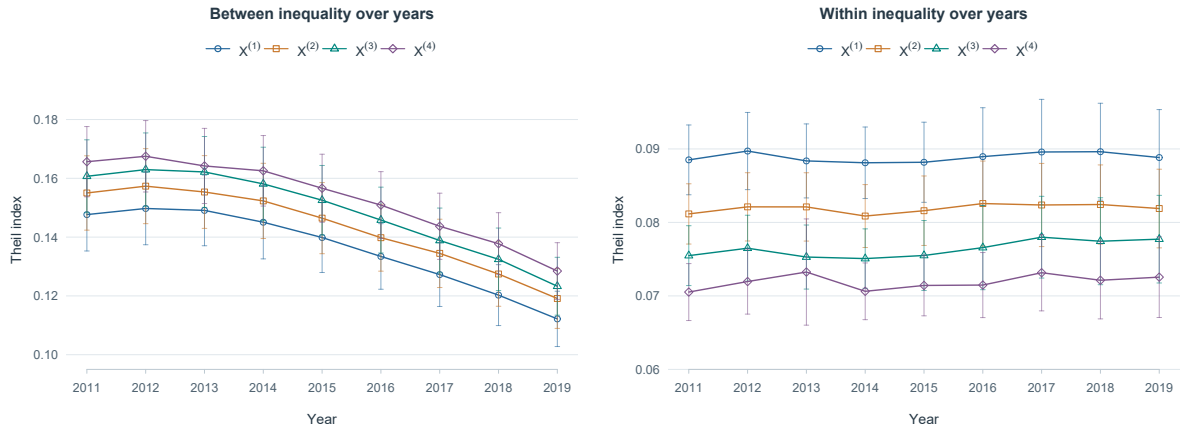
6.1 Aggregate inequality and its conditional structure

The allocation of inequality within and between groups depends on the information used to define the groups. By raising pay for low-wage workers, minimum-wage increases can narrow mean wage differences across worker and job types, across industries and regions, or across employers, depending on where those workers are concentrated. The nested decompositions distinguish these possibilities by adding industry and location, employer scale, and other employer characteristics

to worker and job characteristics in turn, on a common sample. If mean wages converged mainly across employers among otherwise comparable workers, the employer increments would shrink over time; if they converged across worker and job groups, the decline would already appear when groups are defined by worker and job characteristics alone.

Over 2011–2019, wage compression in Portugal appears entirely as lower inequality across the mean wages of groups defined by observed characteristics, with no net decline in the weighted average of wage dispersion within these groups. In the sample with complete information on $X^{(4)}$, the richest of the four conditioning sets, the total Theil index falls from 0.2362 in 2011 to 0.2010 in 2019, a reduction of 0.0352, or approximately 15 percent, with the net decline occurring after 2013 (Table 6). Figure 4 shows that the decline is concentrated in the between component under all four conditioning sets. Under $X^{(4)}$, the between component falls from 0.1657 to 0.1285, an estimated decline of 0.0372 that slightly exceeds the decline in total Theil. The within component, a weighted average of dispersion among workers sharing the observed worker, job, and employer characteristics, moves from 0.0705 to 0.0726, a change of 0.0021 that is not statistically distinguishable from zero. Over 2013–2019, the period spanning the minimum-wage increases, total Theil falls by 0.0364, while the between and within components fall by 0.0357 and 0.0007, respectively: almost the entire decline during the increases occurs in the between component.

Figure 4: Within and between hourly-wage inequality across nested conditioning sets, 2011–2019



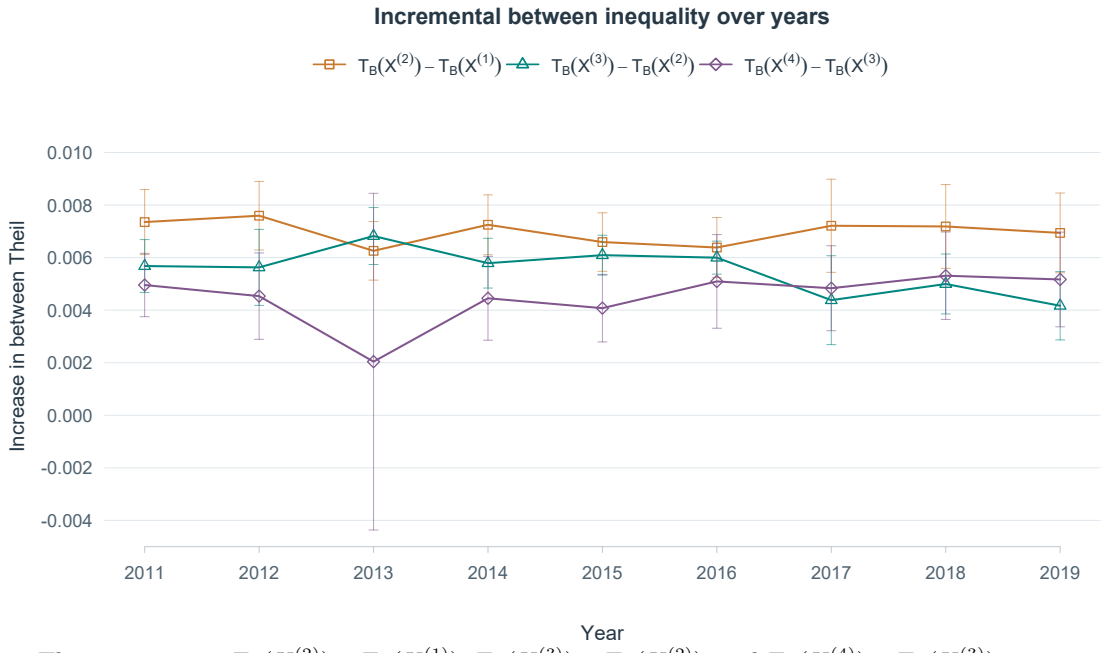
Notes: All four decompositions use the same annual $X^{(4)}$ based sample, without an exposure restriction. The left panel reports $T_B(X^{(k)})$ and the right panel $T_W(X^{(k)})$, for $k = 1, \ldots, 4$. Bars show pointwise 95 percent confidence intervals.

The decline is already visible under $X^{(1)}$, which contains worker and job characteristics. Between inequality under $X^{(1)}$ falls from 0.1477 in 2011 to 0.1122 in 2019, a decline of 0.0355, close to the decline of 0.0372 under $X^{(4)}$. The additional between inequality associated with industry, location, and employer characteristics is 0.0180 in 2011 and 0.0163 in 2019. The annual movements in these additional contributions are small relative to the overall decline in $T_B(X^{(4)})$ (Figure 5).

One possible explanation for almost the entire decline in inequality captured by $X^{(1)}$ is that the increases in the minimum wage and spillovers above the floor generated proportionally higher wage growth among initially lower-paid workers across a range of industries and employer types.

The worker and job characteristics included in $X^{(1)}$ are sufficient to distinguish these groups. Hence, the resulting convergence in mean wages across them would be reflected in a decline in between inequality under $X^{(1)}$. Adding industry, location, and employer characteristics makes little difference to the decline in between inequality. One possible reason is that this convergence occurs in mean wages across worker and job groups, while average pay continues to differ across industries and employer types among otherwise comparable workers.

Figure 5: Additional between inequality associated with successive information blocks, 2011–2019



Notes: The series report $T_B(X^{(2)}) - T_B(X^{(1)})$, $T_B(X^{(3)}) - T_B(X^{(2)})$, and $T_B(X^{(4)}) - T_B(X^{(3)})$, corresponding to industry and location, employer scale, and other employer characteristics. Bars show pointwise 95 percent confidence intervals.

The employer characteristics added in $X^{(3)}$ and $X^{(4)}$ still matter for the level of inequality. Together, these two blocks of employer characteristics add 0.0106 to between inequality in 2011 and 0.0093 in 2019, shifting about 11.5–13 percent of the within component remaining under $X^{(2)}$ into the between component. This share reflects differences in conditional mean wages across observed employer types among workers with similar worker and job characteristics, industry, and location. Estimates show that employer scale contributes the larger increment through 2016, while the remaining employer characteristics contribute slightly more from 2017 onward.[10]

These results complement the literature examining the role of firm pay premia and worker sorting in the evolution of wage inequality (Alvarez et al., 2018; Card et al., 2013; Song et al., 2019). For Portugal, Leitão et al. (2022) attribute around 60 percent of the decline in the variance of log real hourly wages to firms. The nested comparisons here measure a different object. They ask how much of the wage inequality remaining among workers with the same observed worker and job characteristics, industry, and location is associated with differences in conditional mean wages across observed employer types. They trace how the allocation of inequality between the within and between components changes as additional observed characteristics are used to define

[10]The employer-scale increment is related to the employer-size wage differentials studied by Brown and Medoff (1989).

the groups.

The within and between components also depend on the shares of workers in groups defined by the observed characteristics. Therefore, their evolution can reflect changes in the wage distributions within these groups, changes in their employment shares, or both, as emphasized by Lemieux (2006) in the case of residual wage inequality.

Table 6: Contributions of nested variable blocks to annual wage inequality

	2011	2013	2019
Total Theil	0.2362	0.2375	0.2010
	(0.0066)	(0.0065)	(0.0071)
Sequential contributions to between inequality			
Worker and job characteristics: $X^{(1)}$	0.1477	0.1491	0.1122
	(0.0063)	(0.0061)	(0.0048)
Industry and location: $X^{(1)} \to X^{(2)}$	0.0074	0.0063	0.0069
	(0.0006)	(0.0006)	(0.0008)
Employer scale: $X^{(2)} \to X^{(3)}$	0.0057	0.0068	0.0042
	(0.0005)	(0.0006)	(0.0007)
Other employer characteristics: $X^{(3)} \to X^{(4)}$	0.0050	0.0020	0.0052
	(0.0006)	(0.0033)	(0.0009)
Total between, $T_B(X^{(4)})$	0.1657	0.1642	0.1285
	(0.0061)	(0.0065)	(0.0049)
Remaining within, $T_W(X^{(4)})$	0.0705	0.0732	0.0726
	(0.0020)	(0.0037)	(0.0028)
Observations	1,588,097	1,507,072	1,741,556

Notes: All conditioning sets use the same annual $X^{(4)}$ based sample, without an exposure restriction. The first contribution is $T_B(X^{(1)})$; subsequent contributions are the increases in between inequality from adding each information block. The conditional-mean estimates use squared-error XGBoost with five outer and five inner folds. Standard errors are in parentheses: unclustered for total Theil and employer-clustered for components and increments.

A small net change in the estimated within component can conceal different changes in conditional dispersion across groups of workers because it averages conditional Theil indices over workers, using relative conditional mean wages of the group as weights. The conditional-Theil profiles complement this aggregate measure by making differences in dispersion and their evolution directly estimable, while retaining the full set of observed worker and employer characteristics in the definition of conditional inequality. Appendix D reports additional partial-mean profiles of conditional inequality by worker, job, and employer characteristics. The initial- and rolling-exposure analyses next examine these patterns across minimum-wage exposure levels and sex–education groups.

6.2 Minimum-wage exposure and conditional inequality

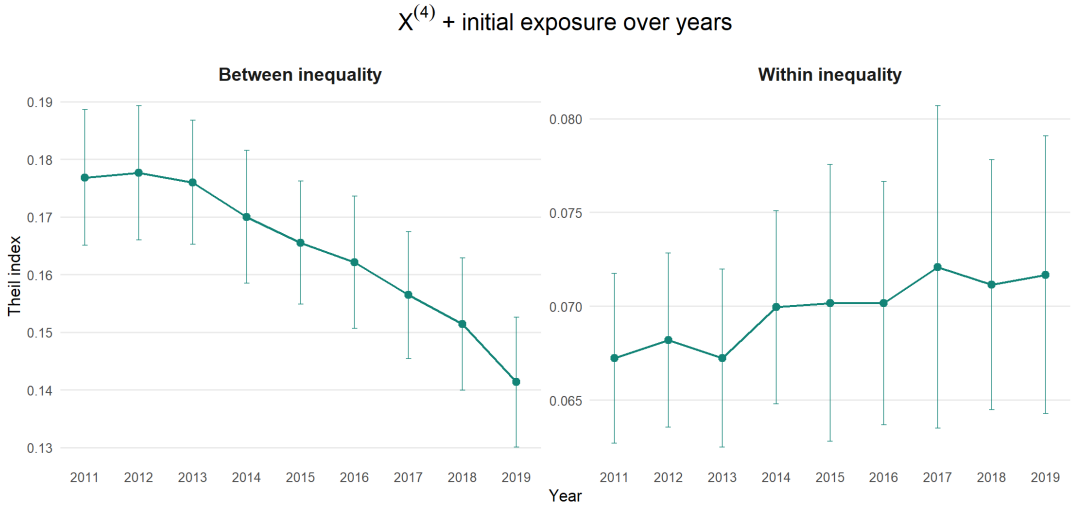
Minimum-wage increases affect employers unevenly, depending on how many of their workers are paid below the new floor and by how much. As noted in Section 5.3, exposure measures the intensity of this shortfall, not the share of employers or workers affected.

This variation calls for examining whether the wage compression among comparable workers is heterogeneous across the employer exposure levels defined in Section 5.3. The initial-exposure

analysis studies conditional inequality over 2011–2019 using exposure to the 2014 increase held fixed across years. It examines changes in conditional dispersion over this period across employers with different initial exposure. The rolling-exposure analysis instead calculates exposure for each increase and compares wage dispersion in the years just before and after the increase, asking whether wage gains near the floor were accompanied by lower conditional dispersion in the short term.

Figure 6 shows that the aggregate pattern of the preceding subsection carries over to the exposure-based samples defined in Section 5.3, although the levels differ. With initial exposure included alongside $X^{(4)}$, the estimated between component falls from 0.1769 in 2011 to 0.1413 in 2019, while the within component shows a small increase from 0.0672 to 0.0717. The decline in the between component therefore more than accounts for the fall in total inequality, from 0.2441 to 0.2130. The rolling-exposure decompositions show a similar pattern across the post-increase distributions: between inequality declines along the sequence of minimum-wage increases, from 0.1700 in 2014 to 0.1384 in 2019, while the five within components lie between 0.0695 and 0.0727 (Figure 6).

Figure 6: Within and between inequality with initial minimum-wage exposure

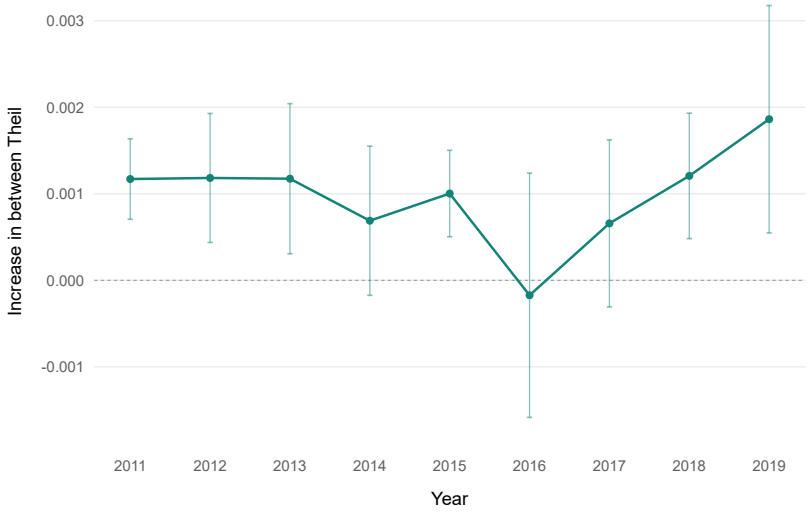


Notes: Components condition on $(X^{(4)}, B)$, where B denotes initial exposure to the 2014 minimum-wage increase. Each decomposition uses its exposure-restricted sample. Bars are nominal 95 percent pointwise intervals.

Figure 7 shows that exposure adds little to the level of between inequality, and its contribution remains small throughout the period. The corresponding rolling-exposure results are reported in Figure 23. When $(X^{(4)}, B)$ is compared with $X^{(4)}$ in the same annual sample, the increment for the initial exposure is 0.0012 in 2011 and 0.0019 in 2019, and for the rolling exposure it ranges from about zero to 0.0016, all below one percent of total unconditional Theil. Hence, once observed worker and employer characteristics are taken into account, exposure accounts for little additional dispersion in mean wages. I therefore use the conditional Theil profiles to ask whether conditional dispersion in wages, rather than mean wages, varies with exposure.

The initial-exposure profiles reveal heterogeneity in conditional dispersion that is not highlighted by the aggregate within component. Figure 8 shows how average conditional hourly-wage inequality varies with employers' initial exposure to the 2014 increase in 2011 and 2019. The

Figure 7: Additional between inequality captured by initial minimum-wage exposure



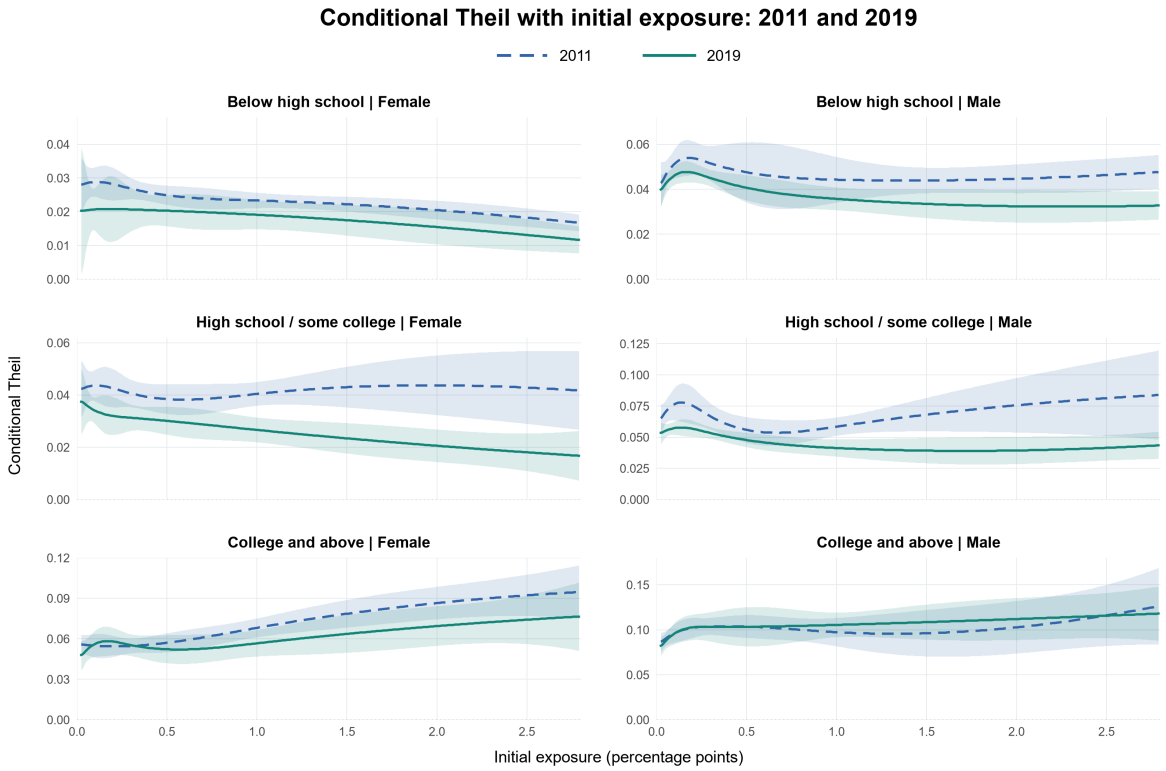
Notes: Each increment compares $X^{(4)}$ and $(X^{(4)}, B)$ for the sample, where B denotes initial exposure to the 2014 minimum-wage increase. Bars are nominal 95 percent pointwise intervals.

columns distinguish women and men, and the rows distinguish the three education categories. The horizontal axis measures the incremental base-pay shortfall implied by the 2014 increase as a percentage of the employer's 2013 base wage bill. Thus, an exposure value of one denotes a shortfall equal to one percent of that wage bill, not exposure affecting one percent of employers. The vertical axis reports the average conditional Theil index: each point on a curve averages the conditional Theil indices, over the workers in the corresponding sex–education group at that exposure level, with lower values indicating less dispersion. It is therefore not the raw Theil index obtained by pooling all workers in that group or the inequality within an individual employer.

Among workers without a college degree, conditional dispersion generally declines with exposure in 2019, although the profiles of men flatten or rise slightly at higher exposure levels. The relationship is less monotonic in 2011, particularly for men with high school or some college, whose profile first decreases and then turns upward at higher exposure levels. Consequently, the gap between the 2011 and 2019 profiles becomes larger towards the upper end of the exposure range for men without a college degree and for women with high school or some college. At the upper end of the range, the gap is largest for women and men with high school or some college. Fitted conditional dispersion nevertheless remains higher for men than for women within both non-college education categories in both years. The 2013 wage distributions of these demographic groups provide context for these patterns. Figure 9 shows that workers without a college degree were several times more likely than college graduates to have monthly base pay below the forthcoming minimum wage. The compression at the lower end of the wage distribution during the minimum-wage increases was therefore more likely to involve these groups.

Figure 8 shows a different pattern among college-educated workers in the lower panels. For women, conditional dispersion rises with exposure over much of the range in both years, and the 2019 profile lies below the 2011 profile over most of the range, with the gap between them

Figure 8: Conditional hourly-wage inequality by initial exposure, sex, and education: 2011 and 2019



Notes: Profiles summarize conditional Theil inequality given $(X^{(4)}, B)$ separately by sex and education, where B is initial exposure to the 2014 minimum-wage increase, measured from 2013 wages as a percentage of the employer's base wage bill. A value of one is a shortfall equal to one percent of the base wage bill, not one percent of employers. Each curve shows average wage dispersion among workers sharing these characteristics at a given exposure level, with lower values indicating less dispersion. Dashed lines denote 2011 and solid lines 2019. Shading shows nominal 95 percent pointwise intervals for each annual profile, using employer-clustered standard errors. The display is restricted up to the worker-weighted 95th percentile in the 2013 reference sample.

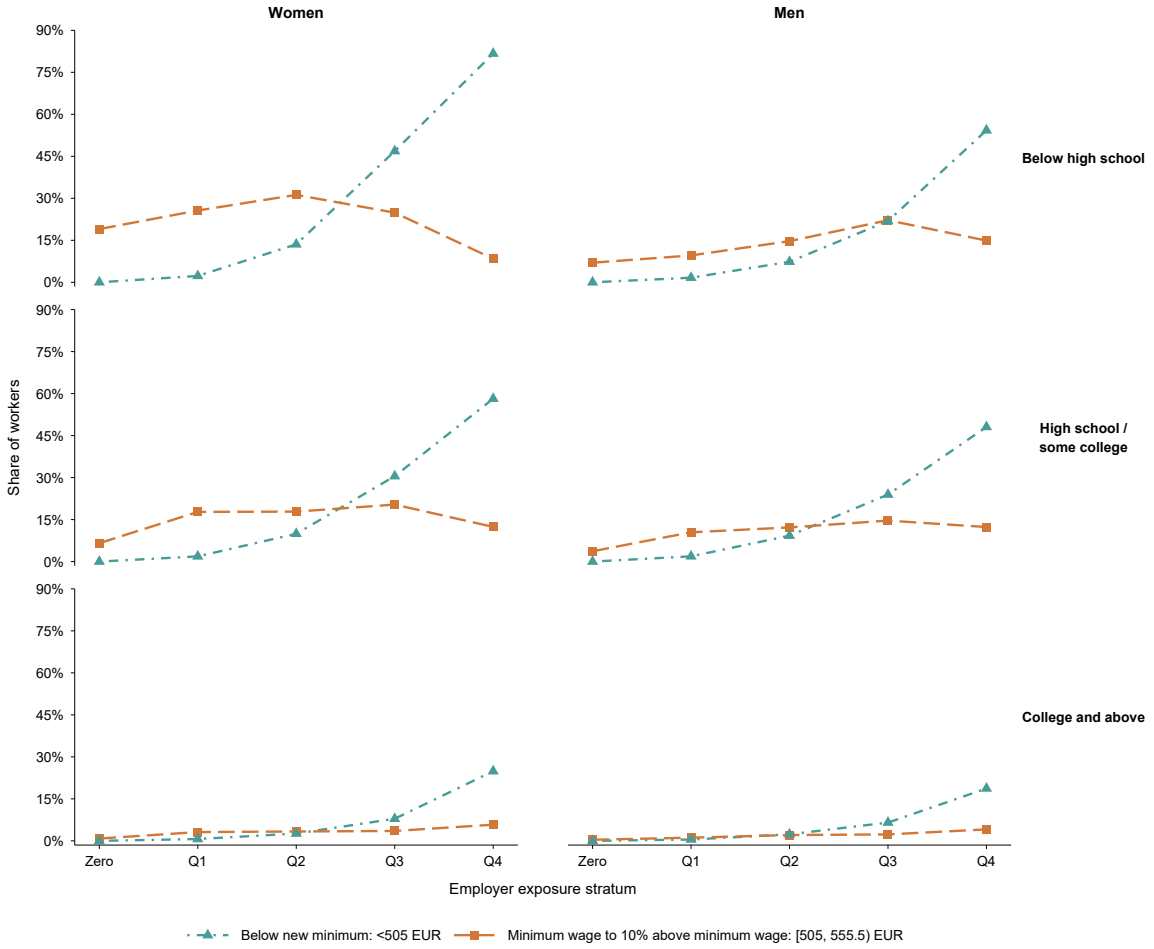
generally widening with exposure. For men, the two profiles are very close to each other, and both rise slightly with exposure.

Taken together, the point estimates show lower conditional dispersion in 2019 than in 2011 over most of the displayed range for five of the six considered groups, with college-educated men following a different pattern. For 2019, they also show a generally negative relationship between exposure and conditional dispersion for workers without a college degree. However, they show a positive relationship with exposure for college-educated workers, whose fitted dispersion is generally the highest. Finally, conditional dispersion remains higher for men than for women within each education category in both years.

These patterns complement the aggregate accounting in Figure 6, where the aggregate within component rises slightly. The declines in the profiles can coexist with this rise because the aggregate component weights conditional inequality by relative conditional mean wages, giving more weight to groups with higher mean wages, and reflects changes in workforce composition.

Workers paid close to the new minimum saw substantial wage gains around each increase. Table 7 groups workers at employers with positive exposure into two bands based on their monthly base pay before each increase. Workers in the first band were paid at or above the

Figure 9: Pre-increase wage positions by employer exposure, sex, and education, 2013



Notes: Shares are calculated using nominal monthly base pay in the 2013 initial-exposure sample. The two series report pay below the forthcoming minimum of 505 euros and pay at or above 505 euros but below 555.5 euros. Zero denotes employers with zero exposure; Q1–Q4 denote quartile groups of positive exposure, defined using common cutoffs across all demographic groups.

old minimum but below the new one, and those in the second were paid at or above the new minimum but less than 110 percent of the new minimum wage. Among workers observed in both years, mean real hourly pay rose by 4.8–7.0 percent in the first band and by 2.6–6.5 percent in the second. Growth was higher in the first band for the first four increases and was the same in both bands for the last increase, in 2018–2019. The disaggregated results by exposure level show that, within the first band, wage growth was lower for workers at more exposed employers (Table 9), while college graduates recorded the largest gains within each exposure category (Tables 11, 13, and 15 in Appendix E).

The rolling-exposure profiles examine changes in conditional dispersion around individual increases. The estimated changes differ across demographic groups, exposure levels, and minimum-wage increases. Men without a college degree show a decline in conditional dispersion at higher exposure during 2013–2014, 2015–2016, and 2017–2018. For these men, the fitted changes are close to zero at higher exposure in 2016–2017. In 2018–2019, the estimates differ across the two education categories: a small negative change for men below high school and a positive but very

Table 7: Wage growth and subsequent observation by pre-increase pay band

	A. Between old and new minima $m_r \leq w_r < m_s$			*B. At or just above new minimum* $m_s \leq w_r < 1.1m_s$		
Comparison	Matched workers	Observed next year (%)	Real hourly-wage growth (%)	Matched workers	Observed next year (%)	Real hourly-wage growth (%)
2013–2014	127,859	72.5	5.0	78,843	77.2	3.2
2015–2016	162,410	72.8	4.8	90,429	78.3	2.6
2016–2017	209,519	72.8	6.2	98,639	79.1	4.2
2017–2018	217,780	72.1	7.0	107,802	78.1	6.2
2018–2019	186,577	69.6	6.5	130,210	75.0	6.5

Notes: For each comparison $e = (r, s)$, pay bands are defined using nominal monthly base pay w_r before the increase; m_r and m_s denote the old and new minimum wages. Both panels use workers with complete information on the variables in $X^{(4)}$ who are employed at firms with positive exposure. Wage growth is calculated as $100(\bar{Y}_{s,V}/\bar{Y}_{r,V} - 1)$, using real regular hourly pay in October-2019 euros for workers with observed wages in both years. The observation rate is the share of the workers in the starting group found in the following year's records.

imprecise change for men with high school or some college (Figures 26 and 30; the other three increases are in Appendix E).

Together, the results add a conditional perspective to the study of wage compression around minimum-wage increases. In this application, the aggregate decline is concentrated in the between component and is already apparent under the worker and job characteristics in $X^{(1)}$. Exposure adds little further between inequality. The fitted profiles show lower conditional dispersion in 2019 than in 2011 over most of the displayed exposure range for five of the six demographic groups. At higher initial exposure, the largest estimated declines are among women and men with high school or some college. Around individual minimum-wage increases, mean wage growth for workers near the wage floor is positive, while men without a college degree show a decline in conditional dispersion at higher exposure in three of the five increases. The profiles therefore reveal differences in conditional wage compression across demographic groups, exposure levels, and increases that the aggregate accounting does not show separately.

7 Conclusion

This paper develops a framework for estimating and conducting inference on conditional Theil inequality and the components of its exact aggregate decomposition. The conditional Theil surface measures dispersion among workers sharing observed worker, job, and employer characteristics. Its wage-share-weighted average is the within component, while the between component measures Theil inequality in conditional mean wages. I derive a Neyman-orthogonal signal for the conditional Theil index and estimate the surface and its profiles by regressing the cross-fitted signal on a growing series basis. I establish pointwise inference under conditions on conditional-moment estimation and series approximation. The aggregate components and their increments across nested conditioning sets have direct locally robust estimators that require only conditional mean wage functions as nonparametric inputs and admit root-n inference under suitable regularity conditions. The estimated within and between components also add exactly to the sample

Theil index. Simulations illustrate the benefits of the orthogonal correction: coverage improves substantially for the conditional surface and fixed-reference partial-mean profiles, although some undercoverage remains. Coverage for the aggregate components depends more strongly on the first-stage learner, with XGBoost performing well for both components.

The framework is used to study wage compression during a sequence of minimum-wage increases in Portugal, where five increases between 2014 and 2019 raised the wage floor by about 24 percent. Three patterns emerge. First, the decline in aggregate inequality is concentrated in the between component, and most of the decline in this component is already recorded when conditioning on worker and job characteristics alone. Industry, location, and observed employer characteristics contribute to the level of between inequality but account for comparatively little of its decline. Second, employer exposure to the minimum-wage increases adds little to between inequality after conditioning on these observed characteristics. Third, the nearly unchanged aggregate within component conceals substantial heterogeneity. Fitted conditional-inequality profiles are generally lower in 2019 than in 2011 among workers without a college degree, with larger estimated declines at higher initial exposure to the 2014 increase among women and men with high school or some college education. College-educated men show little comparable decline. These findings are descriptive: changes in the components and profiles can reflect changes in conditional wage distributions, workforce composition, or both, and do not identify causal effects of the minimum-wage increases.

The framework applies beyond this episode. Under the stated moment and estimation conditions, the same approach can be used to study inequality in other positive outcomes, such as earnings, hours, or firm outcomes, and in other policy settings. The aggregate decomposition and conditional profiles provide a common framework for examining overall inequality and dispersion among units sharing observed characteristics. The approach also suggests extensions to causal questions. Remark 1 outlines an orthogonal signal for conditional inequality treatment effects under additional identification and regularity conditions. Extending this construction to a difference-in-differences design is a direction for future research on the causal effects of wage floors on dispersion among comparable workers.

A Proofs for the Theil estimators and decomposition results

A.1 Proof of Proposition 1

The assumptions imply that $\mu_0(X) > 0$ almost surely and $\mathbb{E}[|\mu_0(X)\log\mu_0(X)|] < \infty$. The latter follows from conditional Jensen's inequality and the assumed integrability of $Y\log Y$. Conditional on X,

$$\mathbb{E}\left[\frac{Y}{\mu}\log\left(\frac{Y}{\mu}\right)\middle|X\right] = \frac{\mu_0(X)}{\mu}T_0(X) + \frac{\mu_0(X)}{\mu}\log\left(\frac{\mu_0(X)}{\mu}\right).$$

Taking expectations and using the definitions of T_W and T_B gives $T = T_W + T_B$. □

A.2 Proof of Corollary 1

For $r \in \{a, b\}$, write $\mu_r = \mathbb{E}[Y \mid X^{(r)}]$. Under the positivity and integrability conditions, $0 < \mu_r < \infty$ almost surely and $\mathbb{E}[\mu_r] = \mu$. Conditional Jensen's inequality, applied to the convex function $u \mapsto u\log u$, which is bounded below by $-e^{-1}$, also gives

$$\mathbb{E}[|\mu_r\log\mu_r|] < \infty, \qquad r \in \{a, b\}.$$

The nesting condition and the tower property imply

$$\mathbb{E}[\mu_b \mid X^{(a)}] = \mu_a.$$

In particular, the cross-logarithmic term is integrable because

$$\mathbb{E}[\mu_b|\log\mu_a|] = \mathbb{E}\Big[\mathbb{E}[\mu_b \mid X^{(a)}]|\log\mu_a|\Big] = \mathbb{E}[\mu_a|\log\mu_a|] < \infty.$$

Here the first equality holds for nonnegative random variables, so it does not presuppose integrability of the left-hand side. Consequently,

$$\mathbb{E}[\mu_b\log\mu_a] = \mathbb{E}[\mu_a\log\mu_a].$$

Applying Proposition 1 to both conditioning sets under the same population distribution shows that

$$T = T_W^{(a)} + T_B^{(a)} = T_W^{(b)} + T_B^{(b)}.$$

Thus,

$$\Delta T_B := T_B^{(b)} - T_B^{(a)} = T_W^{(a)} - T_W^{(b)}.$$

Since $T_B^{(r)} = \mu^{-1}\mathbb{E}[\mu_r\log\mu_r] - \log\mu$, the preceding cross-logarithmic identity gives

$$\begin{aligned}\mu\Delta T_B &= \mathbb{E}[\mu_b\log\mu_b] - \mathbb{E}[\mu_a\log\mu_a] \\ &= \mathbb{E}[\mu_b\log\mu_b] - \mathbb{E}[\mu_b\log\mu_a] \\ &= \mathbb{E}\left[\mu_b\log\left(\frac{\mu_b}{\mu_a}\right)\right].\end{aligned} \tag{43}$$

All terms in this calculation are integrable by the bounds above.

Because $\mathbb{E}[\mu_a - \mu_b] = 0$, equation (43) can also be written as

$$\mu \Delta T_B = \mathbb{E}\left[\mu_b \log\left(\frac{\mu_b}{\mu_a}\right) - \mu_b + \mu_a\right]. \tag{44}$$

For positive u, v,

$$u\log(u/v) - u + v = v\{(u/v)\log(u/v) - (u/v) + 1\} \geq 0,$$

with equality if and only if $u = v$. The integrand in (44) is therefore nonnegative, which proves $\Delta T_B \geq 0$. Since $\mu > 0$, $\Delta T_B = 0$ if and only if this integrand is zero almost surely, or equivalently,

$$\mu_b = \mu_a \quad \text{almost surely.}$$

This is precisely the equality characterization in (4).

To obtain the conditional interpretation, condition (43) on $X^{(a)}$ and use the measurability and positivity of μ_a:

$$\begin{aligned}\Delta T_B &= \frac{1}{\mu}\,\mathbb{E}\left[\mathbb{E}\left[\mu_b \log\left(\frac{\mu_b}{\mu_a}\right)\middle| X^{(a)}\right]\right] \\ &= \mathbb{E}\left[\frac{\mu_a}{\mu}\,\mathbb{E}\left[\frac{\mu_b}{\mu_a}\log\left(\frac{\mu_b}{\mu_a}\right)\middle| X^{(a)}\right]\right].\end{aligned}$$

Since $\mathbb{E}[\mu_b \mid X^{(a)}] = \mu_a$, the inner expectation is the conditional Theil index of μ_b given $X^{(a)}$. It is finite almost surely: its product with μ_a is the conditional expectation of the integrable variable $\mu_b \log(\mu_b/\mu_a)$, and $\mu_a > 0$ almost surely. The weights μ_a/μ have mean one.

Finally, consider any finite nested sequence $X^{[0]}, \ldots, X^{[L]}$ whose initial and final sigma-algebras equal those generated by $X^{(a)}$ and $X^{(b)}$, respectively. The sequential increments telescope:

$$\begin{aligned}\sum_{\ell=1}^{L}\left\{T_B(X^{[\ell]}) - T_B(X^{[\ell-1]})\right\} &= T_B(X^{[L]}) - T_B(X^{[0]}) \\ &= \Delta T_B.\end{aligned}$$

Hence their sum is unchanged by the order of inclusion, provided the population and endpoint information sets are fixed. This identity does not require the individual increments to be invariant to the order. □

A.3 Proof of Proposition 2

Start from the uncorrected moment (8),

$$g_K(W; \beta, \mu) = p_K(X)\left\{\frac{Y\log Y}{\mu(X)} - \log\mu(X) - p_K(X)^\top\beta\right\}.$$

At the truth, $\mathbb{E}[g_K(W; \beta_K, \mu_0)] = 0_K$ because, by the definition of the conditional Theil index, $\mathbb{E}[g_K(W; \beta_K, \mu_0) \mid X] = p_K(X)\{T_0(X) - p_K(X)^\top\beta_K\}$, whose expectation is zero by (7). Consider a path $\mu_t(x)$ with μ_0 at $t = 0$, bounded away from zero for $|t|$ small. Only the terms $Y\log Y/\mu_t(X)$

and $\log \mu_t(X)$ involve the nuisance, so differentiating the population moment gives

$$\begin{aligned}\frac{\partial}{\partial t}\mathbb{E}[g_K(W;\beta_K,\mu_t)]\Big|_{t=0} &= \frac{\partial}{\partial t}\mathbb{E}\left[-\frac{p_K(X)}{\mu_0(X)^2}\,\mu_t(X)\,Y\log Y\right]\Big|_{t=0} - \frac{\partial}{\partial t}\mathbb{E}\left[\frac{p_K(X)}{\mu_0(X)}\,\mu_t(X)\right]\Big|_{t=0} \\ &= \frac{\partial}{\partial t}\mathbb{E}\left[-\frac{p_K(X)}{\mu_0(X)^2}\,\nu_0(X)\,\mu_t(X)\right]\Big|_{t=0} - \frac{\partial}{\partial t}\mathbb{E}\left[\frac{p_K(X)}{\mu_0(X)}\,\mu_t(X)\right]\Big|_{t=0} \qquad (45) \\ &= \frac{\partial}{\partial t}\mathbb{E}\left[\frac{p_K(X)}{\mu_0(X)}\left\{\frac{\nu_0(X)}{\mu_0(X)}+1\right\}\{Y-\mu_t(X)\}\right]\Big|_{t=0},\end{aligned}$$

where the second equality uses $\mathbb{E}[Y\log Y \mid X] = \nu_0(X)$. The boundedness of p_K, the lower bound on μ_0, and $\mathbb{E}[|Y\log Y|] < \infty$ justify the interchange of differentiation and expectation. Hence the first stage correction term is

$$\phi_K(W;\mu,\nu) = -\frac{p_K(X)}{\mu(X)}\{Y-\mu(X)\}\left\{\frac{\nu(X)}{\mu(X)}+1\right\},$$

which has conditional mean zero at the truth, since $\mathbb{E}[Y-\mu_0(X)\mid X]=0$, and whose derivative along μ_t is the negative of (45). Hence the locally robust moment is

$$\psi_K(W;\beta,\mu,\nu) = g_K(W;\beta,\mu) + \phi_K(W;\mu,\nu),$$

which is (12). The correction has conditional mean zero at the true functions, so the locally robust moment retains the mean-zero property of (9). To verify orthogonality with respect to the second nuisance function, fix $\mu=\mu_0$ and consider $\nu_t = \nu_0 + th_\nu$. For bounded h_ν,

$$\frac{d}{dt}\mathbb{E}[\psi_K(W;\beta_K,\mu_0,\nu_t)]\Big|_{t=0} = -\mathbb{E}\left[\frac{p_K(X)}{\mu_0(X)^2}\mathbb{E}[Y-\mu_0(X)\mid X]h_\nu(X)\right] = 0_K.$$

The derivative with respect to μ vanishes by the preceding cancellation. Together these establish the directional derivative condition in (13). □

A.4 Proof of Proposition 3

Normalization and identification. Consider first fixed-order B-splines on quasi-uniform meshes. For coordinate ℓ, let B_{j,m_ℓ} denote the unnormalized B-spline basis with m_ℓ intervals. Normalize each basis function as

$$\widetilde{B}_{j,m_\ell}(u) = \sqrt{m_\ell}\,B_{j,m_\ell}(u).$$

The tensor-product basis is formed from these normalized functions. For balanced meshes, its overall scaling is of order $\sqrt{K}$, where K is the total basis dimension up to the fixed number of discrete strata.

For fixed spline order and uniformly bounded mesh ratio, the normalized B-spline basis has uniformly bounded Lebesgue-Gram eigenvalues. The conditional density bounds transfer these bounds to the population Gram matrix Q_K.

Boundary-corrected orthonormal wavelet spaces satisfy the corresponding Lebesgue-Gram bounds by orthonormality. For shifted Legendre polynomials, $\int_0^1 \phi_j(u)\phi_l(u)\,du = \mathbf{1}\{j=l\}$, and

their tensor products are likewise orthonormal under Lebesgue measure.

With a fixed finite number of discrete strata, the Gram matrix is block diagonal, with blocks weighted by stratum probabilities bounded away from zero. Hence the eigenvalues of Q_K are uniformly bounded above and away from zero.

These standard basis constructions and their tensor-product extensions are discussed in Belloni et al. (2015) and X. Chen and Christensen (2015). For any coefficient vector b, the density bounds transfer the corresponding Lebesgue-Gram bounds to the design-weighted Gram matrix Q_K.

Basis envelopes. A fixed-order normalized B-spline vector has a bounded number of active entries per coordinate, each of order $\sqrt{m_\ell}$. Thus the squared norm of a balanced tensor-product basis is of order K uniformly on the support, giving $\zeta_K \asymp \sqrt{K}$. The partition-of-unity property yields the matching lower bound. Boundary-corrected, compactly supported wavelet spaces satisfy the same projection-kernel bounds, and an orthonormal change of coordinates does not affect the norm of the vector of basis evaluations.

For Legendre polynomials, $|\mathrm{P}_j(v)| \le 1$ on $[-1,1]$ and $\mathrm{P}_j(1) = 1$, so

$$\sup_{u\in[0,1]} \sum_{j=0}^{M-1} \phi_j(u)^2 = \sum_{j=0}^{M-1} (2j+1) = M^2.$$

For a balanced d_x-dimensional tensor product, the squared norm factorizes and the supremum is of order $M^{2d_x} = K^2$, giving $\zeta_K \asymp K$. At each fixed interior point, standard interior Legendre asymptotics yield a squared norm of order K (NIST Digital Library of Mathematical Functions, 2026).

Approximation and projection stability. Tensor-product polynomial approximation on a Hölder ball has best uniform error $O(K^{-s_T/d_x})$. Fixed-order splines and sufficiently regular CDV spaces give $O(K^{-\min\{s_T,s_0\}/d_x})$, where the order convention in Assumption 3(iii) is maintained (Belloni et al., 2015; X. Chen, 2007; X. Chen & Christensen, 2015). These are best-approximation statements, not yet statements about the P_X-weighted projection. For any $v_K \in \mathrm{span}(p_K)$,

$$T_0 - \mathcal{P}_K T_0 = (I - \mathcal{P}_K)(T_0 - v_K), \qquad \|T_0 - \mathcal{P}_K T_0\|_\infty \le L_K \|T_0 - v_K\|_\infty.$$

Taking the infimum proves the projection-error bound. Moreover, by Cauchy–Schwarz and the contraction property of the orthogonal projection in $L_2(P_X)$,

$$|(\mathcal{P}_K f)(x)| \le \{p_K(x)^\top Q_K^{-1} p_K(x)\}^{1/2} \|f\|_2 \le C\zeta_K \|f\|_\infty.$$

Thus $L_K \le 1 + C\zeta_K$ always suffices. B-spline and compactly supported CDV projections are uniformly stable in sup norm under the specified bounded-density and mesh conditions; the wavelet result is Theorem 5.1 of X. Chen and Christensen (2015). This gives $L_K = O(1)$ for the local constructions. No such blanket conclusion is used for Legendre projections. □

Basis-specific rate implications. For notational convenience, define $D_n = r_{\mu,n}^2 + r_{\mu,n} r_{\nu,n}$ and $e_n = r_{\mu,n} + r_{\nu,n}$. The sufficient rate conditions specialize to

Basis	Gram/covariance growth	Nuisance bias	Signal stability
B-splines/CDV	$K^2 \log K/n \to 0$	$\sqrt{nK}\, D_n = o_p(1)$	$\sqrt{K}\, e_n = o_p(1)$
Legendre	$K^3 \log K/n \to 0$	$\sqrt{n}\, K D_n = o_p(1)$	$K e_n = o_p(1)$

Approximation must additionally satisfy $a_{T,K} \to 0$ and, for surface inference, $\sqrt{n/V_K(x)} a_{T,K} \to 0$. In particular, when $L_K = O(1)$ and the evaluation point is interior, $K \asymp n^\kappa$ can satisfy the local-basis conditions if $d_x/(2\bar{s}_T + d_x) < \kappa < 1/2$, requiring $\bar{s}_T > d_x/2$, together with compatible nuisance rates. For Legendre, the universally available bound $L_K \lesssim K$ yields the more conservative sufficient interval $d_x/(2s_T - d_x) < \kappa < 1/3$, which is nonempty if $s_T > 2d_x$. A sharper projector bound or a smaller actual pointwise approximation error can relax this sufficient smoothness threshold. These examples verify compatibility, not optimality, of the displayed second-stage rates; profile inference retains its own approximation condition. None of these statements specifies or establishes a rate for a particular first-stage learner.

A.5 Weighted nuisance control and oracle equivalence

This subsection proves Lemma 1. Each expectation conditional on $\mathcal{D}_{-j}$ is over a fresh observation, with algorithmic randomization included in the training information when present. Define

$$\Delta_j(W) = s(W; \widehat{\mu}_{-j}, \widehat{\nu}_{-j}) - s(W; \mu_0, \nu_0), \tag{46}$$

$$\begin{aligned} m_{K,j} &= \mathbb{E}[p_K(X)\Delta_j(W) \mid \mathcal{D}_{-j}], \\ \Lambda_{n,j}^2 &= \mathbb{E}[\|p_K(X)\|^2 \Delta_j(W)^2 \mid \mathcal{D}_{-j}], \\ B_n &= \max_j \sqrt{n}\, \|m_{K,j}\|, \qquad \Lambda_n = \max_j \Lambda_{n,j}. \end{aligned} \tag{47}$$

These are the basis-weighted conditional-bias and signal-error quantities used in orthogonal-series arguments (Semenova & Chernozhukov, 2021). They are verified from primitive conditions here, rather than imposed in addition to them.

Lemma 2 (Primitive weighted bounds for the Theil signal)**.** *Under Assumption 2,*

$$B_n \le C\sqrt{n}\, \zeta_K D_n, \qquad \Lambda_n \le C \zeta_K e_n, \tag{48}$$

where C is independent of n, K, and the fold. Thus (20) implies

$$B_n = o_p(1), \qquad \Lambda_n = o_p(1).$$

Proof. Fix a fold j, condition on $\mathcal{D}_{-j}$, and work on the stabilization event in Assumption 2(ii)–(iii). For a fixed covariate value x

$$a = \mu_0(x), \quad c = \nu_0(x), \quad m = \widehat{\mu}_{-j}(x), \quad v = \widehat{\nu}_{-j}(x), \quad d = m - a, \quad e = v - c.$$

The conditional moment bound implies that a and $|c|$ are bounded above, and both a and m are bounded below by $\underline{\mu} > 0$. All constants below are uniform in the fold, covariate value, sample, and basis dimension.

Conditional bias. Taking the conditional expectation of (14) and subtracting $T_0(x) = c/a - \log a$ gives the exact identity

$$\begin{aligned} b_j(x) &:= \mathbb{E}[\Delta_j(W) \mid X = x, \mathcal{D}_{-j}] \\ &= \frac{d}{m} - \log\frac{m}{a} - \frac{c}{a}\frac{d^2}{m^2} + \frac{de}{m^2}. \end{aligned}$$

The first two terms can be written as

$$\frac{d}{m} - \log\frac{m}{a} = -d^2 \int_0^1 \frac{t}{(a+td)^2}\, dt.$$

The entire line segment between a and m lies above $\underline{\mu}$, so

$$|b_j(x)| \le C\left(|d|^2 + |de|\right). \tag{49}$$

Thus both first-order conditional mean derivatives vanish, but a quadratic mean-error term and a mixed error term remain. Multiplication by the basis, followed by Cauchy–Schwarz, gives

$$\begin{aligned} \|m_{K,j}\| &\le \zeta_K \mathbb{E}[|b_j(X)| \mid \mathcal{D}_{-j}] \\ &\le C\zeta_K \left\{ \|\widehat{\mu}_{-j} - \mu_0\|_2^2 + \|\widehat{\mu}_{-j} - \mu_0\|_2 \|\widehat{\nu}_{-j} - \nu_0\|_2 \right\}. \end{aligned}$$

Taking the maximum over folds proves the bound for B_n in (48).

Signal stability. Rewrite the scalar signal as

$$s(W; m, v) = \frac{Y \log Y}{m} - Y\left(\frac{v}{m^2} + \frac{1}{m}\right) + \frac{v}{m} + 1 - \log m, \tag{50}$$

where the functions are evaluated at X. The reciprocal maps and logarithm are Lipschitz on $[\underline{\mu}, \infty)$. For example,

$$\left|\frac{v}{m^2} - \frac{c}{a^2}\right| \le \frac{|v-c|}{m^2} + |c|\left|\frac{1}{m^2} - \frac{1}{a^2}\right| \le C(|e| + |d|).$$

The other coefficients in (50) obey analogous bounds. Consequently,

$$|\Delta_j(W)| \le C(1 + Y + |Y \log Y|)(|\widehat{\mu}_{-j}(X) - \mu_0(X)| + |\widehat{\nu}_{-j}(X) - \nu_0(X)|).$$

The conditional second moments implied by (18) yield

$$\begin{aligned} &\mathbb{E}[\Delta_j(W)^2 \mid X, \mathcal{D}_{-j}] \\ &\quad \le C\{|\widehat{\mu}_{-j}(X) - \mu_0(X)|^2 + |\widehat{\nu}_{-j}(X) - \nu_0(X)|^2\}. \end{aligned} \tag{51}$$

Multiplying by $\|p_K(X)\|^2 \le \zeta_K^2$, integrating, and taking square roots proves

$$\Lambda_{n,j} \le C\zeta_K(\|\widehat{\mu}_{-j} - \mu_0\|_2 + \|\widehat{\nu}_{-j} - \nu_0\|_2).$$

This proves the bound for Λ_n. The primitive series rates make both bounds vanish. Only conditional second moments are needed for this nuisance verification. The fourth-moment condition in Assumption 2(ii) is used below to establish pointwise normality and consistency of the feasible covariance estimator.

□

Lemma 3 (Foldwise sample remainder)**.** *Under the sampling and fixed-fold construction of Assumption 2, write $p_i = p_K(X_i)$ and*

$$R_n = \frac{1}{n}\sum_{j=1}^{J}\sum_{i\in I_j} p_i\Delta_j(W_i). \tag{52}$$

For the quantities in (47),

$$\|R_n\| = O_p\left(\frac{B_n + \Lambda_n}{\sqrt{n}}\right). \tag{53}$$

In particular, $B_n + \Lambda_n = o_p(1)$ implies $\sqrt{n}\|R_n\| = o_p(1)$. The argument applies to any fold-measurable signal difference with these bounds, including the scalar case with multiplier one.

Proof. For each fold, conditional on $\mathcal{D}_{-j}$, the evaluation observations are independent and

$$\mathbb{E}\left[\left\|\frac{1}{\sqrt{n}}\sum_{i\in I_j}\{p_i\Delta_j(W_i) - m_{K,j}\}\right\|^2 \middle| \mathcal{D}_{-j}\right] \le \frac{n_j}{n}\Lambda_{n,j}^2.$$

For the conditional mean term, $\|\sum_j (n_j/\sqrt{n})m_{K,j}\| \le B_n$. Apply conditional Markov's inequality to the centered term in each fold and sum over the fixed number of folds. This yields (53). The argument conditions on one training fold at a time, not simultaneously on all fitted nuisances while asserting independence of the full evaluation sample. □

Proof of Lemma 1. The Gram concentration bound (56), which uses only the design and growth conditions, gives $\|\widehat{Q}_K^{-1}\| = O_p(1)$. The exact feasible-minus-oracle identity is

$$\widehat{\beta}_K - \widehat{\beta}_K^o = \widehat{Q}_K^{-1}R_n.$$

Lemmas 2 and 3 therefore give (17). □

Normalization and sufficient rates. For a direction with $V_K(x) \asymp \|p_K(x)\|^2$,

$$\frac{|p_K(x)^\top Q_K^{-1}m_{K,j}|}{\sqrt{V_K(x)/n}} \le \frac{\|p_K(x)\|\|Q_K^{-1}\|}{\sqrt{V_K(x)}}\sqrt{n}\,\|m_{K,j}\| \lesssim B_n.$$

Thus normalization cancels the evaluation norm; it does not also remove the basis envelope in a bound on $m_{K,j}$. The conditions imply the weaker consistency bound

$$\frac{\zeta_K}{\sqrt{n}}(B_n + \Lambda_n) = o_p(1).$$

If both nuisance errors have a common upper rate r_n, $r_n = o_p(n^{-1/4}\zeta_K^{-1/2})$ suffices for both primitive restrictions under the maintained basis growth. This specializes to $o_p((nK)^{-1/4})$ for local bases and $o_p(n^{-1/4}K^{-1/2})$ for Legendre. These are conservative global-norm conditions, not minimal requirements; stronger-norm or direction-specific arguments can yield alternatives.

The restriction $K\zeta_K^2 \log K/n \to 0$ supplies the inverse-replacement and covariance bounds used below. It becomes $K^2 \log K/n \to 0$ for local bases and $K^3 \log K/n \to 0$ for Legendre. Sharper series arguments can allow weaker restrictions (Belloni et al., 2015; X. Chen & Christensen, 2015). Approximation enters the surface linearly: a condition on $\sqrt{n}K^{-2\bar{s}_T/d_x}$ does not replace its target-specific undersmoothing requirement.

A.6 Proof of Theorem 1, part (a)

Let $p_i = p_K(X_i)$, and use the scalar signal s defined in (14). The actual estimator satisfies

$$\widehat{Q}_K\widehat{\beta}_K = \mathbb{E}_n p_i s(W_i; \widehat{\mu}_{-j(i)}, \widehat{\nu}_{-j(i)}), \qquad \widehat{Q}_K = \mathbb{E}_n p_i p_i^\top.$$

Subtracting $\widehat{Q}_K\beta_K$ gives

$$\widehat{Q}_K(\widehat{\beta}_K - \beta_K) = \mathbb{E}_n p_i \varepsilon_{Ki} + R_n, \tag{54}$$

where

$$\varepsilon_{Ki} = s(W_i; \mu_0, \nu_0) - p_i^\top \beta_K$$

and R_n is given by (52). The projection condition implies

$$\mathbb{E}[p_i \varepsilon_{Ki}] = \mathbb{E}[p_K(X_i)\{T_0(X_i) - p_K(X_i)^\top \beta_K\}] = 0_K. \tag{55}$$

The conditional moment bound, positivity, and uniform approximation bound imply

$$\sup_K \operatorname*{ess\,sup}_x \mathbb{E}[\varepsilon_K^4 \mid X = x] < \infty.$$

In particular, $\|\Omega_K\| \lesssim 1$ and $\|\mathbb{E}_n p_i \varepsilon_{Ki}\| = O_p(\sqrt{K/n})$. For the bounded basis and uniformly well-conditioned Q_K, matrix concentration gives

$$\begin{aligned} \|\widehat{Q}_K - Q_K\| &= O_p\left(\zeta_K\sqrt{\frac{\log K}{n}}\right), \\ \|\widehat{Q}_K^{-1}\| = O_p(1), \qquad \|\widehat{Q}_K^{-1} - Q_K^{-1}\| &= O_p\left(\zeta_K\sqrt{\frac{\log K}{n}}\right). \end{aligned} \tag{56}$$

These standard series matrix bounds are applied under the explicit growth restriction (19); see, for example, Belloni et al. (2015). For a deterministic evaluation vector q_K, the inverse-replacement remainder obeys

$$\begin{aligned} &|q_K^\top(\widehat{Q}_K^{-1} - Q_K^{-1})\mathbb{E}_n p_i \varepsilon_{Ki}| \\ &\qquad \leq \|q_K\| \, O_p\left(\frac{\zeta_K\sqrt{K\log K}}{n}\right) = o_p\left(\frac{\|q_K\|}{\sqrt{n}}\right). \end{aligned} \tag{57}$$

The last equality uses $K\zeta_K^2 \log K/n \to 0$. The usual second matrix-concentration term $\zeta_K^2 \log K/n$ is of smaller order under this condition.

For $q_K = p_K(x)$, the population-Gram leading term has mean zero and variance at most $C\|p_K(x)\|^2/n$, so

$$p_K(x)^\top Q_K^{-1}\mathbb{E}_n p_i \varepsilon_{Ki} = O_p\left(\frac{\|p_K(x)\|}{\sqrt{n}}\right).$$

The nuisance remainder is controlled using the empirical inverse itself:

$$\begin{aligned} |p_K(x)^\top \widehat{Q}_K^{-1} R_n| &\leq \|p_K(x)\|\|\widehat{Q}_K^{-1}\|\|R_n\| \\ &= O_p\left(\frac{\|p_K(x)\|}{\sqrt{n}}(B_n + \Lambda_n)\right) = o_p\left(\frac{\|p_K(x)\|}{\sqrt{n}}\right). \end{aligned}$$

Combining these expressions gives

$$\widehat{T}_K(x) - T_0(x) = p_K(x)^\top Q_K^{-1}\mathbb{E}_n p_i \varepsilon_{Ki} - r_K(x) + o_p\left(\frac{\|p_K(x)\|}{\sqrt{n}}\right).$$

The uniform approximation bound proves (22) and pointwise consistency. □

A.7 Proof of Theorem 1, part (b)

Under Assumptions 2–4, Lemmas 2 and 3 give $\sqrt{n}\|R_n\| = o_p(1)$. Therefore

$$\begin{aligned} \frac{|p_K(x)^\top \widehat{Q}_K^{-1} R_n|}{\sqrt{V_K(x)/n}} &\leq \frac{\|p_K(x)\|\|\widehat{Q}_K^{-1}\|}{\sqrt{V_K(x)}}\sqrt{n}\|R_n\| \\ &= o_p(1), \end{aligned}$$

using $V_K(x) \asymp \|p_K(x)\|^2$. This calculation uses the actual variance scale, including at a polynomial boundary; at an interior point it reduces to the familiar $\sqrt{K/n}$ normalization. Equation (57) controls the empirical inverse replacement at the same scale. Thus

$$\widehat{T}_K(x) - T_0(x) = p_K(x)^\top Q_K^{-1}\mathbb{E}_n p_i \varepsilon_{Ki} - r_K(x) + o_p\left(\sqrt{\frac{V_K(x)}{n}}\right).$$

Undersmoothing gives

$$|r_K(x)| \lesssim a_{T,K} = o\left(\sqrt{\frac{V_K(x)}{n}}\right),$$

so

$$\widehat{T}_K(x) - T_0(x) = p_K(x)^\top Q_K^{-1} \mathbb{E}_n p_i \varepsilon_{Ki} + o_p\left(\sqrt{\frac{V_K(x)}{n}}\right).$$

Define

$$\xi_{ni}(x) = \frac{p_K(x)^\top Q_K^{-1} p_K(X_i)\varepsilon_{Ki}}{\sqrt{V_K(x)}}. \tag{58}$$

These variables have mean zero and variance one. It remains to verify the Lindeberg condition.

Let

$$a_K = \frac{Q_K^{-1} p_K(x)}{\sqrt{V_K(x)}}.$$

The eigenvalue bounds on Q_K and $V_K(x) \asymp \|p_K(x)\|^2$ imply $\|a_K\| \lesssim 1$ and $a_K^\top Q_K a_K \lesssim 1$.

By the uniformly bounded conditional fourth moment of ε_K,

$$\begin{aligned}
\mathbb{E}[\xi_{n1}(x)^4] &\lesssim \mathbb{E}[(a_K^\top p_K(X))^4] \\
&\leq \|a_K\|^2 \zeta_K^2 \mathbb{E}[(a_K^\top p_K(X))^2] \\
&= \|a_K\|^2 \zeta_K^2 a_K^\top Q_K a_K \\
&\lesssim \zeta_K^2.
\end{aligned} \tag{59}$$

Consequently, for every $\epsilon > 0$,

$$\begin{aligned}
&\mathbb{E}\left[\xi_{n1}(x)^2 \mathbf{1}\{|\xi_{n1}(x)| > \epsilon\sqrt{n}\}\right] \\
&\quad \leq \frac{\mathbb{E}[\xi_{n1}(x)^4]}{\epsilon^2 n} \lesssim \frac{\zeta_K^2}{\epsilon^2 n} \to 0,
\end{aligned}$$

where the final convergence follows from (19).

The triangular-array central limit theorem therefore gives

$$\frac{1}{\sqrt{n}} \sum_{i=1}^n \xi_{ni}(x) \Rightarrow N(0,1),$$

which proves (24).

Feasible covariance. To verify studentization, rather than infer it from the central limit theorem, define the oracle empirical covariance

$$\widetilde{\Omega}_K = \mathbb{E}_n p_i p_i^\top \varepsilon_{Ki}^2.$$

Let $\|\cdot\|_F$ denote the Frobenius norm. Independence, the uniform conditional fourth moment of ε_K, and $\mathbb{E}\|p_K(X)\|^2 \asymp K$ give

$$\begin{aligned}
\mathbb{E}\|\widetilde{\Omega}_K - \Omega_K\|_F^2 &\leq \frac{1}{n}\mathbb{E}[\|p_K(X)\|^4 \varepsilon_K^4] \\
&\leq \frac{C\zeta_K^2}{n}\mathbb{E}\|p_K(X)\|^2 \lesssim \frac{K\zeta_K^2}{n} \to 0.
\end{aligned} \tag{60}$$

Hence $\|\widetilde{\Omega}_K - \Omega_K\| = o_p(1)$ and $\|\widetilde{\Omega}_K\| = O_p(1)$. By (54), $\|\widehat{\beta}_K - \beta_K\| = O_p(\sqrt{K/n})$. Write

$$d_i = \widehat{\varepsilon}_{Ki} - \varepsilon_{Ki} = \Delta_{j(i)}(W_i) - p_i^\top(\widehat{\beta}_K - \beta_K), \qquad H_n = \mathbb{E}_n p_i p_i^\top d_i^2.$$

The matrix H_n is positive semidefinite and

$$\begin{aligned}\|H_n\| &\le 2\mathbb{E}_n\|p_i\|^2\Delta_{j(i)}(W_i)^2 \\ &\quad + 2\zeta_K^2\|\widehat{\beta}_K - \beta_K\|^2\|\widehat{Q}_K\| = o_p(1).\end{aligned}$$

For the first term, apply conditional Markov's inequality within each fold using $\Lambda_{n,j}^2 = o_p(1)$, and then sum over the fixed number of folds. The second term is $O_p(\zeta_K^2 K/n) = o_p(1)$. Matrix Cauchy–Schwarz now gives

$$\|\widehat{\Omega}_K - \widetilde{\Omega}_K\| \le \|H_n\| + 2\sqrt{\|H_n\|\,\|\widetilde{\Omega}_K\|} = o_p(1). \tag{61}$$

Together with (56) and (60), this proves

$$\left\|\widehat{Q}_K^{-1}\widehat{\Omega}_K\widehat{Q}_K^{-1} - Q_K^{-1}\Omega_K Q_K^{-1}\right\| = o_p(1). \tag{62}$$

Since $\|p_K(x)\|^2/V_K(x) \lesssim 1$, $\widehat{V}_K(x)/V_K(x) \to_p 1$. Slutsky's theorem proves (25). □

A.8 Partial-mean lemma and proof of Corollary 2

Fix a reported a satisfying the conditions of Corollary 2. If the reference distribution is empirical and constructed from an independent reference sample, condition on that sample throughout, treating its realized distribution as fixed. Write $\bar{p} = \bar{p}_K(a)$, $\bar{r} = \bar{r}_K(a)$, $\bar{V} = \bar{V}_K(a)$, and $p_i = p_K(X_i)$. The variance-order condition in the corollary gives

$$c\|\bar{p}\|^2 \le \bar{V} \le C\|\bar{p}\|^2.$$

Because the sieve span contains constants, there is a vector c_K satisfying $c_K^\top p_K(x) = 1$, $c_K^\top Q_K c_K = 1$, and hence $\|c_K\| \asymp 1$. Thus $1 = c_K^\top\bar{p} \le \|c_K\|\|\bar{p}\|$, so $\|\bar{p}\| \gtrsim 1$. Also $\|\bar{p}\| \le \zeta_K$, giving $1 \lesssim \bar{V} \lesssim \zeta_K^2$. A sharper profile-specific bound can hold, but is not needed for this argument.

Lemma 4 (Linear representation of the partial-mean profile)**.** *Under the conditions of Corollary 2,*

$$\widehat{\theta}_K(a) - \theta_0(a) = \bar{p}_K(a)^\top Q_K^{-1}\mathbb{E}_n p_i\varepsilon_{Ki} - \bar{r}_K(a) + o_p\left(\sqrt{\frac{\bar{V}_K(a)}{n}}\right),$$

where the leading term has mean zero and variance exactly $\bar{V}_K(a)/n$.

Proof. The support restriction permits integration of the sieve approximation over $F_Z^\star$, giving

$\bar{p}^\top \beta_K = \theta_0(a) - \bar{r}$. Applying the normal-equation identity (54),

$$\begin{aligned}\widehat{\theta}_K(a) - \theta_0(a) = \bar{p}^\top Q_K^{-1} \mathbb{E}_n p_i \varepsilon_{Ki} - \bar{r} \\ + \bar{p}^\top (\widehat{Q}_K^{-1} - Q_K^{-1}) \mathbb{E}_n p_i \varepsilon_{Ki} + \bar{p}^\top \widehat{Q}_K^{-1} R_n.\end{aligned}$$

The leading term has the stated mean and variance by (55). Equation (57), now with $q_K = \bar{p}$, gives an inverse-replacement error

$$o_p(\|\bar{p}\|/\sqrt{n}) = o_p\left(\sqrt{\bar{V}/n}\right).$$

For the nuisance term, use $\sqrt{n}\|R_n\| = o_p(1)$ from Lemma 3:

$$\frac{|\bar{p}^\top \widehat{Q}_K^{-1} R_n|}{\sqrt{\bar{V}/n}} \leq \frac{\|\bar{p}\| \|\widehat{Q}_K^{-1}\|}{\sqrt{\bar{V}}} \sqrt{n}\|R_n\| = o_p(1).$$

No additional basis factor is introduced by averaging once the direction is normalized by its own variance. Combining these bounds proves the lemma. □

Proof of Corollary 2. The leading variance is at most $C\zeta_K^2/n \to 0$, and $|\bar{r}| \leq \|r_K\|_\infty \lesssim a_{T,K} \to 0$, so the lemma first gives consistency. For normality, the target-specific undersmoothing requirement implies

$$\frac{\sqrt{n}\,|\bar{r}_K(a)|}{\sqrt{\bar{V}_K(a)}} \lesssim \sqrt{\frac{n}{\bar{V}_K(a)}} a_{T,K} \to 0.$$

To verify Lindeberg, define

$$\bar{\xi}_{ni}(a) = \frac{\bar{p}_K(a)^\top Q_K^{-1} p_K(X_i) \varepsilon_{Ki}}{\sqrt{\bar{V}_K(a)}}, \qquad b_K = \frac{Q_K^{-1} \bar{p}_K(a)}{\sqrt{\bar{V}_K(a)}}.$$

The eigenvalue and variance bounds imply $\|b_K\| \lesssim 1$ and $b_K^\top Q_K b_K \lesssim 1$. By the bounded conditional fourth moment of ε_K,

$$\begin{aligned}\mathbb{E}[\bar{\xi}_{n1}(a)^4] &\lesssim \mathbb{E}[(b_K^\top p_K(X))^4] \\ &\leq \|b_K\|^2 \zeta_K^2 \mathbb{E}[(b_K^\top p_K(X))^2] \lesssim \zeta_K^2.\end{aligned}$$

For every $\epsilon > 0$,

$$\mathbb{E}[\bar{\xi}_{n1}(a)^2 \mathbf{1}\{|\bar{\xi}_{n1}(a)| > \epsilon\sqrt{n}\}] \leq \frac{\mathbb{E}[\bar{\xi}_{n1}(a)^4]}{\epsilon^2 n} \lesssim \frac{\zeta_K^2}{\epsilon^2 n} \to 0$$

by (19). The triangular-array CLT therefore applies to the leading term. Finally, (62) and $\|\bar{p}\|^2/\bar{V} \lesssim 1$ imply $\widehat{\bar{V}}_K(a)/\bar{V}_K(a) \to_p 1$, proving the studentized result. □

A.9 Proof of Corollary 3

Proof. By (27), the true orthogonal signal satisfies

$$\mathbb{E}[s(W; \mu_0, \nu_0) \mid A] = \tau_0(A).$$

The first-stage nuisance functions are conditional on the full vector X, whereas the reporting basis q_L depends only on A. Write $\varepsilon_{Li} = s(W_i; \mu_0, \nu_0) - q_L(A_i)^\top \gamma_L$ and $\Delta_{j(i)}(W_i) = \widehat{T}_i - s(W_i; \mu_0, \nu_0)$. The second-stage normal equations yield

$$\widehat{\gamma}_L - \gamma_L = \widehat{Q}_{A,L}^{-1} \left\{ \mathbb{E}_n[q_L(A_i)\varepsilon_{Li}] + \mathbb{E}_n[q_L(A_i)\Delta_{j(i)}(W_i)] \right\}.$$

The projection identity implies $\mathbb{E}[q_L(A)\varepsilon_L] = 0_L$.

Because A is a subvector of X, the conditional-bias and signal-stability bounds in the proof of Lemma 1 apply with $q_L(A)$ in place of $p_K(X)$. Uniformly over folds, the two resulting bounds are, up to constants,

$$\sqrt{n}\, \|\mathbb{E}[q_L(A)\Delta_j(W) \mid \mathcal{D}_{-j}]\| \lesssim \sqrt{n}\, \zeta_{A,L}(r_{\mu,n}^2 + r_{\mu,n} r_{\nu,n}),$$
$$\mathbb{E}\left[\|q_L(A)\|^2 \Delta_j(W)^2 \mid \mathcal{D}_{-j} \right]^{1/2} \lesssim \zeta_{A,L}(r_{\mu,n} + r_{\nu,n}).$$

Both quantities are $o_p(1)$ under the stated reporting-basis first-stage rate conditions. The foldwise remainder argument therefore gives

$$\sqrt{n} \left\| \mathbb{E}_n[q_L(A_i)\Delta_{j(i)}(W_i)] \right\| = o_p(1).$$

Thus the feasible reporting coefficients are oracle-equivalent.

The conditional moment assumptions for the true signal and the uniform approximation bound for τ_0 imply a bounded conditional fourth moment for the oracle regression error given A. The Gram-matrix, projection, consistency, and feasible-variance arguments of Theorem 1 now apply with (A, τ_0, q_L, L) in place of (X, T_0, p_K, K). The variance condition and the requirement that approximation bias be negligible relative to $\sqrt{V_L(a)/n}$ give the pointwise normal and studentized limits, including (28). Throughout, μ_0 and ν_0 remain conditional moments given X, not moments conditional only on A. □

A.10 Fixed-basis inference for conditional-average profiles

Proposition 5 (Inference for a fixed reporting basis)**.** *Suppose Assumption 2 holds and the reporting basis $q(A) \in \mathbb{R}^L$ has fixed dimension L, is bounded, and has nonsingular population Gram matrix $Q_q = \mathbb{E}[q(A)q(A)^\top]$. Define the population projection coefficient*

$$\gamma_q = Q_q^{-1}\mathbb{E}[q(A)\tau_0(A)], \qquad \varepsilon_q = s(W; \mu_0, \nu_0) - q(A)^\top \gamma_q,$$

and let $\Omega_q = \mathbb{E}[q(A)q(A)^\top \varepsilon_q^2]$ and $V_q(a) = q(a)^\top Q_q^{-1}\Omega_q Q_q^{-1} q(a)$. Suppose the cross-fitted nuisance errors satisfy

$$\sqrt{n}\,(r_{\mu,n}^2 + r_{\mu,n} r_{\nu,n}) = o_p(1), \qquad r_{\mu,n} + r_{\nu,n} = o_p(1).$$

Let $\widehat{Q}_q = \mathbb{E}_n[q(A_i)q(A_i)^\top]$ and $\widehat{\gamma}_q = \widehat{Q}_q^{-1}\mathbb{E}_n[q(A_i)\widehat{T}_i]$. Then

$$\sqrt{n}(\widehat{\gamma}_q - \gamma_q) = \frac{1}{\sqrt{n}} \sum_{i=1}^n Q_q^{-1} q(A_i)\varepsilon_{q,i} + o_p(1) \Rightarrow N(0, Q_q^{-1}\Omega_q Q_q^{-1}). \tag{63}$$

Let $\widehat{\varepsilon}_{q,i} = \widehat{T}_i - q(A_i)^\top \widehat{\gamma}_q$, $\widehat{\Omega}_q = \mathbb{E}_n[q(A_i)q(A_i)^\top \widehat{\varepsilon}^2_{q,i}]$, *and* $\widehat{V}_q(a) = q(a)^\top \widehat{Q}_q^{-1}\widehat{\Omega}_q\widehat{Q}_q^{-1}q(a)$. *Then* $\widehat{V}_q(a) \to_p V_q(a)$ *and, if* $V_q(a) > 0$,

$$\frac{q(a)^\top(\widehat{\gamma}_q - \gamma_q)}{\sqrt{\widehat{V}_q(a)/n}} \Rightarrow N(0,1). \tag{64}$$

The inferential target is the finite-basis projection $q(a)^\top\gamma_q$, *not necessarily* $\tau_0(a)$. *No series undersmoothing condition is imposed.*

Proof. Write $q_i = q(A_i)$, $\bar{q} = \sup_a \|q(a)\| < \infty$, $s_i = s(W_i; \mu_0, \nu_0)$, and $\widehat{T}_i = s_i + \Delta_{j(i)}(W_i)$, with Δ_j as in (46). The normal equations give

$$\widehat{\gamma}_q - \gamma_q = \widehat{Q}_q^{-1}\left\{\mathbb{E}_n[q_i\varepsilon_{q,i}] + R_n^q\right\}, \qquad R_n^q = \mathbb{E}_n[q_i\Delta_{j(i)}(W_i)]. \tag{65}$$

Oracle term. Because q is bounded and the observations are i.i.d., $\widehat{Q}_q \to_p Q_q$ and $\widehat{Q}_q^{-1} = Q_q^{-1} + o_p(1)$. The identity (27) gives $\mathbb{E}[q(A)s(W;\mu_0,\nu_0)] = Q_q\gamma_q$, and hence $\mathbb{E}[q(A)\varepsilon_q] = 0_L$. The assumed conditional fourth moments and bounded q give $\mathbb{E}[\|q(A)\varepsilon_q\|^2] < \infty$. The multivariate central limit theorem yields

$$n^{-1/2}\sum_{i=1}^n q_i\varepsilon_{q,i} \Rightarrow N(0,\Omega_q).$$

Nuisance remainder. The conditional-bias and signal-stability bounds from Lemma 2 apply with p_K replaced by q and ζ_K by $\bar{q}$. Uniformly over folds,

$$\begin{aligned}\sqrt{n}\,\|\mathbb{E}[q(A)\Delta_j(W) \mid \mathcal{D}_{-j}]\| &\lesssim \bar{q}\sqrt{n}(r_{\mu,n}^2 + r_{\mu,n}r_{\nu,n}),\\ \mathbb{E}[\|q(A)\|^2\Delta_j(W)^2 \mid \mathcal{D}_{-j}]^{1/2} &\lesssim \bar{q}(r_{\mu,n} + r_{\nu,n}).\end{aligned}$$

The fixed-basis nuisance-rate conditions make both bounds $o_p(1)$. The foldwise argument therefore gives $\sqrt{n}\|R_n^q\| = o_p(1)$. Combining this with the oracle term and Slutsky's lemma establishes (63).

Feasible variance. Define $\widehat{\varepsilon}_{q,i} = \widehat{T}_i - q_i^\top\widehat{\gamma}_q$ and $d_i = \widehat{\varepsilon}_{q,i} - \varepsilon_{q,i}$. Then

$$d_i = \Delta_{j(i)}(W_i) - q_i^\top(\widehat{\gamma}_q - \gamma_q).$$

The oracle covariance $\widetilde{\Omega}_q = \mathbb{E}_n[q_iq_i^\top\varepsilon_{q,i}^2]$ converges to Ω_q by the law of large numbers. Moreover,

$$\begin{aligned}\|\mathbb{E}_n[q_iq_i^\top d_i^2]\| &\le 2\bar{q}^2\mathbb{E}_n[\Delta_{j(i)}(W_i)^2] + 2\bar{q}^4\|\widehat{\gamma}_q - \gamma_q\|^2\\ &= o_p(1).\end{aligned}$$

The first term follows from the foldwise signal-stability bound and conditional Markov's inequality; the second follows from (63). The same matrix Cauchy–Schwarz argument as in (61) gives $\widehat{\Omega}_q \to_p \Omega_q$ and hence $\widehat{V}_q(a) \to_p V_q(a)$. Slutsky's lemma establishes (64). No undersmoothing condition is needed because the target is the fixed-dimensional population projection. □

A.11 Conditional inequality treatment effects

Remark 1 (Conditional inequality treatment effects). The construction also suggests an extension to causal comparisons. Let $D \in \{0,1\}$ denote treatment, with positive potential outcomes $Y(0), Y(1)$, and let $Y = Y(D)$ be observed. Write $T_0^{(d)}(x)$ for the Theil index of $Y(d)$ conditional on $X = x$ and $\Delta_0(x) = T_0^{(1)}(x) - T_0^{(0)}(x)$. Assume $\mathbb{E}[Y(d) \mid X, D] = \mathbb{E}[Y(d) \mid X]$ and $\mathbb{E}[Y(d)\log Y(d) \mid X, D] = \mathbb{E}[Y(d)\log Y(d) \mid X]$, along with positivity of $p_d(x) = \Pr(D = d \mid X = x)$ and suitable moment and conditional-mean bounds. Define $\mu_d(x) = \mathbb{E}[Y \mid X = x, D = d]$, $\nu_d(x) = \mathbb{E}[Y \log Y \mid X = x, D = d]$, and $m_d(x) = \nu_d(x)/\mu_d(x) - \log \mu_d(x)$. Let $s_d(W; \mu_d, \nu_d)$ be the scalar signal (14) formed using arm-d conditional moments. Consider

$$\psi_d(W) = m_d(X) + \frac{\mathbf{1}\{D = d\}}{p_d(X)}\{s_d(W; \mu_d, \nu_d) - m_d(X)\}.$$

At the true nuisance functions, $\mathbb{E}[\psi_d(W) \mid X] = T_0^{(d)}(X)$, and the conditional expectation is Neyman-orthogonal with respect to the arm-specific conditional moments and propensity score. Regressing cross-fitted $\psi_1 - \psi_0$ on a reporting basis $q_L(A)$ can estimate the conditional-average profile $\mathbb{E}[\Delta_0(X) \mid A = a]$ under corresponding basis, variance, approximation, positivity, and nuisance-rate conditions. In addition to the quadratic and product errors in the arm-specific moments, the first-stage bias contains products of propensity error with conditional-moment error; signal stability also requires control of propensity-estimation error. Separate unweighted regressions within treatment arms instead average over the arm-specific distributions of the remaining characteristics.

A.12 Proof of Proposition 4

Hold the unconditional mean μ fixed when differentiating with respect to the candidate conditional mean m. Let $m_t = \mu_0 + th$, where h is bounded and m_t remains positive for sufficiently small $|t|$.

Within component. At the truth, $\mathbb{E}[g_W(W; \mu_0, \mu)] = 0$ by (29). Differentiation under the expectation gives

$$\frac{d}{dt}\mathbb{E}[g_W(W; m_t, \mu)]\Big|_{t=0} = -\frac{1}{\mu}\mathbb{E}\left[\frac{Y}{\mu_0(X)}h(X)\right] = -\frac{\mathbb{E}[h(X)]}{\mu},$$

where the last equality uses $\mathbb{E}[Y \mid X] = \mu_0(X)$. For $\varphi_W(W; m) = -\{Y - m(X)\}/\mu$, the derivative of its expectation is $\mathbb{E}[h(X)]/\mu$, which cancels the derivative of the plug-in moment. Since $\mathbb{E}[\varphi_W(W; \mu_0)] = 0$, the corrected within moment has mean zero and is Neyman-orthogonal at the truth.

Between component. By (30), $\mathbb{E}[g_B(W; \mu_0, \mu)] = 0$. The derivative of the plug-in moment is

$$\frac{d}{dt}\mathbb{E}[g_B(W; m_t, \mu)]\Big|_{t=0} = \frac{1}{\mu}\mathbb{E}\left[\left\{1 + \log\left(\frac{\mu_0(X)}{\mu}\right)\right\}h(X)\right].$$

The correction

$$\varphi_B(W; m) = \frac{Y - m(X)}{\mu}\left\{1 + \log\left(\frac{m(X)}{\mu}\right)\right\}$$

has mean zero at $m = \mu_0$, and its derivative in expectation is

$$\frac{1}{\mu}\mathbb{E}\left[\frac{Y-\mu_0(X)}{\mu_0(X)}h(X) - \left\{1+\log\left(\frac{\mu_0(X)}{\mu}\right)\right\}h(X)\right] \\ = -\frac{1}{\mu}\mathbb{E}\left[\left\{1+\log\left(\frac{\mu_0(X)}{\mu}\right)\right\}h(X)\right],$$

where the conditional mean of $Y - \mu_0(X)$ is zero. This cancels the derivative of g_B. Expanding $g_c + \varphi_c$ yields exactly the two moments in (32), establishing the mean-zero and orthogonality claims of Proposition 4. □

A.13 Proof of Theorem 2: within component

Notation for Appendices A.13–A.15. For a positive function m on $\mathcal{X}$, define the numerator signals

$$\varrho_W(W;m) = Y\log\left(\frac{Y}{m(X)}\right) - \{Y - m(X)\}, \qquad \varrho_B(W;m) = Y\log m(X) + \{Y - m(X)\},$$

which add to $Y\log Y$ for every m. Their means at the truth are $\theta_W = \mathbb{E}[\varrho_W(W;\mu_0)] = \mu T_W$ and $\theta_B = \mathbb{E}[\varrho_B(W;\mu_0)] = \mathbb{E}[\mu_0(X)\log\mu_0(X)] = \mu(T_B + \log\mu)$, by (29), (30), and $\mathbb{E}[Y - \mu_0(X) \mid X] = 0$. The locally robust moments of Proposition 4 are the normalized signals, $\psi_W(W;\mu_0,\mu) = \varrho_W(W;\mu_0)/\mu - T_W$ and $\psi_B(W;\mu_0,\mu) = \{\varrho_B(W;\mu_0) - Y\log\mu\}/\mu - T_B$, so the signals inherit their orthogonality: for $\mu_t = \mu_0 + th$ with bounded h, $\partial_t\mathbb{E}[\varrho_c(W;\mu_t)]|_{t=0} = 0$, $c \in \{W, B\}$. With $\widehat{\theta}_c = \mathbb{E}_n\varrho_c(W_i;\widehat{\mu})$, the estimators (33) and (34) are $\widehat{T}_W^{LR} = \widehat{\theta}_W/\bar{Y}$ and $\widehat{T}_B^{LR} = \widehat{\theta}_B/\bar{Y} - \log\bar{Y}$. The influence functions χ_W, χ_B, and χ_Δ are all defined in (37), and $\widehat{\chi}_{W,i}$, $\widehat{\chi}_{B,i}$, $\widehat{\chi}_{\Delta,i}$ are the estimated versions defined in the statements.

Oracle equivalence. Fix a fold j and write $a = \mu_0(X)$ and $m = \widehat{\mu}_{-j}(X)$. Conditional on X and the training data $\mathcal{D}_{-j}$, the bias of the fitted signal is

$$\mathbb{E}[\varrho_W(W;m) - \varrho_W(W;a) \mid X, \mathcal{D}_{-j}] = a\log\left(\frac{a}{m}\right) - a + m, \tag{66}$$

by $\mathbb{E}[Y \mid X] = a$. The right-hand side is the generalized Kullback–Leibler divergence between the positive scalars a and m: it is nonnegative, vanishes only at $m = a$, and a second-order expansion in m around a bounds it by $a(m-a)^2/(2\xi^2)$ for some ξ between a and m. Under Assumption 2(ii)–(iii), $a \le C_\mu$ and $\xi \ge \underline{\mu}$, so the bias lies between 0 and $C(m-a)^2$ with $C = C_\mu/(2\underline{\mu}^2)$. The same bounds give $|\varrho_W(W;m) - \varrho_W(W;a)| \le (1 + Y/\underline{\mu})|m-a|$, and with the conditional moment bound (18), $\mathbb{E}[\{\varrho_W(W;m) - \varrho_W(W;a)\}^2 \mid \mathcal{D}_{-j}] \le C\|m-a\|_2^2$. Averaged over fold j, the bias is therefore at most $Cr_{\mu,n}^2 = o_p(n^{-1/2})$ by (38), and the centered part has conditional variance at most $C\|m-a\|_2^2/n_j = o_p(n^{-1})$. Conditional Markov's inequality in each of the J folds, the scalar case of Lemma 3 with multiplier one, gives

$$\widehat{\theta}_W = \mathbb{E}_n\varrho_W(W_i;\mu_0) + o_p(n^{-1/2}). \tag{67}$$

Consistency and asymptotic linearity. The law of large numbers and (67) give $\widehat{\theta}_W \to_p \theta_W$,

and $\bar{Y} \to_p \mu$, so $\widehat{T}_W^{LR} \to_p \theta_W/\mu = T_W$. The delta method applied to $(\widehat{\theta}_W, \bar{Y}) \mapsto \widehat{\theta}_W/\bar{Y}$ gives

$$\sqrt{n}\{\widehat{T}_W^{LR} - T_W\} = \frac{1}{\sqrt{n}} \sum_{i=1}^{n} \left[\frac{\varrho_W(W_i; \mu_0) - \theta_W}{\mu} - \frac{\theta_W}{\mu^2}(Y_i - \mu) \right] + o_p(1),$$

and the summand equals $\chi_W(W_i; X)$ in (37) after substituting $\theta_W = \mu T_W$. The second moment of χ_W is finite by (18), so the univariate central limit theorem gives the normal limit for the leading term in (39) when $\sigma_W^2 > 0$. The studentized limit (40) follows after variance consistency and Slutsky's theorem, as shown below.

Feasible variance. The signal-difference bound and $r_{\mu,n} = o_p(1)$ give

$$\mathbb{E}_n\{\varrho_W(W_i; \widehat{\mu}) - \varrho_W(W_i; \mu_0)\}^2 = o_p(1).$$

Together with $\bar{Y} \to_p \mu$, $\widehat{T}_W^{LR} \to_p T_W$, and the finite second moments of Y and $\varrho_W(W; \mu_0)$, this implies $\mathbb{E}_n(\widehat{\chi}_{W,i} - \chi_W(W_i; X))^2 = o_p(1)$. Cauchy–Schwarz and the law of large numbers then give $\widehat{\sigma}_W^2 \to_p \sigma_W^2$, whether or not the limit is zero. Slutsky's theorem justifies studentization when the limit is positive. □

A.14 Proof of Theorem 2: between component

For any positive m and a,

$$\varrho_B(W; m) - \varrho_B(W; a) = Y \log\left(\frac{m}{a}\right) - (m - a) = -\{\varrho_W(W; m) - \varrho_W(W; a)\},$$

because the two signals add to $Y \log Y$. The conditional bias of the fitted between signal is therefore the negative of (66) and lies between $-C(m-a)^2$ and 0, and the mean-square signal bound of Appendix A.13 applies unchanged. The foldwise argument and (38) give

$$\widehat{\theta}_B = \mathbb{E}_n \varrho_B(W_i; \mu_0) + o_p(n^{-1/2}). \tag{68}$$

The remainders in (67) and (68) are equal and opposite in every sample, consistent with (35).

Consistency follows from $\widehat{\theta}_B \to_p \theta_B$ and $\bar{Y} \to_p \mu$, since $\theta_B/\mu - \log\mu = T_B$. The delta method applied to $(\widehat{\theta}_B, \bar{Y}) \mapsto \widehat{\theta}_B/\bar{Y} - \log\bar{Y}$ gives

$$\sqrt{n}\{\widehat{T}_B^{LR} - T_B\} = \frac{1}{\sqrt{n}} \sum_{i=1}^{n} \left[\frac{\varrho_B(W_i; \mu_0) - \theta_B}{\mu} - \left(\frac{\theta_B}{\mu^2} + \frac{1}{\mu}\right)(Y_i - \mu) \right] + o_p(1),$$

and the summand equals $\chi_B(W_i; X)$ in (37) after substituting $\theta_B = \mu(T_B + \log\mu)$; the term $1/\mu$ in the bracket is the contribution of the normalization $-\log\bar{Y}$. The feasible-variance argument of Appendix A.13 applies with $\widehat{\theta}_B$, $\widehat{T}_B^{LR}$, and ϱ_B in place of their within counterparts and gives $\mathbb{E}_n(\widehat{\chi}_{B,i} - \chi_B(W_i; X))^2 = o_p(1)$, hence $\widehat{\sigma}_B^2 \to_p \sigma_B^2$. The central limit theorem and Slutsky's theorem give the normal and studentized limits when $\sigma_B^2 > 0$. No basis-dependent condition enters the argument. □

A.15 Proof of Corollary 4

Apply Theorem 2 to the two conditioning sets on the same sample with the same fold partition. Each representation (39) has an $o_p(1)$ remainder after root-n scaling, so their difference does too, which gives (41) with influence function $\chi_B(W; X^{(b)}) - \chi_B(W; X^{(a)})$. Writing $\mu_a = \mu_a(X^{(a)})$ and $\mu_b = \mu_b(X^{(b)})$ and substituting $T_B^{(b)} - T_B^{(a)} = \Delta T_B$, the terms in $\log \mu$ and in μ/μ cancel, leaving the expression for χ_Δ in (37). Consistency of $\widehat{\Delta T}_B^{LR}$ follows from the consistency of the two between estimators.

The moment bounds give a finite variance,

$$\sigma_\Delta^2 = \mathrm{Var}\{\chi_B(W; X^{(b)})\} + \mathrm{Var}\{\chi_B(W; X^{(a)})\} - 2\,\mathrm{Cov}\{\chi_B(W; X^{(b)}), \chi_B(W; X^{(a)})\}.$$

The mean-square consistency of each estimated influence function, established in Appendix A.14, implies that of their difference $\widehat{\chi}_{\Delta,i}$. Cauchy–Schwarz and the law of large numbers give $\widehat{\sigma}_\Delta^2 \to_p \sigma_\Delta^2$, and the central limit theorem with Slutsky's theorem gives (42) when $\sigma_\Delta^2 > 0$. The argument uses the joint asymptotic linearity of the two estimators on a common sample, not their independence; adding the two separate variances would drop the covariance term.

At $\Delta T_B = 0$, Corollary 1 gives $\mu_a = \mu_b$ almost surely, so χ_Δ is identically zero and the root-n representation has a degenerate limit. The corollary therefore says nothing about Wald inference at or near this null. □

B Extension to the mean log deviation

This appendix extends the framework to the mean log deviation (MLD, or GE(0)), using the general conditioning vector X and the cross-fitting and series notation from Section 3. It collects the conditional target, exact decomposition, orthogonal signal, index-specific conditions, estimators, and proof details. The simulation in Section 4 reports no MLD results.

B.1 Conditional target and exact decomposition

Define the conditional log-wage moment

$$\eta_0(x) = \mathbb{E}[\log Y \mid X = x].$$

The conditional GE(0), or mean log deviation (MLD), surface is

$$L_0(x) = \log \mu_0(x) - \eta_0(x) = \log \mu_0(x) - \mathbb{E}[\log Y \mid X = x].$$

For the same conditioning set X, the exact decomposition of the unconditional MLD $L = \log \mu - \mathbb{E}[\log Y]$ is

$$L = L_W(X) + L_B(X),$$

where

$$L_W(X) = \mathbb{E}[L_0(X)], \qquad L_B(X) = \log \mu - \mathbb{E}[\log \mu_0(X)]. \tag{69}$$

Equivalently,

$$L_W(X) = \mathbb{E}[\log \mu_0(X) - \log Y], \qquad L_B(X) = \log \mu - \mathbb{E}[\log \mu_0(X)].$$

The MLD within component averages the conditional surface with population shares rather than the wage-share weights in (1). The between component likewise uses population-share weighting of log conditional mean wages. This provides a complementary inequality specification with greater lower-tail sensitivity, not a requirement that it reproduce the Theil results.

Annual MLD objects are defined by evaluating the same identities under $\mathbb{E}_t$:

$$\begin{aligned} L_t(s) &= \log \mu_t(s) - \mathbb{E}_t[\log Y_t \mid S_t = s], \\ L_t &= \log \mu_t - \mathbb{E}_t[\log Y_t] = L_{W,t}(S_t) + L_{B,t}(S_t), \\ L_{W,t}(S_t) &= \mathbb{E}_t[L_t(S_t)], \qquad L_{B,t}(S_t) = \log \mu_t - \mathbb{E}_t[\log \mu_t(S_t)]. \end{aligned}$$

The empirical conditioning sets, annual sample rules, and time-indexing conventions are those of Section 5.

B.2 Series representation and locally robust signal

For GE(0), I use the same basis and write

$$L_0(x) = p_K(x)^\top \gamma_K + \ell_K(x), \qquad \gamma_K \in \mathbb{R}^K, \quad \ell_K : \mathcal{X} \to \mathbb{R},$$

where

$$\mathbb{E}\left[p_K(X)\{L_0(X) - p_K(X)^\top \gamma_K\}\right] = 0_K, \qquad \gamma_K = Q_K^{-1}\mathbb{E}[p_K(X)L_0(X)]. \tag{70}$$

For GE(0)/MLD, the non-robust conditional moment is

$$m_K^L(W;\gamma,\mu) = p_K(X)\left[\log\mu(X) - \log Y - p_K(X)^\top\gamma\right].$$

At (γ_K, μ_0), this moment has mean zero because

$$\mathbb{E}[m_K^L(W;\gamma_K,\mu_0)] = \mathbb{E}\left[p_K(X)\{L_0(X) - p_K(X)^\top\gamma_K\}\right] = 0_K.$$

Proposition 6 (Conditional MLD signal identity and orthogonality)**.** *Suppose μ_0 is bounded away from zero, $\mathbb{E}[Y^2 + |\log Y|^2] < \infty$, and p_K is bounded with nonsingular Q_K. The correction and resulting moment are*

$$\phi_K^L(W;\mu) = \frac{p_K(X)}{\mu(X)}\{Y - \mu(X)\},$$

$$\psi_K^L(W;\gamma,\mu) = p_K(X)\left[\log\mu(X) - \log Y + \frac{Y-\mu(X)}{\mu(X)} - p_K(X)^\top\gamma\right].$$

The scalar signal $s_L(W;\mu) = \log\mu(X) - \log Y + (Y-\mu(X))/\mu(X)$ has conditional expectation $L_0(X)$ at μ_0, and its conditional-mean derivative at μ_0 is zero in every bounded direction preserving positivity. Consequently $\mathbb{E}[\psi_K^L(W;\gamma_K,\mu_0)] = 0_K$, and the derivative of this population moment in the nuisance is zero. The proof is in Appendix B.7.

B.3 MLD-specific regularity and weighted nuisance verification

Use Assumption 2 with its outcome-moment restriction replaced by the MLD bound below. For the second-stage MLD regression, retain the design, basis, and identification parts of Assumption 3 and the growth conditions of Assumption 4(i), with the MLD target substitutions stated next. Require

$$\operatorname*{ess\,sup}_{x\in\mathcal{X}} \mathbb{E}[Y^4 + |\log Y|^4 \mid X = x] \le M. \tag{71}$$

In place of the Theil smoothness and approximation requirements in Assumption 3, impose $L_0 \in \mathcal{H}^{s_L}(\mathcal{X})$ and $\sup_x |\ell_K(x)| \lesssim a_{L,K}$, where $a_{L,K} = L_K K^{-\bar{s}_L/d_x}$, with $\bar{s}_L = s_L$ for Legendre and $\bar{s}_L = \min\{s_L, s_0\}$ for splines/wavelets. Require $a_{L,K} \to 0$. The MLD signal requires no estimate of η_0 or ν_0: it uses $\log Y$ directly and only the conditional-mean nuisance. For an MLD-only analysis, replace the Theil nuisance rates by

$$\sqrt{n}\,\zeta_K r_{\mu,n}^2 = o_p(1), \qquad \zeta_K r_{\mu,n} = o_p(1). \tag{72}$$

The common Theil rates in (20) imply these conditions when the indices are estimated together. The scalar MLD result below requires only (38), which the first restriction implies; the second-stage conditions are not needed for standalone scalar estimation.

Write

$$s_L(W;\mu) = \log\mu(X) - \log Y + \frac{Y-\mu(X)}{\mu(X)},$$

and define $\Delta_j^L, m_{K,j}^L, B_n^L, \Lambda_{n,j}^L, \Lambda_n^L$ by replacing s with s_L in (46)–(47). These definitions condition on each fold's training sample and integrate over a fresh observation, exactly as for Theil.

Lemma 5 (Weighted nuisance bounds for MLD)**.** *Under the stabilization and moment conditions just stated,*

$$B_n^L \le C\sqrt{n}\,\zeta_K r_{\mu,n}^2, \qquad \Lambda_n^L \le C\zeta_K r_{\mu,n}.$$

Consequently, (72) *implies* $B_n^L + \Lambda_n^L = o_p(1)$ *for either basis family.*

Proof. Fix a training fold and write $a = \mu_0(x)$, $m = \widehat{\mu}_{-j}(x)$, and $d = m - a$. The exact conditional signal bias is

$$\mathbb{E}[s_L(W;m) - s_L(W;a) \mid X = x, \mathcal{D}_{-j}] = \log(m/a) - d/m = d^2 \int_0^1 \frac{t}{(a+td)^2}\,dt.$$

The fixed positive lower bound therefore gives a bound Cd^2. Also, $|s_L(W;m) - s_L(W;a)| \le C(1+Y)|d|$. Multiplying the bias by the basis and applying the conditional second-moment bound to the signal difference proves the two inequalities, as in Appendix A.5. The foldwise argument in (53) then applies with the MLD signal. □

For pointwise inference use the MLD oracle residual and variance,

$$\begin{aligned}
\varepsilon_K^L &= s_L(W;\mu_0) - p_K(X)^\top \gamma_K, \\
\Omega_K^L &= \mathbb{E}[p_K(X)p_K(X)^\top(\varepsilon_K^L)^2], \\
V_K^L(x) &= p_K(x)^\top Q_K^{-1}\Omega_K^L Q_K^{-1} p_K(x).
\end{aligned}$$

Require $V_K^L(x) \asymp \|p_K(x)\|^2$, the corresponding triangular-array Lindeberg condition, and the approximation restriction

$$\sqrt{n/V_K^L(x)}\,a_{L,K} \to 0.$$

These are the MLD substitutions for the target-specific conditions in Theorem 1; the remaining sampling and series restrictions have the scope specified above.

B.4 Estimator of the conditional GE(0)/MLD surface

Given a cross-fitted first-stage estimate $\widehat{\mu} : \mathcal{X} \to (0,\infty)$, define the estimated locally robust GE(0)/MLD moment

$$\widehat{\psi}_{Ki}^L(\gamma) = p_K(X_i)\left[\log\widehat{\mu}(X_i) - \log Y_i - p_K(X_i)^\top\gamma\right] + \frac{p_K(X_i)}{\widehat{\mu}(X_i)}\{Y_i - \widehat{\mu}(X_i)\}.$$

The locally robust estimator of the conditional MLD surface solves

$$\mathbb{E}_n\widehat{\psi}_{Ki}^L(\widehat{\gamma}_K) = 0_K, \qquad \widehat{\gamma}_K \in \mathbb{R}^K,$$

or equivalently

$$\widehat{\gamma}_K = \widehat{Q}_K^{-1}\mathbb{E}_n p_K(X_i)\left[\log\widehat{\mu}(X_i) - \log Y_i + \frac{Y_i - \widehat{\mu}(X_i)}{\widehat{\mu}(X_i)}\right].$$

The estimated conditional GE(0)/MLD surface is

$$\widehat{L}_K(x) = p_K(x)^\top \widehat{\gamma}_K, \qquad x \in \mathcal{X}.$$

For variance estimation, use the MLD residual

$$\widehat{\varepsilon}^L_{Ki} = \log \widehat{\mu}(X_i) - \log Y_i - p_K(X_i)^\top \widehat{\gamma}_K + \frac{Y_i - \widehat{\mu}(X_i)}{\widehat{\mu}(X_i)},$$

and replace $\widehat{\varepsilon}_{Ki}$ by $\widehat{\varepsilon}^L_{Ki}$ in (21). Under the MLD specialization in Appendix B.3, parts (a) and (b) of Theorem 1 apply with $(T_0, \beta_K, r_K, s_T, a_{T,K}, \varepsilon_K, \Omega_K, V_K)$ replaced by $(L_0, \gamma_K, \ell_K, s_L, a_{L,K}, \varepsilon^L_K, \Omega^L_K, V^L_K)$. Lemma 5 supplies the first-stage bounds that Lemma 1 supplies in the Theil case, and (71) the additional moment condition. The proofs in Appendices A.6–A.7 then apply to this signal with those substitutions.

B.5 Standardized MLD profiles

For the same fixed reference distribution and support rule as in (5), define

$$\theta^L_0(a) = \int L_0(a, z)\, dF^\star_Z(z), \qquad \widehat{\theta}^L_K(a) = \bar{p}_K(a)^\top \widehat{\gamma}_K.$$

Use $\bar{V}^L_K(a) = \bar{p}_K(a)^\top Q_K^{-1} \Omega^L_K Q_K^{-1} \bar{p}_K(a)$ and its empirical counterpart. Corollary 2 and its proof apply after making the MLD substitutions, imposing the MLD weighted nuisance bounds, and replacing the profile undersmoothing condition by $\sqrt{n/\bar{V}^L_K(a)}\, a_{L,K} \to 0$. The residual variance and Lindeberg conditions must likewise be imposed for ε^L_K. The fixed-reference interpretation and the need for a separate argument when the population reference distribution is estimated remain unchanged.

B.6 Locally robust within and between GE(0)/MLD estimators

For the components in (69), the locally robust signals are

$$\varphi^L_W(W; \mu_0) = \log \mu_0(X) - \log Y + \frac{Y - \mu_0(X)}{\mu_0(X)},$$

$$\varphi^L_B(W; \mu_0) = \log \mu_0(X) + \frac{Y - \mu_0(X)}{\mu_0(X)}.$$

The same correction term appears in both equations because both MLD components depend on μ_0 through $\log \mu_0$. Its pathwise derivative cancels the derivative of the plug-in term $\log \mu_0$, while its conditional mean is zero at the truth.

Given a cross-fitted estimate $\widehat{\mu}$, define

$$\widehat{L}^{LR}_W(X) = \mathbb{E}_n \left[\log \widehat{\mu}(X_i) - \log Y_i + \frac{Y_i - \widehat{\mu}(X_i)}{\widehat{\mu}(X_i)} \right],$$

$$\widehat{L}^{LR}_B(X) = \log \bar{Y} - \mathbb{E}_n \left[\log \widehat{\mu}(X_i) + \frac{Y_i - \widehat{\mu}(X_i)}{\widehat{\mu}(X_i)} \right].$$

For nested conditioning sets $X^{(a)} \subset X^{(b)}$, the incremental GE(0) between component is

$$\widehat{\Delta} L_B^{LR}(X^{(b)}, X^{(a)}) = \widehat{L}_B^{LR}(X^{(b)}) - \widehat{L}_B^{LR}(X^{(a)}).$$

The population-share weighting of MLD differs from the wage-share weighting of Theil; agreement is not imposed, and differences between the indices require interpretation rather than being treated as a failed robustness check.

The scalar estimators also satisfy exact sample accounting,

$$\widehat{L}_W^{LR} + \widehat{L}_B^{LR} = \log \bar{Y} - \mathbb{E}_n \log Y_i.$$

To state their inference, define

$$q_L(W) = \log \mu_0(X) + \frac{Y - \mu_0(X)}{\mu_0(X)}, \qquad \eta_\mu = \mathbb{E}[\log \mu_0(X)] = \mathbb{E}[q_L(W)].$$

The component influence functions are

$$\begin{aligned} \psi_{L_W}(W) &= q_L(W) - \log Y - L_W(X), \\ \psi_{L_B}(W) &= \frac{Y - \mu}{\mu} - \{q_L(W) - \eta_\mu\}. \end{aligned} \tag{73}$$

They reflect the MLD estimators' own normalization and are not obtained by replacing the index symbol in the Theil influence functions. For estimation, let $\widehat{q}_{L,i} = \varphi_B^L(W_i; \widehat{\mu})$ and $\widehat{\eta}_\mu = \mathbb{E}_n \widehat{q}_{L,i}$, and use

$$\begin{aligned} \widehat{\psi}_{L_W,i} &= \widehat{q}_{L,i} - \log Y_i - \widehat{L}_W^{LR}, \\ \widehat{\psi}_{L_B,i} &= \frac{Y_i - \bar{Y}}{\bar{Y}} - (\widehat{q}_{L,i} - \widehat{\eta}_\mu). \end{aligned} \tag{74}$$

Proposition 7 (Direct MLD component inference)**.** *Suppose Assumption 2 holds with its conditional moment restriction replaced by (71), and impose only (38) on the nuisance error. No series, support-density, or smoothness restrictions are needed for this result. With* $\Psi_L = (\psi_{L_W}, \psi_{L_B})^\top$ *and* $\Sigma_L = \mathbb{E}[\Psi_L \Psi_L^\top]$*,*

$$\sqrt{n} \begin{pmatrix} \widehat{L}_W^{LR} - L_W \\ \widehat{L}_B^{LR} - L_B \end{pmatrix} = \frac{1}{\sqrt{n}} \sum_{i=1}^{n} \Psi_L(W_i) + o_p(1) \Rightarrow N(0, \Sigma_L). \tag{75}$$

Both estimators are consistent. The empirical covariance of the estimated influence vectors in (74) converges to Σ_L*, which may be singular. Each component or fixed linear contrast with positive limiting variance has a studentized standard normal limit. For two nested conditioning sets satisfying the same conditions, the influence function of* $\widehat{\Delta} L_B^{LR}$ *is* $\psi_{L_B}^{(b)} - \psi_{L_B}^{(a)}$*; use its paired empirical variance, and require that variance to be positive for Wald inference. The proof is in Appendix B.8.*

At the nested MLD null, conditional Jensen's inequality for log implies that $\Delta L_B = 0$ only when the richer and coarser means coincide almost surely. Their between-component influence functions then coincide as well. As for Theil, a positive-variance Wald approximation does not

test that null.

B.7 Proof of Proposition 6

Start from the non-robust MLD moment

$$m_K^L(W;\gamma,\mu) = p_K(X)\left[\log\mu(X) - \log Y - p_K(X)^\top\gamma\right].$$

At the truth, $\mathbb{E}[m_K^L(W;\gamma_K,\mu_0)] = 0_K$ because γ_K is defined by the projection condition in (70). Consider a path $\mu_t(x) = \mu_0(x) + th(x)$. Differentiating the population moment gives

$$\frac{\partial}{\partial t}\mathbb{E}[m_K^L(W;\gamma_K,\mu_t)]\Big|_{t=0} = \mathbb{E}\left[\frac{p_K(X)h(X)}{\mu_0(X)}\right]. \tag{76}$$

Now define the correction term

$$\phi_K^L(W;\mu_t) = \frac{p_K(X)}{\mu_t(X)}\{Y - \mu_t(X)\}.$$

At the truth, $\mathbb{E}[\phi_K^L(W;\mu_0) \mid X] = 0_K$. Its pathwise derivative at $t = 0$ is

$$\frac{\partial}{\partial t}\phi_K^L(W;\mu_t)\Big|_{t=0} = -\frac{p_K(X)Yh(X)}{\mu_0(X)^2}.$$

Taking conditional expectations given X gives

$$\mathbb{E}\left[\frac{\partial}{\partial t}\phi_K^L(W;\mu_t)\Big|_{t=0}\Big|X\right] = -\frac{p_K(X)h(X)}{\mu_0(X)},$$

which cancels (76). Therefore

$$\psi_K^L(W;\gamma_K,\mu_0) = m_K^L(W;\gamma_K,\mu_0) + \phi_K^L(W;\mu_0)$$

has mean zero at the truth and zero nuisance derivative. At the scalar level, $\mathbb{E}[s_L(W;\mu_0) \mid X] = \log\mu_0(X) - \eta_0(X) = L_0(X)$, and $\partial_m\{\log m - \eta_0(X) + (\mu_0(X) - m)/m\}|_{m=\mu_0(X)} = 0$. The lower bound and moment conditions justify the indicated differentiation. This establishes the conditional and unconditional assertions in Proposition 6. □

B.8 Proof of Proposition 7

For a training fold, write $a = \mu_0(X)$, $m = \widehat{\mu}_{-j}(X)$, and $d = m - a$. The feasible log-mean signal is $q_L(W;m) = \log m + (Y - m)/m$. Its exact conditional mean error is

$$\mathbb{E}[q_L(W;m) - q_L(W;a) \mid X, \mathcal{D}_{-j}] = \log(m/a) - \frac{d}{m} = d^2\int_0^1 \frac{t}{(a+td)^2}\,dt.$$

It is bounded by Cd^2, and $|q_L(W;m) - q_L(W;a)| \le C(1+Y)|d|$. The conditional moment bound therefore yields a mean-square signal error bounded by $C\|\widehat{\mu}_{-j} - \mu_0\|_2^2$. The scalar version of

Lemma 3, with multiplier one, and (38) give

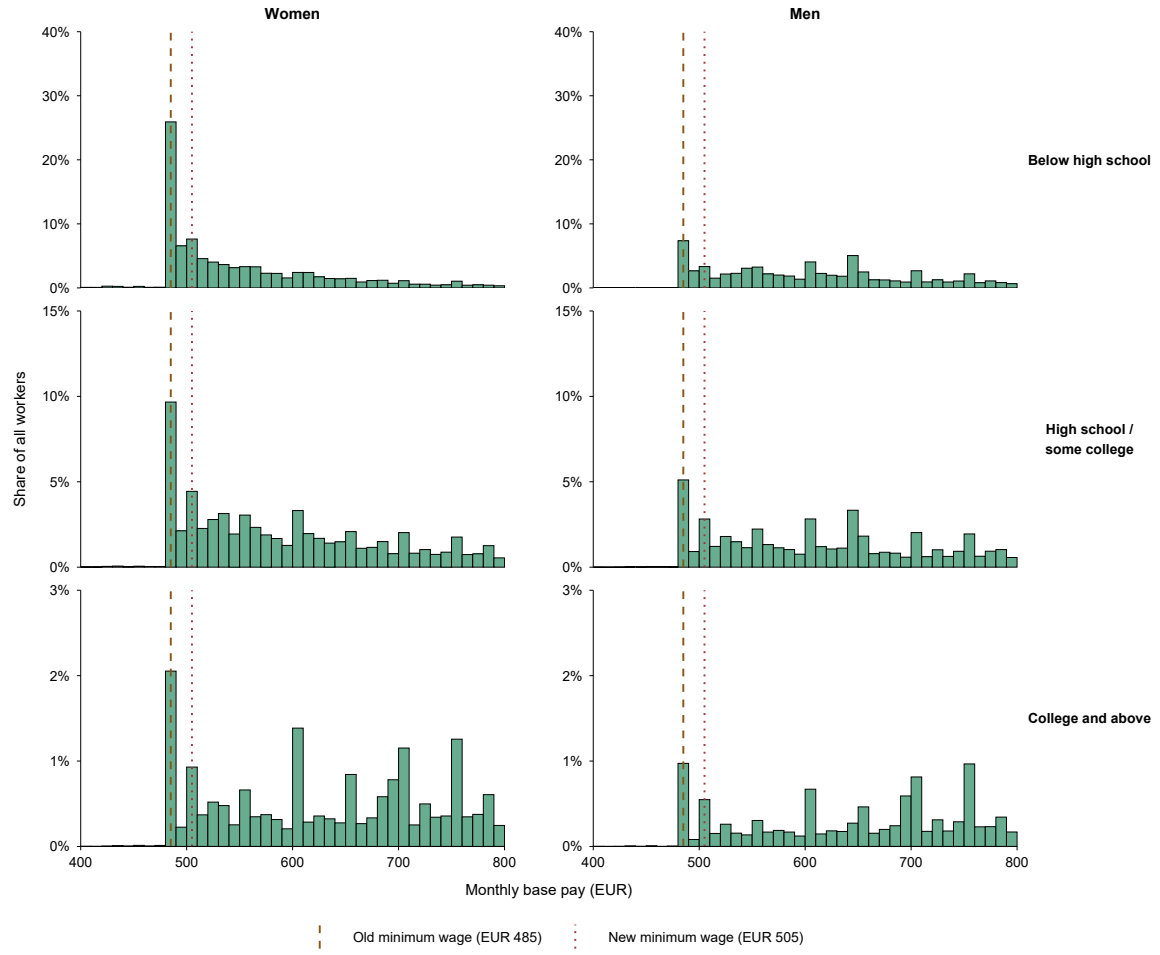
$$\mathbb{E}_n \widehat{q}_{L,i} = \mathbb{E}_n q_L(W_i) + o_p(n^{-1/2}). \tag{77}$$

Subtracting the sample mean of $\log Y$ proves the within-component expansion. Expanding $\log \bar{Y}$ at μ and subtracting (77) gives the between-component expansion. Stacking these two expansions gives the influence functions in (73) and the multivariate central limit in (75). The finite second moments needed by that limit follow from (71) and the mean bounds.

Conditional Markov's inequality within folds gives $\mathbb{E}_n(\widehat{q}_{L,i} - q_L(W_i))^2 = o_p(1)$. Consistency of the component estimates, $\bar{Y}$, and $\widehat{\eta}_\mu$ then implies empirical mean-square consistency of the estimated influence vectors. Cauchy–Schwarz and the law of large numbers prove covariance consistency. Studentization of a contrast follows by Slutsky only when its variance is positive. For nested conditioning sets, apply these representations jointly and subtract the between components on the same observations, retaining the covariance term exactly as in Appendix A.15. Finally, $\mathbb{E}[\log \mu_a - \log \mu_b] \geq 0$ under nesting, and strict concavity of log gives equality only when $\mu_a = \mu_b$ almost surely; this verifies the stated null-degeneracy qualification. □

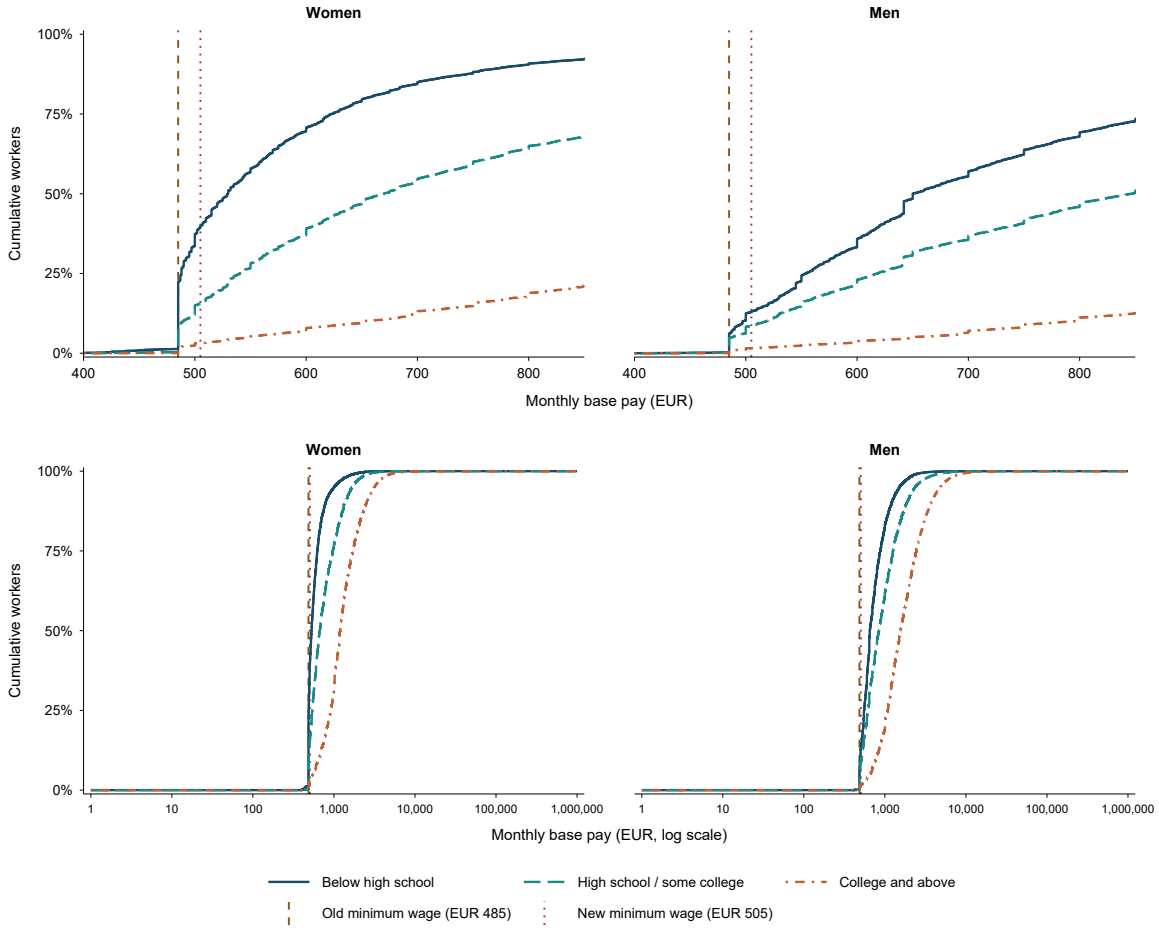
C Pre-increase wage distributions by education and sex

Figure 10: Pre-increase monthly base-pay distributions by education and sex, 2013



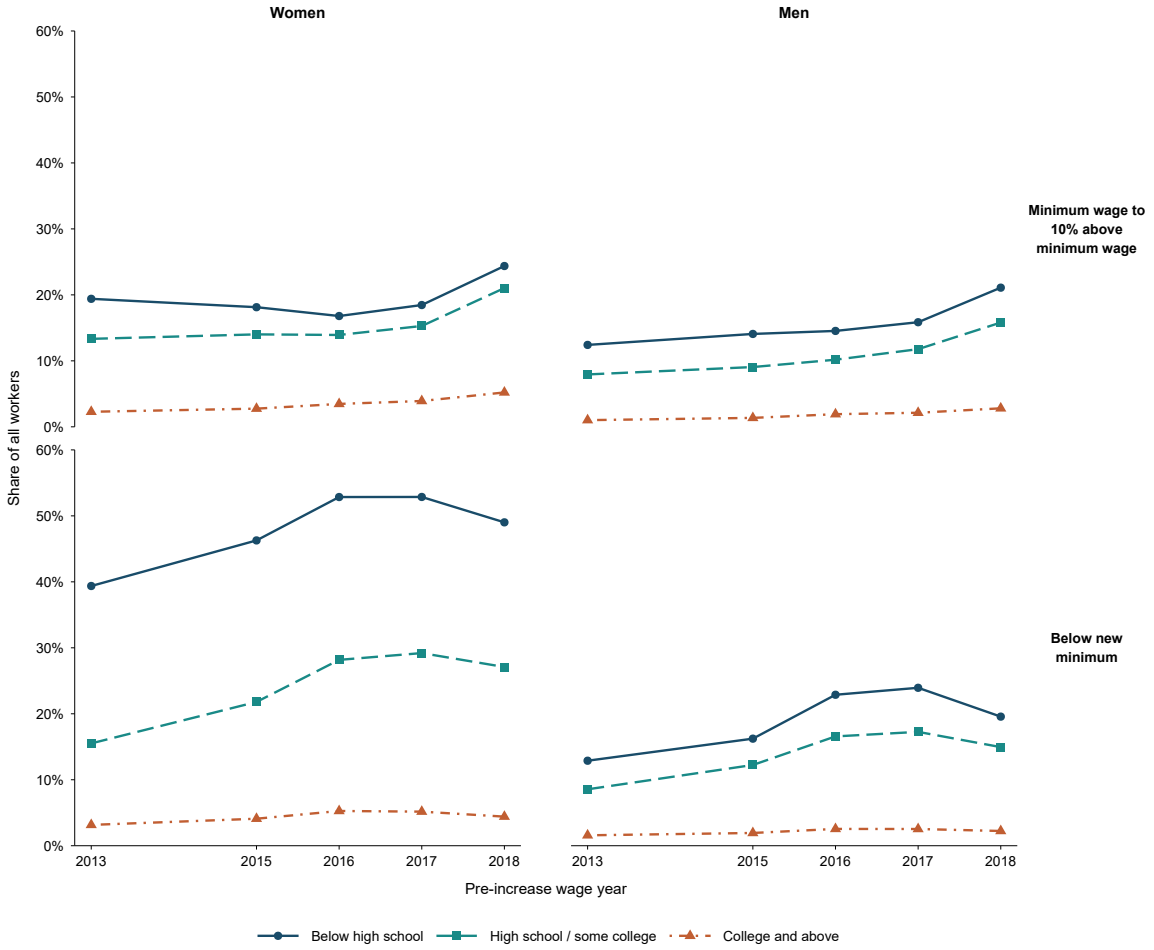
Notes: The histograms use nominal monthly base pay in the 2013 initial-exposure sample, including workers at employers with zero and positive exposure. Each bar reports the share of workers in the corresponding sex–education group within a 10-euro pay interval. The display covers pay from 400 euros to below 800 euros; shares retain the full group denominator and are not renormalized over the displayed range. Dashed and dotted vertical lines mark the old minimum of 485 euros and the forthcoming minimum of 505 euros, respectively. Vertical scales differ across education groups but are the same for women and men within each education group.

Figure 11: Cumulative distributions of pre-increase monthly base pay by education and sex, 2013



Notes: Curves report cumulative shares of workers by nominal monthly base pay within each sex–education group in the 2013 initial-exposure sample, including workers at employers with zero and positive exposure. The upper panels show the distribution near the minimum wage; the lower panels show the full range using a logarithmic wage axis. All shares use the full group denominator. Dashed and dotted vertical lines mark the old minimum of 485 euros and the forthcoming minimum of 505 euros, respectively.

Figure 12: Pre-increase wage-band shares by education and sex across the five minimum-wage increases



Notes: For each comparison $e = (r, s)$, w_r denotes nominal monthly base pay before the increase, while m_r and m_s denote the old and new minimum wages. The upper row reports the share of workers with $m_s \leq w_r < 1.1m_s$, and the lower row reports the share with $w_r < m_s$. Each share uses all workers in the corresponding sex–education group in that comparison's pre-increase sample, including workers at employers with zero and positive exposure. The horizontal axis reports the pre-increase years 2013, 2015, 2016, 2017, and 2018, corresponding to increases in 2014, 2016, 2017, 2018, and 2019. Each comparison uses its own sample and minimum-wage thresholds.

D Partial-mean profiles of conditional wage inequality

Figure 13: Partial-mean profiles by log real sales per worker, 2011, 2013, and 2019

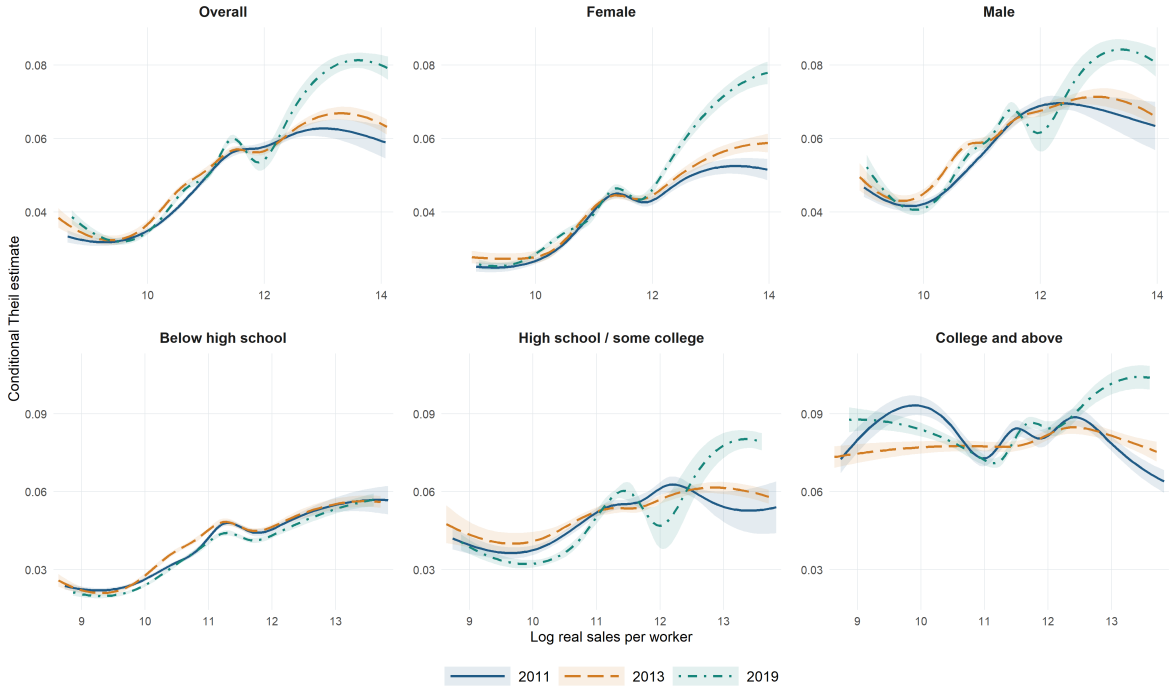


Figure 14: Partial-mean profiles by number of establishments, 2011, 2013, and 2019

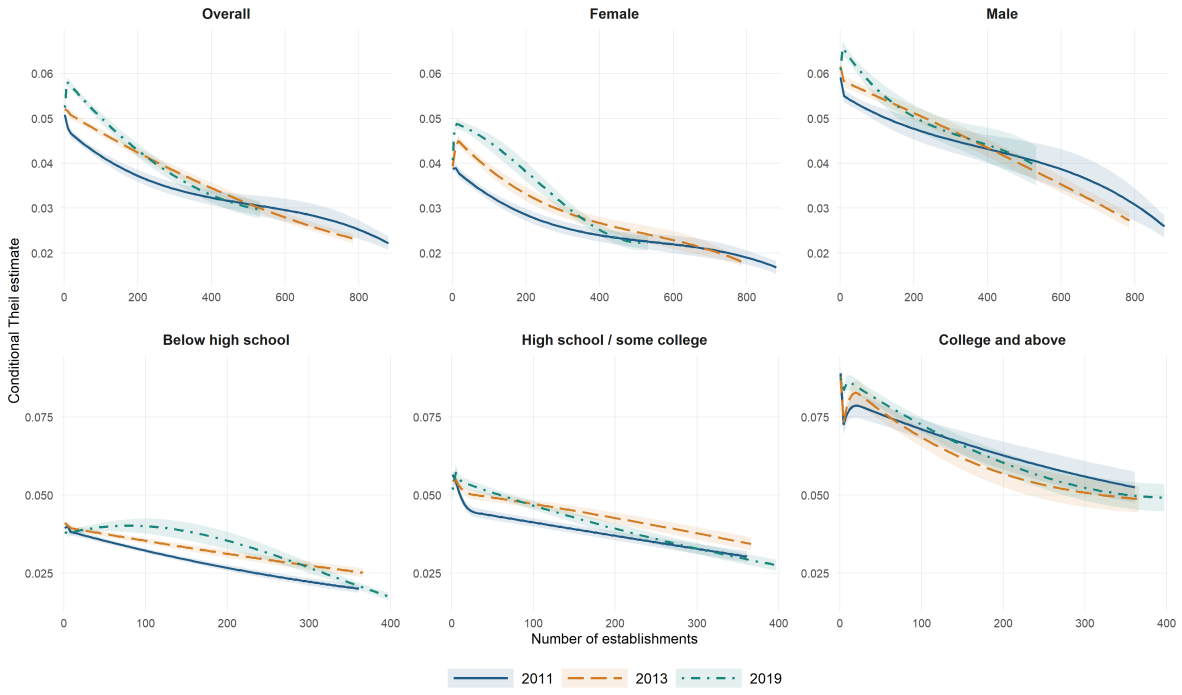


Figure 15: Partial-mean profiles by log employer employment, 2011, 2013, and 2019

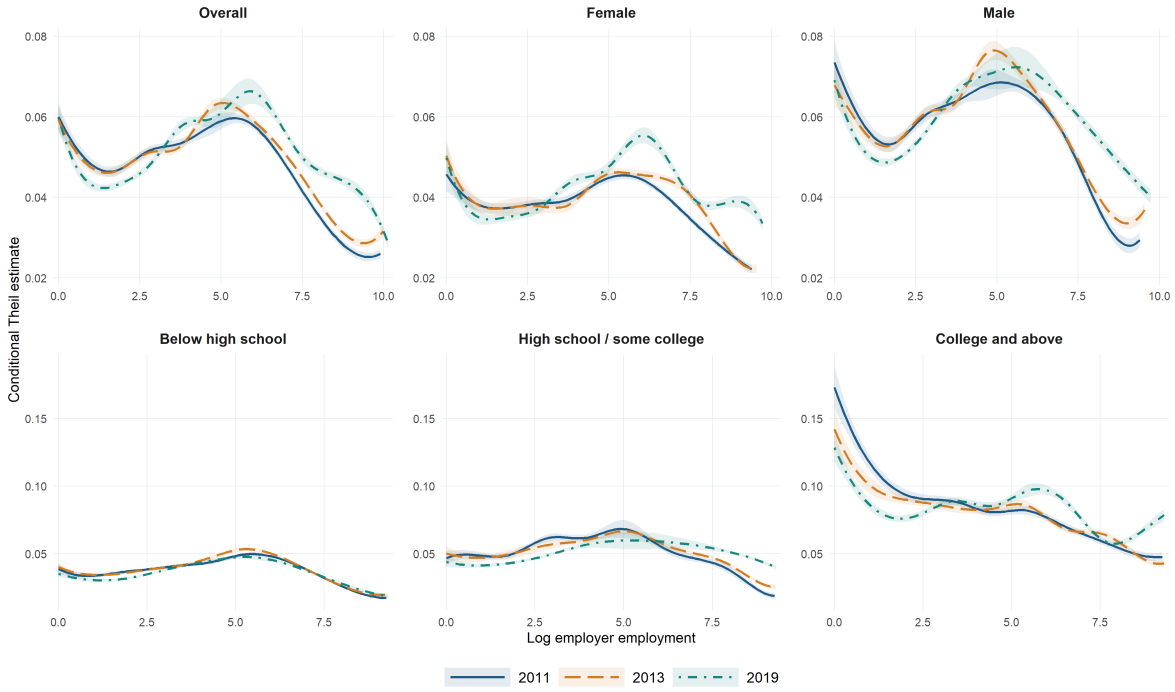


Figure 16: Age profiles of conditional inequality by sex and occupation group, 2011, 2013, and 2019

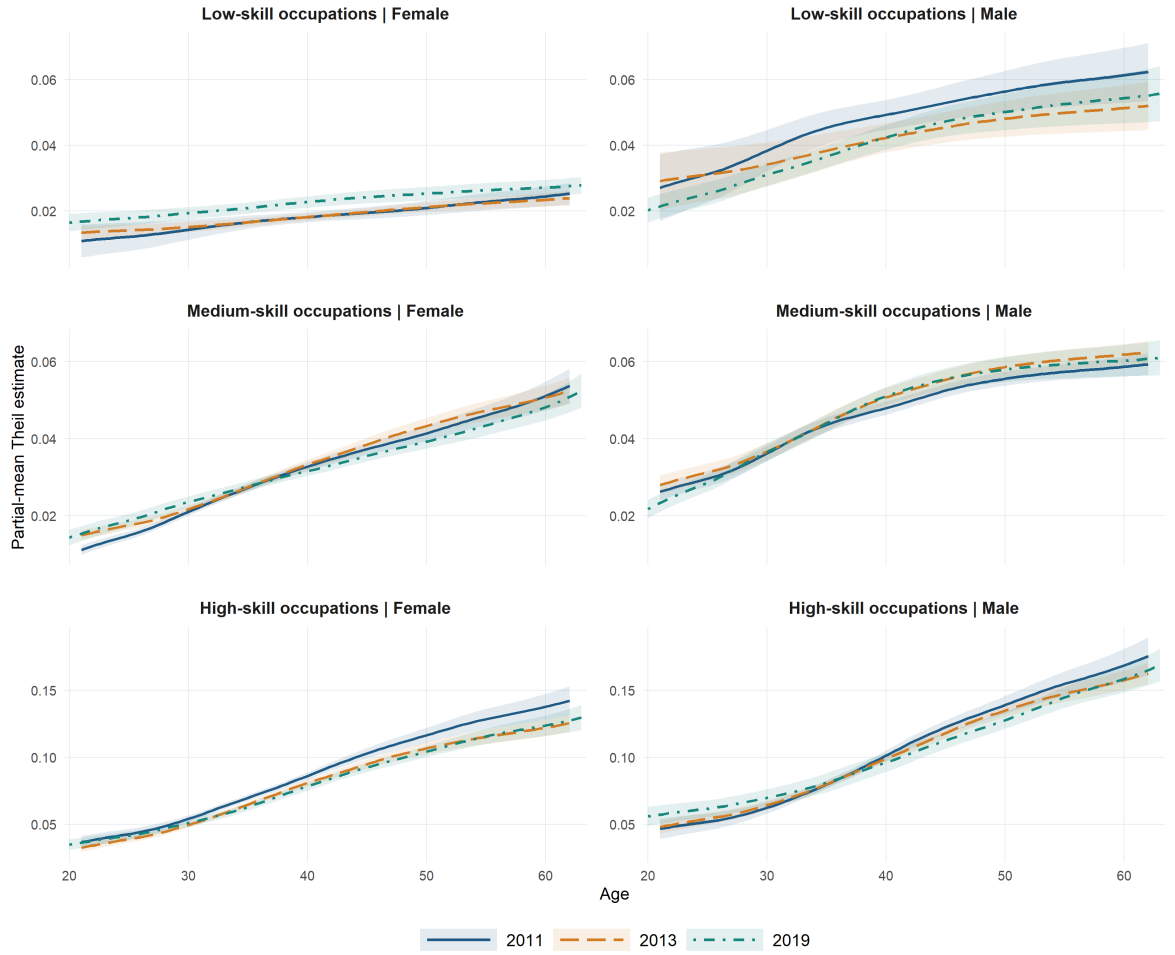


Figure 17: Age profiles of conditional inequality by sex and contract status, 2011, 2013, and 2019

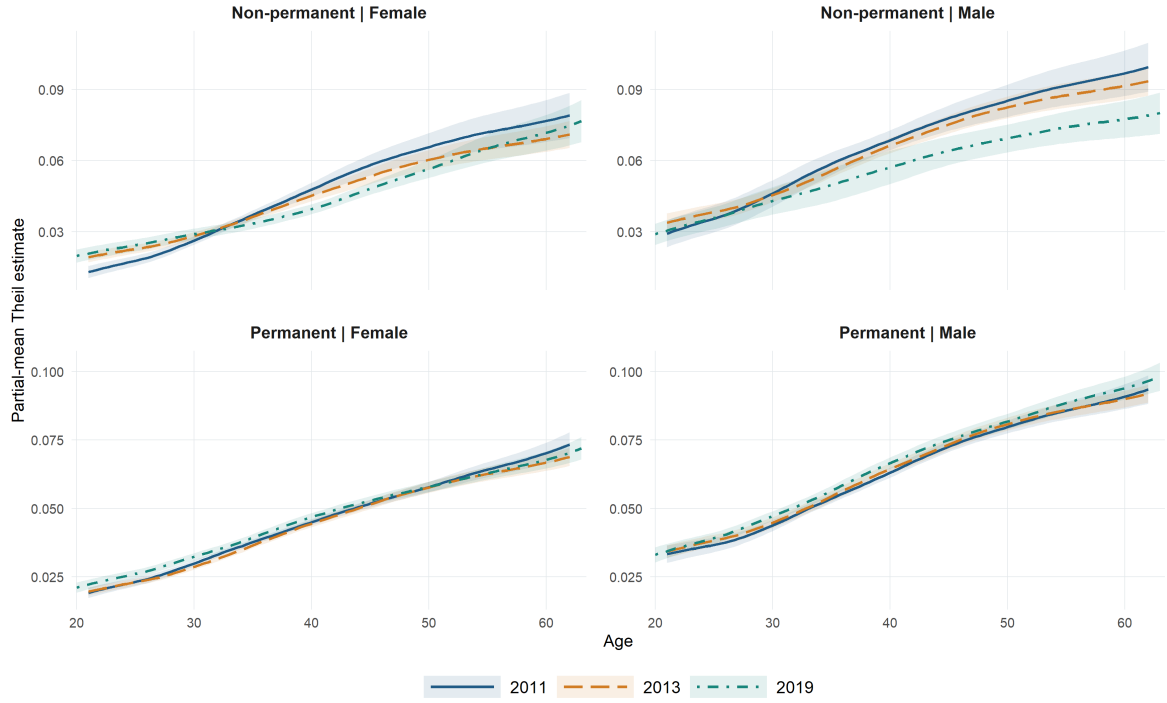


Figure 18: Age profiles of conditional inequality by education and occupation group, 2011, 2013, and 2019

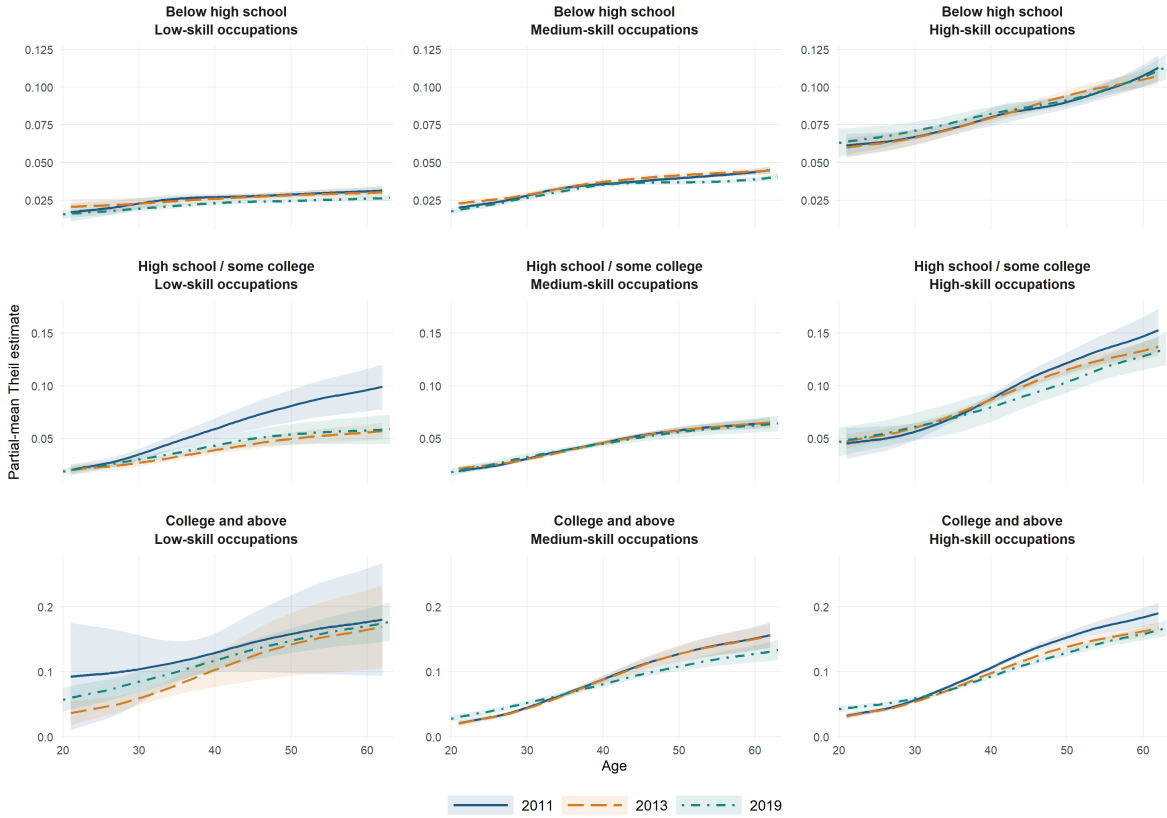


Figure 19: Age profiles of conditional inequality by education and contract status, 2011, 2013, and 2019

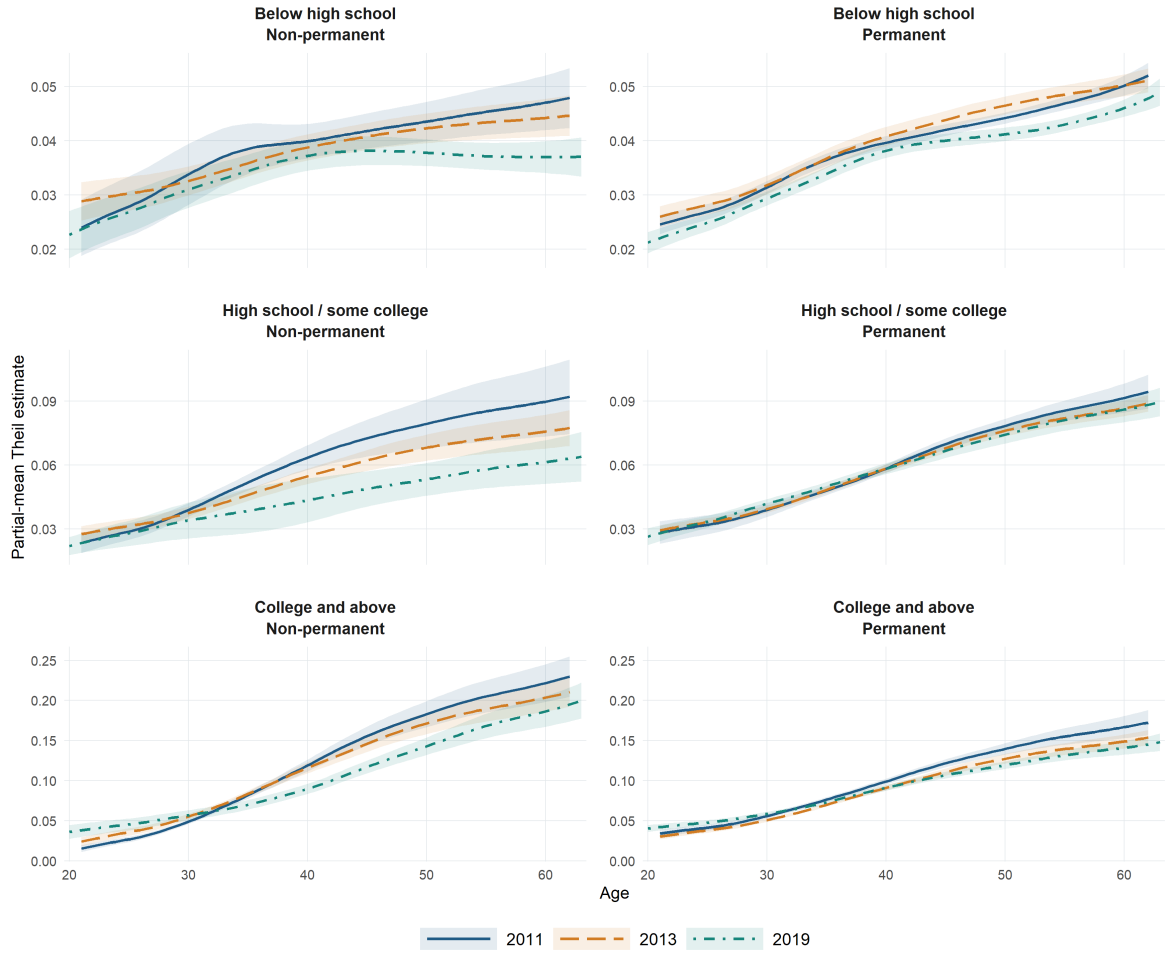
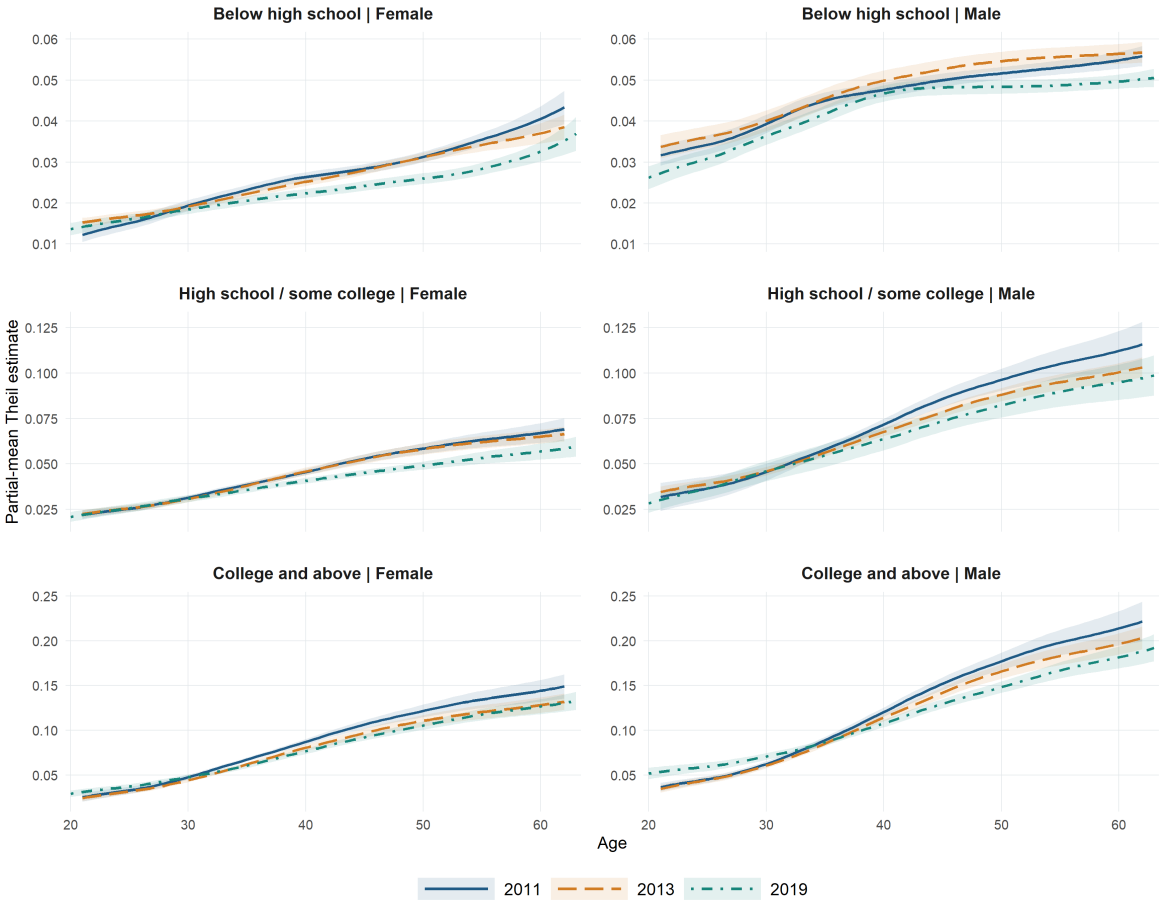


Figure 20: Age profiles of conditional inequality by education and sex, 2011, 2013, and 2019



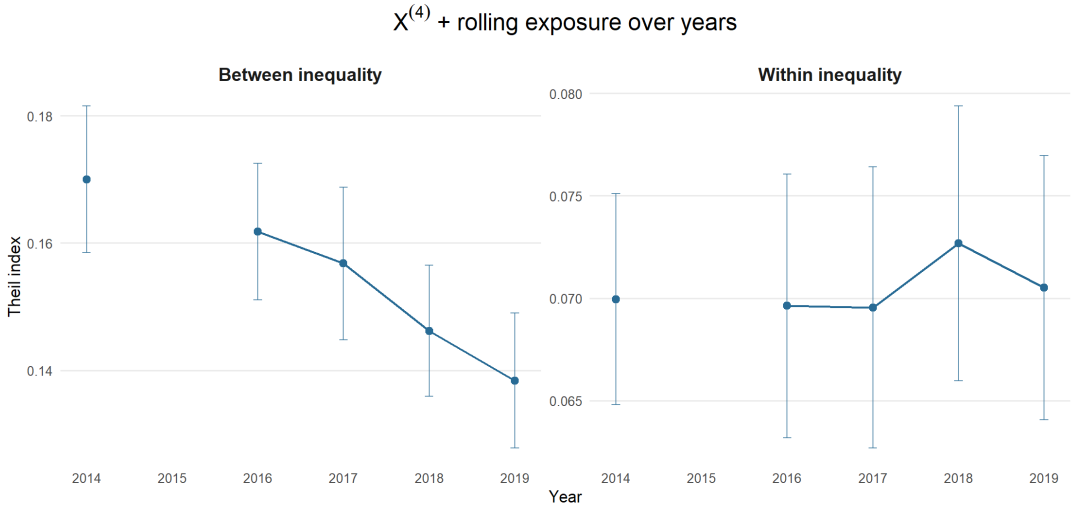
E Additional exposure analyses and wage adjustment

This appendix provides implementation details for the conditional-moment estimates, reports the rolling-exposure within–between decomposition and supplementary initial- and rolling-exposure profiles, and presents wage growth and subsequent observation for workers initially paid between the old and new minimum-wage thresholds.

First-stage implementation. Conditional moments are estimated with gradient-boosted trees (T. Chen & Guestrin, 2016) on the conditioning variables of Table 5, with categorical variables encoded within each training sample. The conditional means in the exposure analyses use a gamma objective, which keeps fitted means positive; all other moments use squared error. The number of boosting rounds is chosen by early stopping on employer-grouped validation folds within each training sample. Cross-fitting uses five employer-level folds for the aggregate decompositions and for the exposure analyses. Fitted means are bounded below by the 0.1 percent quantile of the training-sample wages. The reporting basis for the exposure profiles consists of a natural cubic spline in B, with the number of interior knots $J_n \asymp n^{0.3}$, where n is the estimation-sample size. Knots are placed at quantiles of positive exposure in each estimation year. The spline basis and a separate indicator for zero exposure are each interacted with the six sex–education groups.

E.1 Rolling-exposure inequality decomposition

Figure 21: Within and between inequality with rolling minimum-wage exposure



Notes: Components condition on $(X^{(4)}, B)$, where B denotes rolling exposure to each minimum-wage increase. The panels show the five post-increase observations. Each decomposition uses its exposure-restricted sample. Bars are nominal 95 percent pointwise intervals.

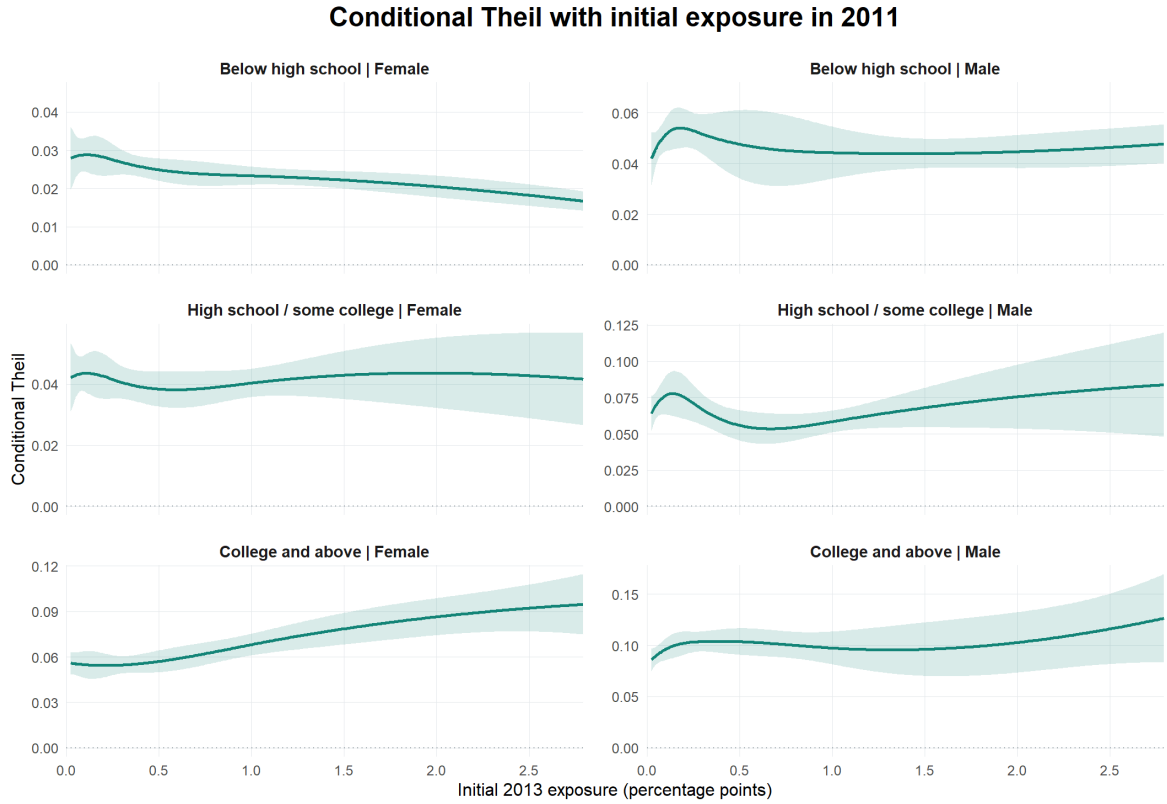
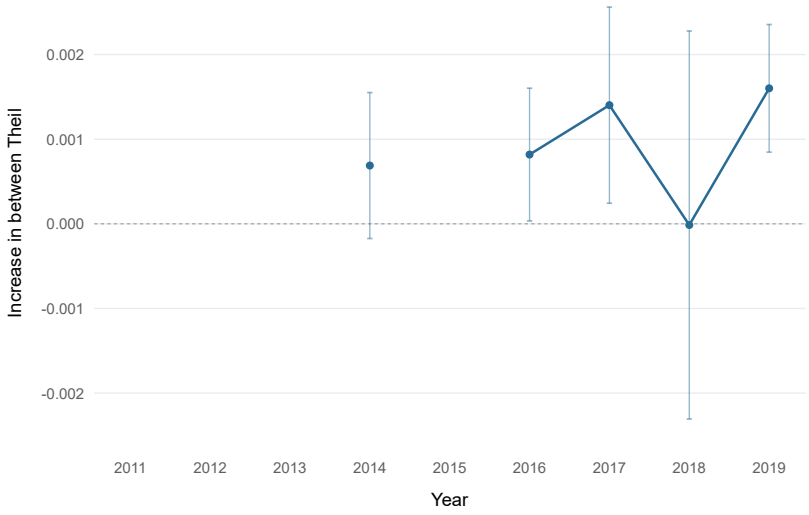


Figure 22: Conditional hourly-wage inequality by initial exposure, 2011

Notes: Shaded area shows nominal 95 percent pointwise intervals. Profiles summarize conditional Theil given $(X^{(4)}, B)$ separately by sex and education. Here B denotes initial exposure to 2014 minimum wage increase. Shading shows nominal 95 percent pointwise .

Figure 23: Additional between inequality captured by rolling minimum-wage exposure



Notes: Each increment compares $X^{(4)}$ and $(X^{(4)}, B)$ for the same sample, where B denotes rolling exposure. Estimates refer to the post-increase year of each comparison. Bars are nominal 95 percent pointwise intervals.

E.2 Additional exposure profiles

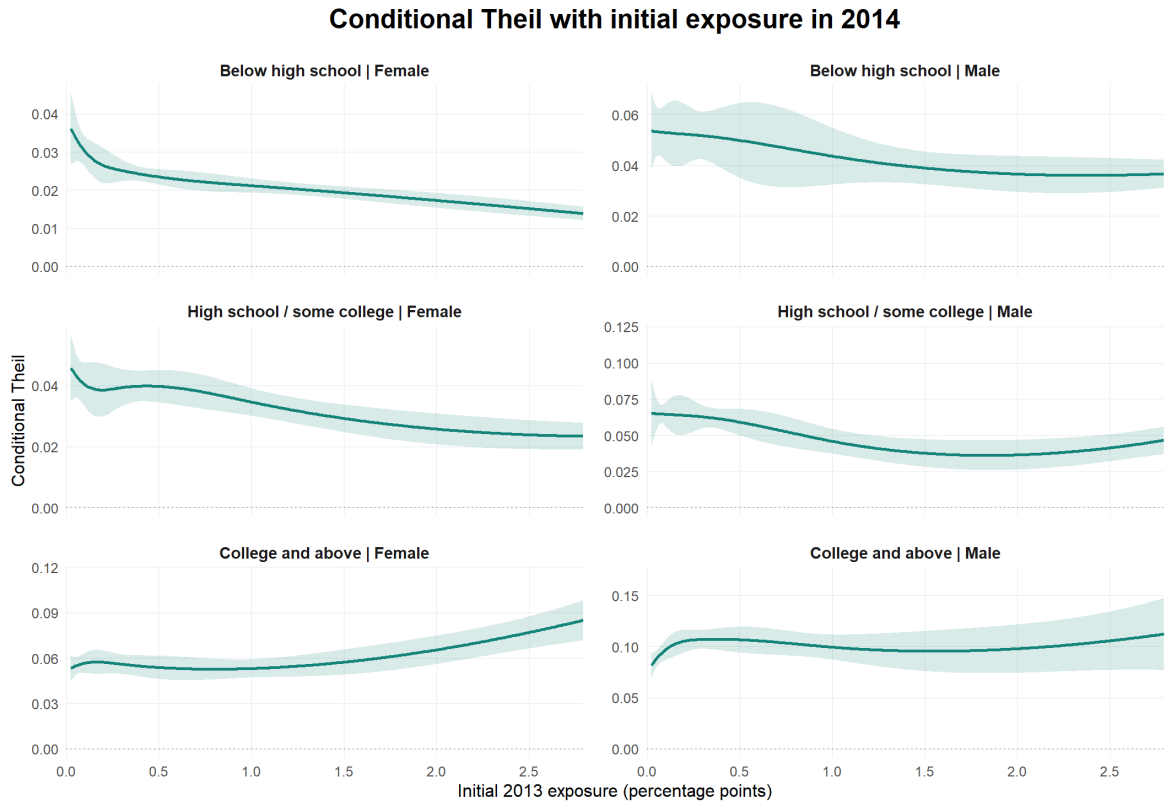


Figure 24: Conditional hourly-wage inequality by initial exposure, 2014

Notes: Shaded area shows nominal 95 percent pointwise intervals. Profiles summarize conditional Theil given $(X^{(4)}, B)$ separately by sex and education. Here B denotes initial exposure to 2014 minimum wage increase. Shading shows nominal 95 percent pointwise .

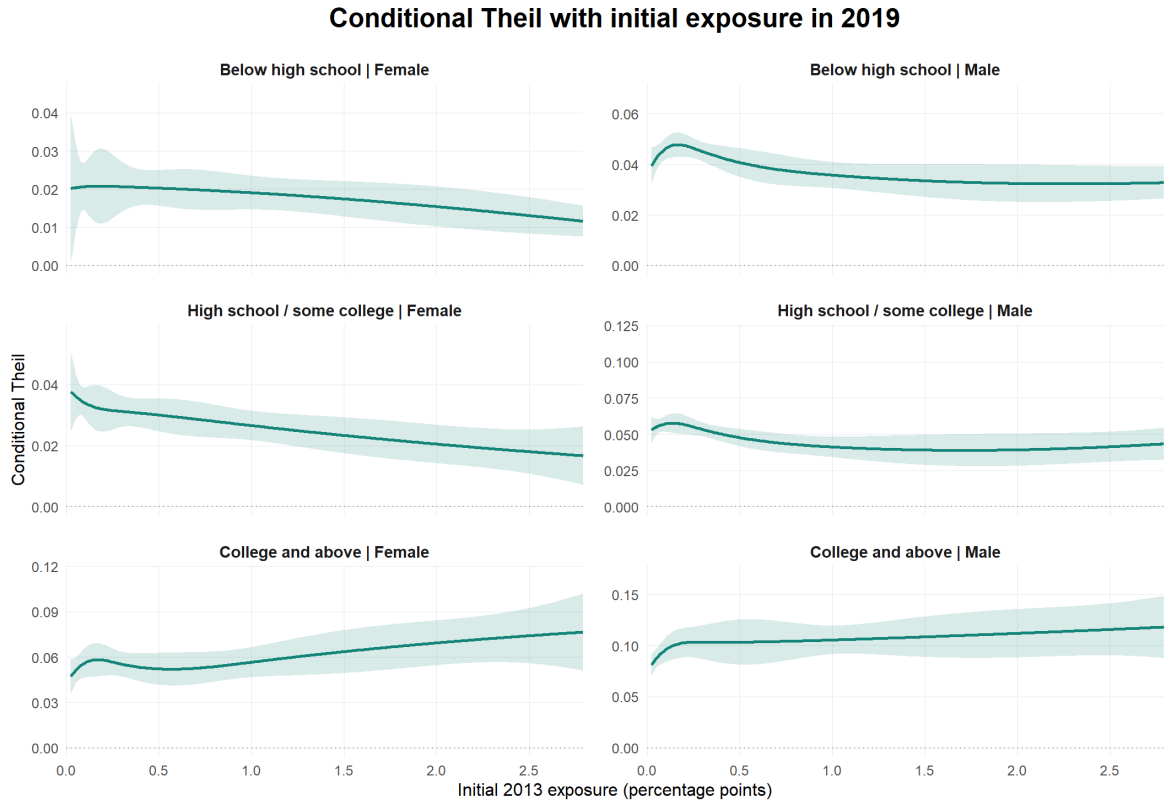


Figure 25: Conditional hourly-wage inequality by initial exposure, sex, and education, 2019
Notes: Shaded area shows nominal 95 percent pointwise intervals . Profiles summarize conditional Theil given $(X^{(4)}, B)$ separately by sex and education. Here B denotes initial exposure to 2014 minimum wage increase. Shading shows nominal 95 percent pointwise .

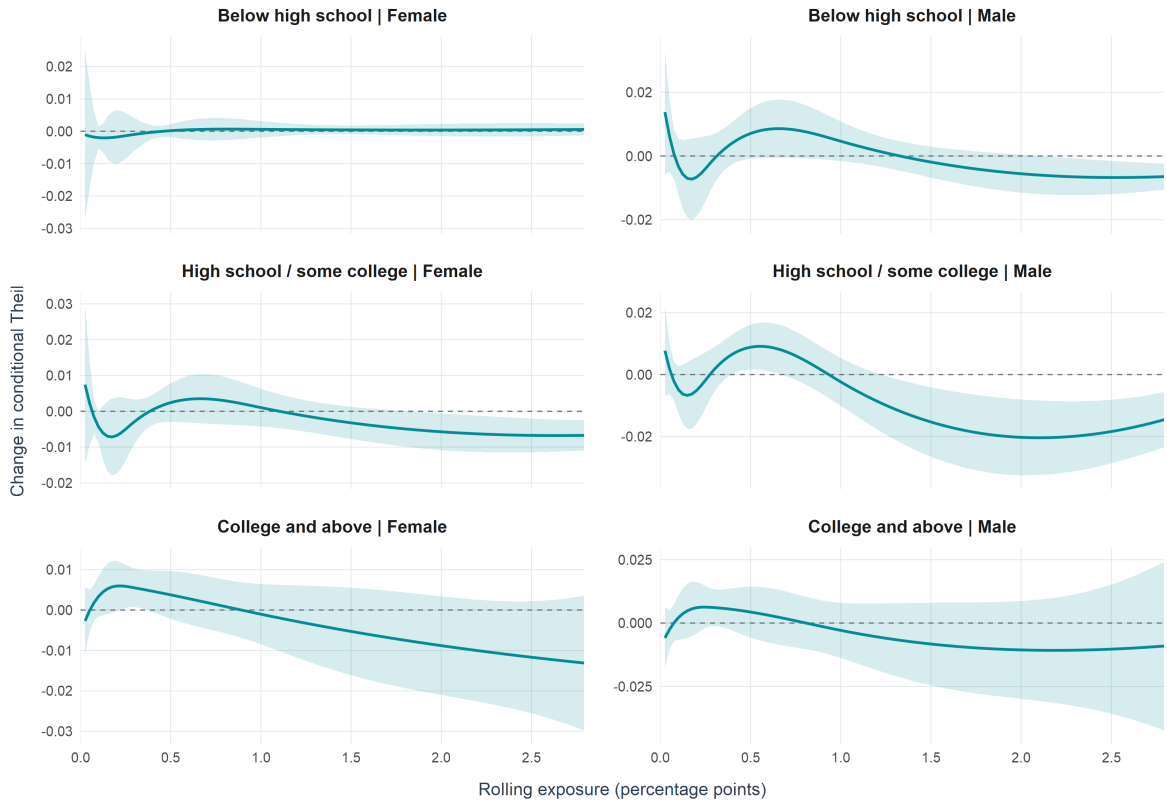
Table 8: Conditional-Theil profiles at one percent employer exposure

	Initial exposure: levels			Rolling exposure: changes	
	2011	2014	2019	2013–2014	2018–2019
Women: below high school	0.0233	0.0212	0.0191	0.0006	-0.0021
	(0.0012)	(0.0010)	(0.0023)	(0.0013)	(0.0024)
Women: high school/some college	0.0403	0.0346	0.0266	0.0010	0.0018
	(0.0024)	(0.0023)	(0.0025)	(0.0027)	(0.0037)
Women: college and above	0.0681	0.0532	0.0567	-0.0010	-0.0015
	(0.0036)	(0.0031)	(0.0051)	(0.0038)	(0.0030)
Men: below high school	0.0443	0.0436	0.0357	0.0046	0.0026
	(0.0052)	(0.0058)	(0.0026)	(0.0032)	(0.0020)
Men: high school/some college	0.0585	0.0460	0.0414	-0.0024	-0.0012
	(0.0038)	(0.0044)	(0.0036)	(0.0039)	(0.0047)
Men: college and above	0.0974	0.0995	0.1055	-0.0029	0.0019
	(0.0081)	(0.0062)	(0.0071)	(0.0055)	(0.0040)

Notes: All entries evaluate a fitted profile at $B = 1$: the incremental base-pay shortfall equals one percent of the total monthly base pay used in the exposure calculation. Initial-exposure columns use the fixed 2013 measure. Rolling columns report the later year minus the earlier year using each comparison's own pre-increase measure and samples. Standard errors are in parentheses. Initial-level standard errors are employer-clustered, equivalent to the saved employer-plus-worker-minus-intersection calculation because workers are unique within each year. Rolling-change standard errors retain cross-year covariance under that joint clustering rule. The initial-level standard errors are not standard errors for changes across these three years. Entries summarize conditional inequality given the full observed information, not raw Theil indices of demographic groups.

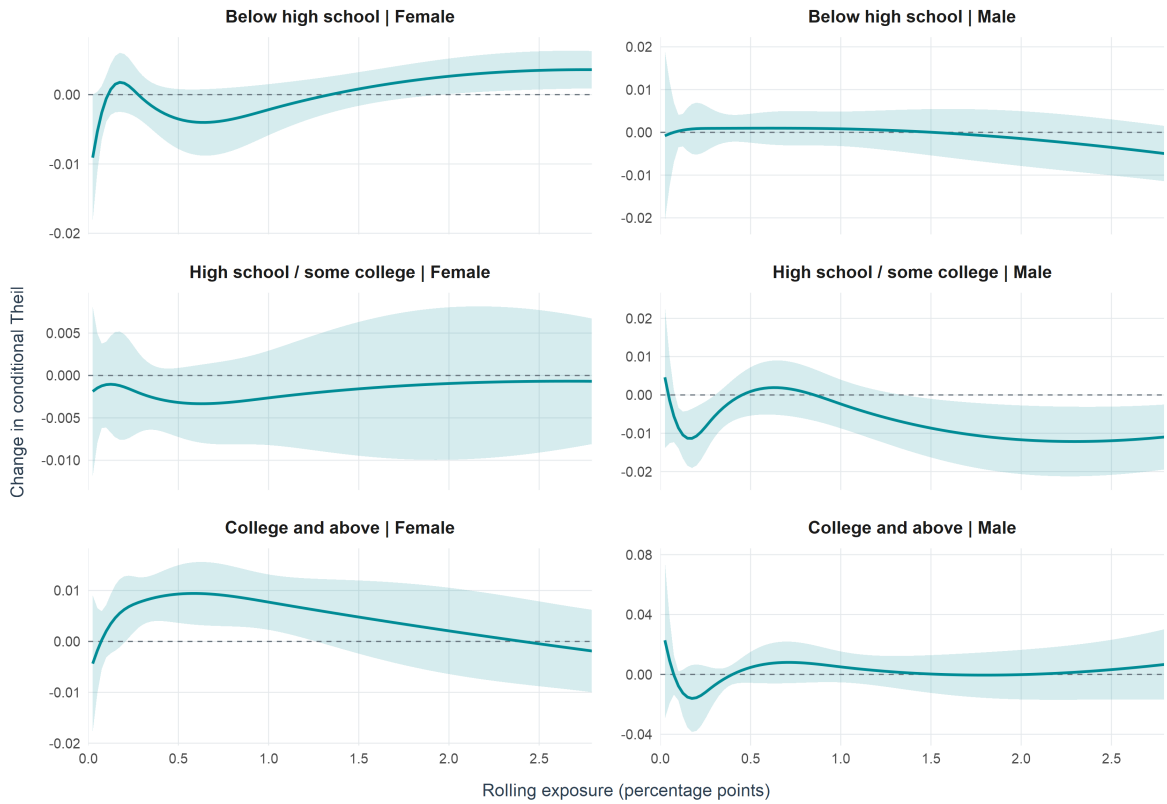
E.3 Rolling changes

Figure 26: Rolling-exposure change in conditional Theil, 2014 minus 2013



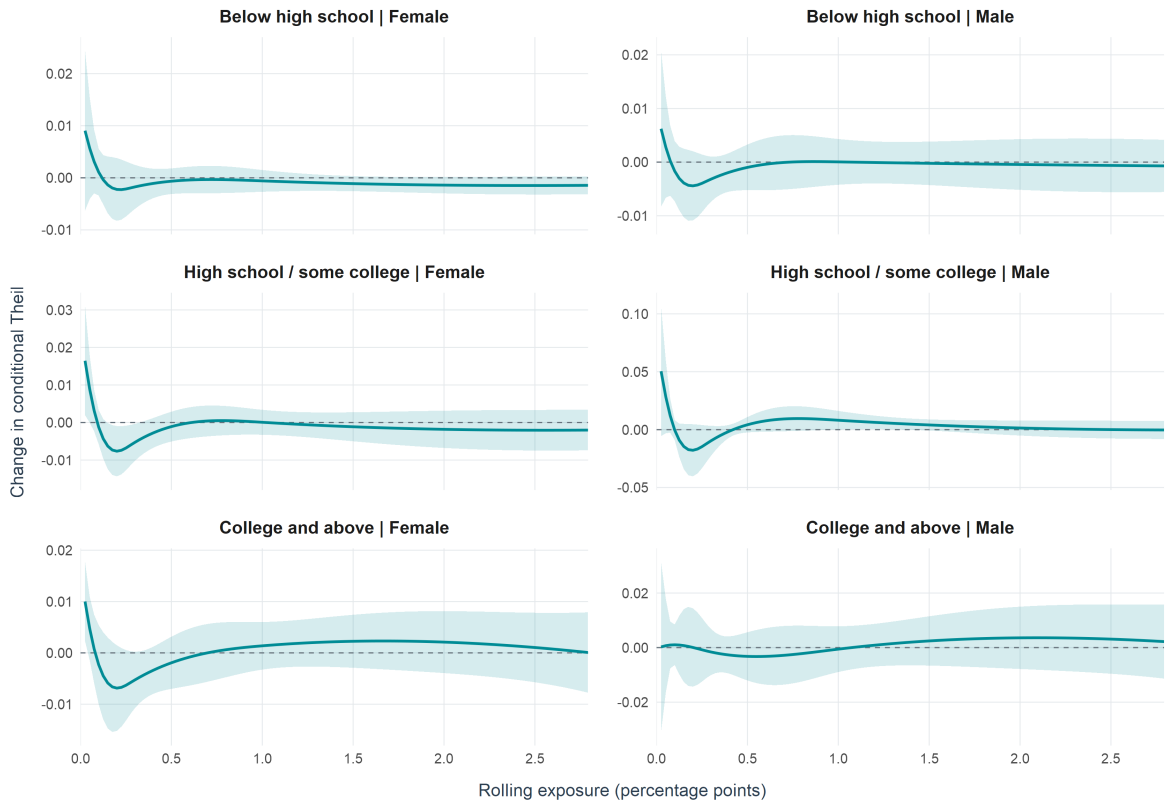
Notes: Shaded areas show nominal 95 percent pointwise intervals. Zero exposure is omitted from the display. Profiles summarize conditional Theil given $(X^{(4)}, B)$ separately by sex and education. Here B denotes exposure to the 2014 minimum-wage increase.

Figure 27: Rolling-exposure change in conditional Theil, 2016 minus 2015



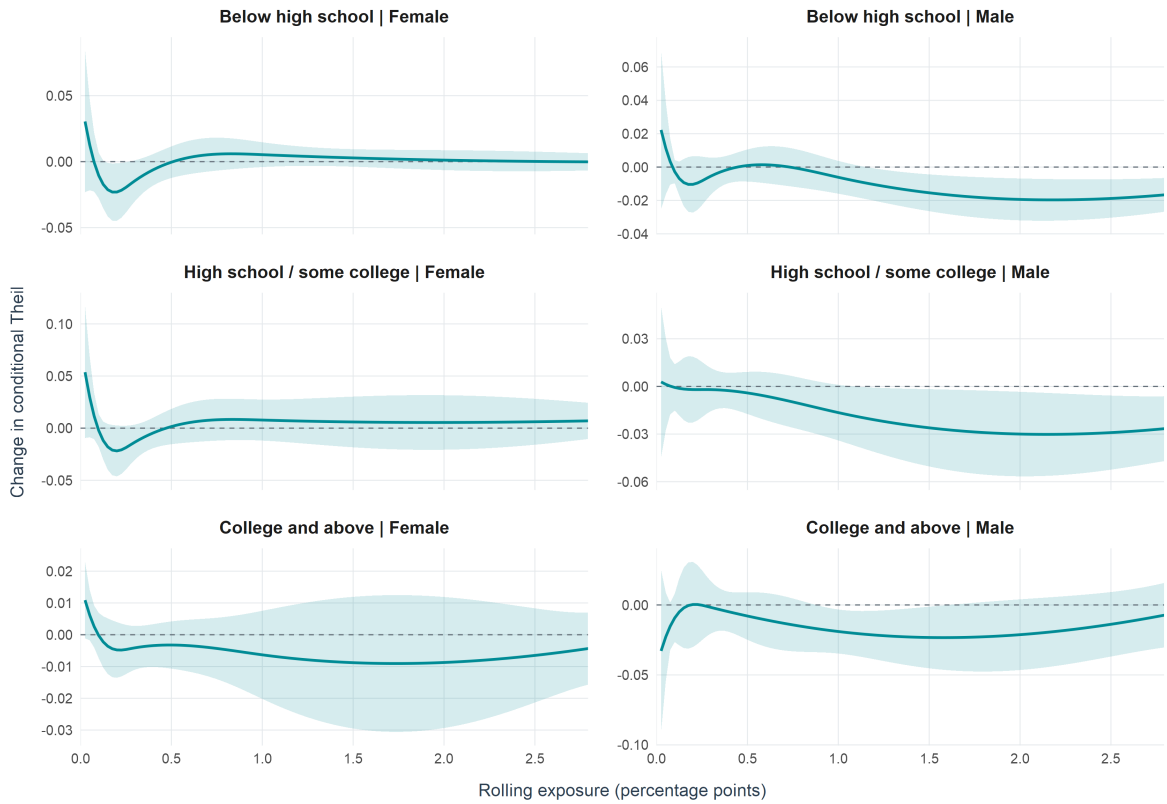
Notes: Shaded area shows nominal 95 percent pointwise intervals . Zero exposure is omitted from the display. Profiles summarize conditional Theil given $(X^{(4)}, B)$ separately by sex and education. Here B denotes exposure to 2016 minimum wage increase. Shading shows nominal 95 percent pointwise .

Figure 28: Rolling-exposure change in conditional Theil, 2017 minus 2016



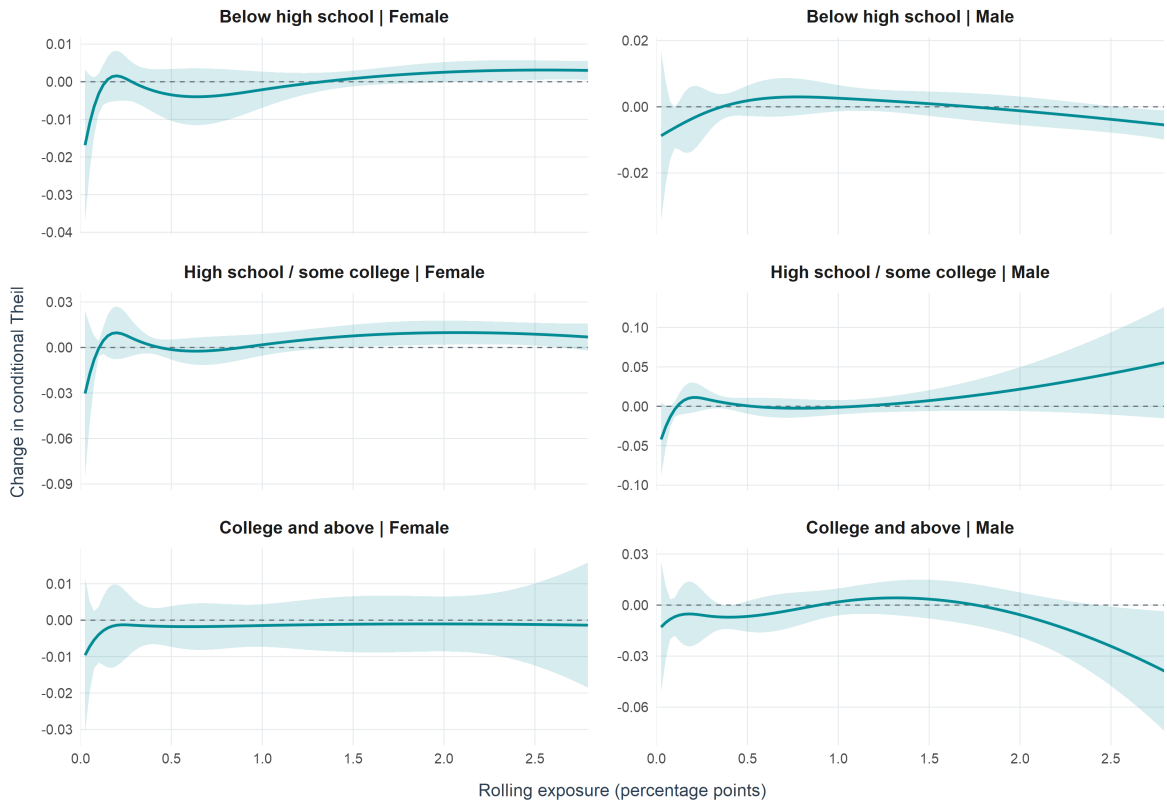
Notes: Shaded area shows nominal 95 percent pointwise intervals . Zero exposure is omitted from the display. Profiles summarize conditional Theil given $(X^{(4)}, B)$ separately by sex and education. Here B denotes exposure to 2017 minimum wage increase. Shading shows nominal 95 percent pointwise .

Figure 29: Rolling-exposure change in conditional Theil, 2018 minus 2017



Notes: Shaded area shows nominal 95 percent pointwise intervals . Zero exposure is omitted from the display. Profiles summarize conditional Theil given $(X^{(4)}, B)$ separately by sex and education. Here B denotes exposure to 2018 minimum wage increase. Shading shows nominal 95 percent pointwise .

Figure 30: Rolling-exposure change in conditional Theil, 2019 minus 2018



Notes: Shaded areas show nominal 95 percent pointwise intervals. Zero exposure is omitted from the display. Profiles summarize conditional Theil given $(X^{(4)}, B)$ separately by sex and education. Here B denotes exposure to the 2019 minimum-wage increase.

E.4 Wage growth by employer exposure and education

The tables report descriptive wage growth and subsequent observation for workers in the pre-increase sample with complete information on the variables in $X^{(4)}$ with measured rolling exposure. Exposure is constructed for employers with at least ten workers satisfying the construction restrictions before the increase. For comparison $e = (r, s)$, w_r denotes nominal monthly base pay before the increase, and m_r and m_s are the old and new minimum wages. Exposure B is expressed as a percentage of the pre-increase base pay used in its construction. Pay bands and employer exposure are fixed using pre-increase records; workers retain that exposure if they change employer. Education is also measured before the increase and held fixed, with women and men pooled.

"Observations" counts workers with valid real regular hourly wages in both years, denoted N_V. Wage growth is $100(\bar{Y}_{s,V}/\bar{Y}_{r,V} - 1)$, with both means calculated for those same workers. The observation rate is $100N_O/N_0$, where N_0 is the full starting group and N_O counts its members found in the following October's native main-job records, without requiring complete information on $X^{(4)}$ in that year. In all reported rows, $N_O = N_V$. Missing follow-up is not assigned a wage. Each comparison follows its own starting group.

Table 9: Wage growth by employer exposure: workers initially paid between the old and new minima

Comparison	Exposure B	Observed next year (%)	Observations	Real hourly-wage growth (%)
2013–2014	$0 \leq B < 0.5$	74.0	28,924	5.8
	$0.5 \leq B < 1$	70.7	23,867	4.8
	$B \geq 1$	72.5	75,068	4.7
2015–2016	$0 \leq B < 0.5$	74.9	26,037	5.4
	$0.5 \leq B < 1$	73.5	24,971	5.0
	$B \geq 1$	72.1	111,402	4.7
2016–2017	$0 \leq B < 0.5$	75.0	30,347	7.1
	$0.5 \leq B < 1$	74.9	29,284	6.6
	$B \geq 1$	71.9	149,888	5.9
2017–2018	$0 \leq B < 0.5$	73.5	32,145	7.9
	$0.5 \leq B < 1$	76.1	37,463	7.6
	$B \geq 1$	70.9	148,172	6.6
2018–2019	$0 \leq B < 0.5$	72.5	36,381	7.6
	$0.5 \leq B < 1$	69.6	32,843	6.4
	$B \geq 1$	68.7	117,353	6.1

Notes: Initial pay satisfies $m_r \leq w_r < m_s$. The lowest exposure category includes zero by definition, but no workers in this pay band have zero exposure. The observations therefore partition the corresponding positive-exposure group in Table 7.

Table 10: Wage growth by employer exposure: workers initially paid at or just above the new minimum

Comparison	Exposure B	Observed next year (%)	Observations	Real hourly-wage growth (%)
2013–2014	$0 \leq B < 0.5$	77.5	72,367	3.2
	$0.5 \leq B < 1$	76.2	13,303	3.5
	$B \geq 1$	76.9	12,175	3.6
2015–2016	$0 \leq B < 0.5$	78.5	67,171	2.8
	$0.5 \leq B < 1$	78.6	17,813	1.9
	$B \geq 1$	77.2	21,689	3.2
2016–2017	$0 \leq B < 0.5$	79.1	63,298	4.7
	$0.5 \leq B < 1$	80.1	18,940	3.4
	$B \geq 1$	78.2	29,785	3.9
2017–2018	$0 \leq B < 0.5$	78.1	73,160	6.8
	$0.5 \leq B < 1$	80.0	21,284	4.9
	$B \geq 1$	76.0	31,437	6.1
2018–2019	$0 \leq B < 0.5$	75.6	113,007	6.4
	$0.5 \leq B < 1$	74.3	25,518	6.8
	$B \geq 1$	73.1	25,528	7.0

Notes: Initial pay satisfies $m_s \leq w_r < 1.1m_s$. The lowest exposure category includes workers at zero-exposure employers. This sample is therefore broader than the positive-exposure group in Table 7.

Table 11: Wage growth by exposure: below high school, initially between the old and new minima

Comparison	Exposure B	Observed next year (%)	Observations	Real hourly-wage growth (%)
2013–2014	$0 \leq B < 0.5$	75.8	20,332	4.9
	$0.5 \leq B < 1$	74.1	17,810	4.0
	$B \geq 1$	74.3	62,938	4.6
2015–2016	$0 \leq B < 0.5$	75.3	15,694	4.3
	$0.5 \leq B < 1$	74.3	17,333	4.0
	$B \geq 1$	73.4	85,856	3.9
2016–2017	$0 \leq B < 0.5$	75.4	17,480	5.8
	$0.5 \leq B < 1$	75.1	17,950	5.8
	$B \geq 1$	73.1	112,580	5.2
2017–2018	$0 \leq B < 0.5$	73.6	18,213	6.9
	$0.5 \leq B < 1$	76.2	22,402	7.0
	$B \geq 1$	71.3	107,786	5.9
2018–2019	$0 \leq B < 0.5$	72.6	20,425	6.8
	$0.5 \leq B < 1$	69.3	19,582	5.9
	$B \geq 1$	69.1	82,424	5.4

Notes: Initial pay satisfies $m_r \leq w_r < m_s$. Education is measured before the increase; women and men are pooled. No workers in this pay band have zero exposure.

Table 12: Wage growth by exposure: below high school, initially at or just above the new minimum

Comparison	Exposure B	Observed next year (%)	Observations	Real hourly-wage growth (%)
2013–2014	$0 \leq B < 0.5$	78.0	51,235	2.5
	$0.5 \leq B < 1$	77.5	10,098	2.7
	$B \geq 1$	77.2	9,341	3.5
2015–2016	$0 \leq B < 0.5$	79.2	44,675	2.1
	$0.5 \leq B < 1$	78.9	13,247	1.3
	$B \geq 1$	78.0	15,534	2.2
2016–2017	$0 \leq B < 0.5$	79.5	39,708	3.9
	$0.5 \leq B < 1$	80.0	12,446	2.9
	$B \geq 1$	78.8	20,826	3.1
2017–2018	$0 \leq B < 0.5$	78.4	44,300	6.1
	$0.5 \leq B < 1$	79.7	14,122	4.7
	$B \geq 1$	76.5	20,563	5.4
2018–2019	$0 \leq B < 0.5$	75.7	67,124	5.3
	$0.5 \leq B < 1$	73.8	15,518	6.1
	$B \geq 1$	74.0	16,086	5.8

Notes: Initial pay satisfies $m_s \leq w_r < 1.1m_s$. Education is measured before the increase; women and men are pooled. The lowest exposure category includes workers at zero-exposure employers.

Table 13: Wage growth by exposure: high school or some college, initially between the old and new minima

Comparison	Exposure B	Observed next year (%)	Observations	Real hourly-wage growth (%)
2013–2014	$0 \leq B < 0.5$	73.7	7,396	6.6
	$0.5 \leq B < 1$	63.7	5,334	6.6
	$B \geq 1$	65.4	10,772	4.9
2015–2016	$0 \leq B < 0.5$	74.8	8,810	5.5
	$0.5 \leq B < 1$	72.2	6,717	6.8
	$B \geq 1$	68.4	22,385	6.3
2016–2017	$0 \leq B < 0.5$	74.6	10,946	8.0
	$0.5 \leq B < 1$	74.6	9,693	7.1
	$B \geq 1$	68.6	32,889	7.8
2017–2018	$0 \leq B < 0.5$	73.5	12,006	8.0
	$0.5 \leq B < 1$	75.8	12,850	7.8
	$B \geq 1$	69.5	35,675	8.0
2018–2019	$0 \leq B < 0.5$	72.2	13,756	7.6
	$0.5 \leq B < 1$	70.1	11,508	6.8
	$B \geq 1$	67.9	31,171	7.4

Notes: Initial pay satisfies $m_r \leq w_r < m_s$. Education is measured before the increase; women and men are pooled. No workers in this pay band have zero exposure.

Table 14: Wage growth by exposure: high school or some college, initially at or just above the new minimum

Comparison	Exposure B	Observed next year (%)	Observations	Real hourly-wage growth (%)
2013–2014	$0 \leq B < 0.5$	76.6	18,724	3.9
	$0.5 \leq B < 1$	72.7	2,849	5.0
	$B \geq 1$	76.0	2,446	3.6
2015–2016	$0 \leq B < 0.5$	77.5	19,649	3.7
	$0.5 \leq B < 1$	78.1	4,073	3.1
	$B \geq 1$	75.4	5,222	4.7
2016–2017	$0 \leq B < 0.5$	78.6	20,032	5.3
	$0.5 \leq B < 1$	80.6	5,487	3.8
	$B \geq 1$	77.0	7,431	4.9
2017–2018	$0 \leq B < 0.5$	78.0	24,416	6.7
	$0.5 \leq B < 1$	80.7	6,081	4.6
	$B \geq 1$	75.1	9,062	7.0
2018–2019	$0 \leq B < 0.5$	75.5	38,825	6.9
	$0.5 \leq B < 1$	75.0	8,497	7.4
	$B \geq 1$	71.5	7,951	8.1

Notes: Initial pay satisfies $m_s \leq w_r < 1.1m_s$. Education is measured before the increase; women and men are pooled. The lowest exposure category includes workers at zero-exposure employers.

Table 15: Wage growth by exposure: college and above, initially between the old and new minima

Comparison	Exposure B	Observed next year (%)	Observations	Real hourly-wage growth (%)
2013–2014	$0 \leq B < 0.5$	54.1	1,196	14.9
	$0.5 \leq B < 1$	53.3	723	12.3
	$B \geq 1$	56.9	1,358	6.6
2015–2016	$0 \leq B < 0.5$	71.6	1,533	14.9
	$0.5 \leq B < 1$	68.7	921	10.6
	$B \geq 1$	66.0	3,161	12.5
2016–2017	$0 \leq B < 0.5$	73.4	1,921	13.7
	$0.5 \leq B < 1$	74.8	1,641	11.4
	$B \geq 1$	67.4	4,419	10.8
2017–2018	$0 \leq B < 0.5$	73.4	1,926	15.9
	$0.5 \leq B < 1$	77.2	2,211	12.6
	$B \geq 1$	70.8	4,711	10.8
2018–2019	$0 \leq B < 0.5$	73.5	2,200	13.7
	$0.5 \leq B < 1$	69.8	1,753	9.5
	$B \geq 1$	67.6	3,758	11.6

Notes: Initial pay satisfies $m_r \leq w_r < m_s$. Education is measured before the increase; women and men are pooled. No workers in this pay band have zero exposure.

Table 16: Wage growth by exposure: college and above, initially at or just above the new minimum

Comparison	Exposure B	Observed next year (%)	Observations	Real hourly-wage growth (%)
2013–2014	$0 \leq B < 0.5$	75.1	2,408	9.1
	$0.5 \leq B < 1$	72.2	356	11.5
	$B \geq 1$	73.1	388	5.9
2015–2016	$0 \leq B < 0.5$	75.1	2,847	8.9
	$0.5 \leq B < 1$	73.7	493	6.7
	$B \geq 1$	74.9	933	11.2
2016–2017	$0 \leq B < 0.5$	76.8	3,558	11.5
	$0.5 \leq B < 1$	79.4	1,007	7.3
	$B \geq 1$	76.0	1,528	9.8
2017–2018	$0 \leq B < 0.5$	76.3	4,444	13.7
	$0.5 \leq B < 1$	79.8	1,081	8.9
	$B \geq 1$	74.4	1,812	10.6
2018–2019	$0 \leq B < 0.5$	74.8	7,058	14.1
	$0.5 \leq B < 1$	75.8	1,503	11.0
	$B \geq 1$	72.0	1,491	13.8

Notes: Initial pay satisfies $m_s \leq w_r < 1.1m_s$. Education is measured before the increase; women and men are pooled. The lowest exposure category includes workers at zero-exposure employers.

Table 17: Wage growth from the new minimum to less than 20 percent above it

Comparison	Observed next year (%)	Observations	Real hourly-wage growth (%)
2013–2014	78.6	141,270	2.7
2015–2016	79.3	159,081	2.7
2016–2017	80.3	173,726	3.5
2017–2018	79.1	186,887	5.5
2018–2019	76.7	203,199	6.4

Notes: Initial pay satisfies $m_s \leq w_r < 1.2m_s$, and only workers at employers with positive exposure are included. This band includes $m_s \leq w_r < 1.1m_s$ in Table 7; it is an overlapping extension, not a separate band covering only 10–20 percent above the new minimum.

F Data construction

This appendix documents the construction of the worker–year records, the analysis samples, and the wage variables used in Section 5. The order of operations is important: one job is selected for each worker and year before the subsequent sample restrictions are imposed.

F.1 Source files and institutional coverage

The application uses annual worker records from the Portuguese *Quadros de Pessoal* (QP), linked to the corresponding firm and establishment records for the reference years 2011–2019. QP is compiled by the Gabinete de Estratégia e Planeamento from mandatory employer reports submitted through Annex A of the *Relatório Único*. The reference month is October. The linked files contain worker remuneration, hours, demographic and job characteristics, together with firm characteristics, establishment location, and persistent worker and employer identifiers. Public Administration employees covered by civil-service regimes are outside the coverage used here (Card et al., 2016; Cardoso & Portugal, 2005; Gabinete de Estratégia e Planeamento, 2016, 2021).

F.2 Selection of one job per worker-year

Records with a missing worker identifier are removed. For each remaining worker-year, the job records are ordered using the following priorities:

1. wage-earner status;
2. positive, nonmissing regular monthly pay and positive, nonmissing normal monthly hours;
3. higher normal monthly hours;
4. higher regular monthly pay;
5. higher total earnings;
6. higher monthly base pay; and
7. ascending firm identifier and then ascending establishment identifier to resolve any remaining ties.

The first record in this ordering is retained, and worker-year uniqueness is verified after selection. The selection is performed before the remaining analysis-sample restrictions. Thus, the rule does not select the highest-paying job from the subset of records that already satisfy all later restrictions; it first defines one main job for every worker-year and then applies the sample criteria below.

F.3 Analysis populations

After the main-job selection, the basic annual sample retains native wage earners aged 18–64 who work full time in mainland Portugal, have complete remuneration information, positive

regular monthly pay and normal monthly hours, and a matched employer record. Workers in agriculture and workers identified as apprentices are excluded. Figure 3 uses this basic sample.

The nested decompositions and conditional-inequality estimates additionally require complete information on the variables in Table 5. In particular, the common sample used to compare the four nested conditioning sets is the annual $X^{(4)}$-complete-case sample. Holding this sample fixed across conditioning sets ensures that each incremental between component reflects a change in conditioning information rather than a change in observations. The exposure analyses impose the further requirement that the worker's current employer have a measured initial or rolling exposure value. Workers at employers without a qualifying exposure measure are excluded only from those exposure analyses.

The construction therefore proceeds in the following order:

1. link the annual worker records to the firm and establishment files;
2. select one main job per worker-year using the rule in Appendix F.2;
3. impose the basic worker, job, pay, hours, geographic, and employer restrictions;
4. impose complete-case requirements for the conditioning information when estimating the nested decompositions and conditional profiles; and
5. attach employer exposure and impose the exposure-availability restriction only for the exposure exercises.

F.4 Hourly wage measures and price adjustment

Let w_{it}^{base} denote monthly base pay, w_{it}^{reg} regular monthly supplements, and h_{it} reported normal monthly hours in the selected job. Regular monthly pay is

$$R_{it} = w_{it}^{\text{base}} + w_{it}^{\text{reg}}.$$

The nominal regular hourly wage is R_{it}/h_{it}. The denominator is the reported monthly normal-hours field (`hnormais`); it is not constructed from contractual weekly hours. Normal weekly hours remain available as a separate conditioning characteristic. Overtime pay and irregular bonuses are not included in the numerator.

Nominal wages are converted to October-2019 euros using the October value of the all-items consumer price index for Portugal published by Statistics Portugal. Let $P_{t,\text{Oct}}$ denote that index and define

$$c_t = \frac{P_{2019,\text{Oct}}}{P_{t,\text{Oct}}}.$$

The main outcome is therefore

$$Y_{it} = c_t \frac{w_{it}^{\text{base}} + w_{it}^{\text{reg}}}{h_{it}}.$$

The CPI extraction uses INE indicator 0014639, with Portugal, all items, and October observations selected (Instituto Nacional de Estatística, n.d.). Because the same positive deflator applies to every worker within a year, the annual Theil index and its within–between decomposition are unchanged by the nominal-to-real conversion. Real wages are nevertheless required for

comparisons of wage levels and wage growth across years.

F.5 Exposure linkage and follow-up samples

Employer exposure is constructed from pre-increase monthly base pay as described in Section 5.3. Initial exposure uses the 2013 employer wage structure for the 2014 increase and is held fixed in the 2011–2019 analysis. Rolling exposure is recomputed in the year preceding each increase and is held fixed across the two annual observations in that comparison. In the annual profile samples, workers receive the exposure of their current employer. The annual samples need not contain the same workers over time.

The separate wage-growth analysis begins with workers in the pre-increase sample and follows them into the next annual file. Those workers retain their pre-increase employer's exposure classification if they move. A worker counts as observed in the following year when they are found in that year's selected main-job records; complete information on $X^{(4)}$ is not required again for this observation indicator. Wage growth is calculated only for workers with observed real regular hourly wages in both years, and missing follow-up wages are not imputed.

References

Alvarez, J., Benguria, F., Engbom, N., & Moser, C. (2018). Firms and the decline in earnings inequality in Brazil. *American Economic Journal: Macroeconomics*, *10*(1), 149–189.

Autor, D. H., Manning, A., & Smith, C. L. (2016). The contribution of the minimum wage to U.S. wage inequality over three decades: A reassessment. *American Economic Journal: Applied Economics*, *8*(1), 58–99.

Belloni, A., Chernozhukov, V., Chetverikov, D., & Kato, K. (2015). Some new asymptotic theory for least squares series: Pointwise and uniform results. *Journal of Econometrics*, *186*(2), 345–366.

Bergolo, M. L., Ceni, R., Fondo, M., & Vergara, D. (2025). *Minimum wages and the distribution of firm wage premia* (Working Paper No. 34188). NBER. doi: 10.3386/w34188

Bhardwaj, V., & Díaz-Villarejo, D. (2026). *Gini index regression* (Working paper). Universidad Carlos III de Madrid.

Bourguignon, F. (1979). Decomposable income inequality measures. *Econometrica*, *47*(4), 901–920.

Brown, C., & Medoff, J. (1989). The employer size-wage effect. *Journal of Political Economy*, *97*(5), 1027–1059.

Cameron, A. C., Gelbach, J. B., & Miller, D. L. (2011). Robust inference with multiway clustering. *Journal of Business & Economic Statistics*, *29*(2), 238–249.

Card, D., Cardoso, A. R., Heining, J., & Kline, P. (2018). Firms and labor market inequality: Evidence and some theory. *Journal of Labor Economics*, *36*(S1), S13–S70.

Card, D., Cardoso, A. R., & Kline, P. (2016). Bargaining, sorting, and the gender wage gap: Quantifying the impact of firms on the relative pay of women. *Quarterly Journal of Economics*, *131*(2), 633–686.

Card, D., Heining, J., & Kline, P. (2013). Workplace heterogeneity and the rise of West German wage inequality. *Quarterly Journal of Economics*, *128*(3), 967–1015. doi: 10.1093/qje/qjt006

Cardoso, A. R., & Portugal, P. (2005). Contractual wages and the wage cushion under different bargaining settings. *Journal of Labor Economics*, *23*(4), 875–902.

Chen, T., & Guestrin, C. (2016). XGBoost: A scalable tree boosting system. In *Proceedings of the 22nd ACM SIGKDD international conference on knowledge discovery and data mining* (pp. 785–794). New York: ACM.

Chen, X. (2007). Large sample sieve estimation of semi-nonparametric models. In J. J. Heckman & E. E. Leamer (Eds.), *Handbook of econometrics* (Vol. 6B, pp. 5549–5632). Amsterdam: Elsevier.

Chen, X., & Christensen, T. M. (2015). Optimal uniform convergence rates and asymptotic normality for series estimators under weak dependence and weak conditions. *Journal of Econometrics*, *188*(2), 447–465.

Chernozhukov, V., Chetverikov, D., Demirer, M., Duflo, E., Hansen, C., Newey, W., & Robins, J. (2018). Double/debiased machine learning for treatment and structural parameters. *The Econometrics Journal*, *21*(1), C1–C68.

Chernozhukov, V., Escanciano, J. C., Ichimura, H., Newey, W., & Robins, J. (2022). Locally robust semiparametric estimation. *Econometrica*, *90*(4), 1501–1535.

Chiang, H. D., Kato, K., Ma, Y., & Sasaki, Y. (2022). Multiway cluster robust double/debiased machine learning. *Journal of Business & Economic Statistics*, *40*(3), 1046–1056. doi: 10.1080/07350015.2021.1895815

DiNardo, J., Fortin, N. M., & Lemieux, T. (1996). Labor market institutions and the distribution of wages, 1973–1992: A semiparametric approach. *Econometrica*, *64*(5), 1001–1044.

Draca, M., Machin, S., & Van Reenen, J. (2011). Minimum wages and firm profitability. *American Economic Journal: Applied Economics*, *3*(1), 129–151.

Dustmann, C., Lindner, A., Schönberg, U., Umkehrer, M., & vom Berge, P. (2022). Reallocation effects of the minimum wage. *Quarterly Journal of Economics*, *137*(1), 267–328.

Engbom, N., & Moser, C. (2022). Earnings inequality and the minimum wage: Evidence from Brazil. *American Economic Review*, *112*(12), 3803–3847.

Escanciano, J. C., & Terschuur, J. R. (2026). *Debiased machine learning U-Statistics.* arXiv:2206.05235. (June 9, 2026 version)

Foster, J. E., & Ok, E. A. (1999). Lorenz dominance and the variance of logarithms. *Econometrica*, *67*(4), 901–907. doi: 10.1111/1468-0262.00057

Gabinete de Estratégia e Planeamento. (2016). *Quadros de Pessoal 2015* (Coleção Estatísticas). Lisboa: Ministério do Trabalho, Solidariedade e Segurança Social. Retrieved from https://catesoc.gep.mtsss.gov.pt/uploads/eecb57bb116c7fd22bfcd8ced818e0e7.pdf

Gabinete de Estratégia e Planeamento. (2021). *Quadros de Pessoal 2019* (Statistical report). Lisboa: Ministério do Trabalho, Solidariedade e Segurança Social. Retrieved from https://www.gep.mtsss.gov.pt/documents/10182/10928/qp2019pub.pdf

Garnero, A., Kampelmann, S., & Rycx, F. (2015). Minimum wage systems and earnings

inequalities: Does institutional diversity matter? *European Journal of Industrial Relations*, *21*(2), 115–130.

Instituto Nacional de Estatística. (n.d.). *Índice de preços no consumidor (IPC, base - 2025) por localização geográfica e consumo individual por objetivo; mensal.* Statistics Portugal database, indicator 0014639. Retrieved from https://www.ine.pt/xurl/indx/0014639/PT (Selection used: Portugal; Total (all items); October observations, 2011–2019)

International Labour Organization. (2024). *Global wage report 2024–25: Is wage inequality decreasing globally?* Geneva: International Labour Organization. doi: 10.54394/CJQU6666

Katz, L. F., & Murphy, K. M. (1992). Changes in relative wages, 1963–1987: Supply and demand factors. *Quarterly Journal of Economics*, *107*(1), 35–78.

Krueger, A. B., & Summers, L. H. (1988). Efficiency wages and the inter-industry wage structure. *Econometrica*, *56*(2), 259–293.

Lee, D. S. (1999). Wage inequality in the United States during the 1980s: Rising dispersion or falling minimum wage? *Quarterly Journal of Economics*, *114*(3), 977–1023.

Leitão, M., Montaña, J., & Silva, J. (2022). *The role of firms in wage inequality dynamics* (Discussion Paper No. 17327). CEPR.

Lemieux, T. (2006). Increasing residual wage inequality: Composition effects, noisy data, or rising demand for skill? *American Economic Review*, *96*(3), 461–498.

Liang, Y. (2026). *Unpacking wage inequality: Minimum wage effects on within- and between-firm disparities.* (Working paper, May 2026)

Linton, O., & Nielsen, J. P. (1995). A kernel method of estimating structured nonparametric regression based on marginal integration. *Biometrika*, *82*(1), 93–100.

Machin, S., Manning, A., & Rahman, L. (2003). Where the minimum wage bites hard: Introduction of minimum wages to a low wage sector. *Journal of the European Economic Association*, *1*(1), 154–180.

Moretti, E. (2011). Local labor markets. In O. Ashenfelter & D. Card (Eds.), *Handbook of labor economics* (Vol. 4B, pp. 1237–1313). Amsterdam: Elsevier.

Newey, W. K. (1994a). The asymptotic variance of semiparametric estimators. *Econometrica*, *62*(6), 1349–1382.

Newey, W. K. (1994b). Kernel estimation of partial means and a general variance estimator. *Econometric Theory*, *10*(2), 233–253.

NIST Digital Library of Mathematical Functions. (2026). *Section 18.15(iii): Legendre.* Asymptotic Approximations for Classical Orthogonal Polynomials. Retrieved 2026-09-14, from https://dlmf.nist.gov/18.15.iii (Accessed September 14, 2026)

Oka, T., & Yamada, K. (2023). Heterogeneous impact of the minimum wage: Implications for changes in between- and within-group inequality. *Journal of Human Resources*, *58*(1), 335–362. doi: 10.3368/jhr.58.3.0719-10339R1

Oliveira, C. (2023). The minimum wage and the wage distribution in Portugal. *Labour Economics*, *85*, 102459. doi: 10.1016/j.labeco.2023.102459

Oliveira, C. (2024). Income and wage inequality in democratic Portugal, 1974–2020. *Fiscal Studies*, *45*(3), 393–414. doi: 10.1111/1475-5890.12391

Semenova, V., & Chernozhukov, V. (2021). Debiased machine learning of conditional average treatment effects and other causal functions. *The Econometrics Journal*, *24*(2), 264–289.

Shorrocks, A. F. (1980). The class of additively decomposable inequality measures. *Econometrica*, *48*(3), 613–625.

Song, J., Price, D. J., Guvenen, F., Bloom, N., & von Wachter, T. (2019). Firming up inequality. *Quarterly Journal of Economics*, *134*(1), 1–50. doi: 10.1093/qje/qjy025

Theil, H. (1967). *Economics and information theory*. Amsterdam: North-Holland.